Quadratic orbit maps and three-Engel Lie algebras in characteristic two

Wencong Huang, Baohua Li

Version 2026-11-02.2

Abstract

The equation $f \circ f = f + \mathrm{id}$ couples three-cycles to binary addition. We prove that every finite quadratic solution has a nilpotent polar Lie algebra and describe the whole category by ordinary $\mathbb{F}_4$-Lie algebras. Explicit mutual changes of operations identify its image with the three-Engel algebras satisfying a sextic identity, preserving all linear intertwiners, centers and lower-central terms. Every solution admits a linearizing change of coordinates that is quadratic in both directions. On the full ordinary three-Engel category the same construction gives cubic orbit maps, with third additive difference exactly the sextic obstruction. The canonical scalar kernel is a natural image of ordinary second Lie homology. Every object has a class- and head-rank-preserving central repair whose minimum ordinary dimension cost is the dimension of this image; the minimum extension class is unique over the given bottom. For compact indecomposable bottoms at every even point rank we determine the sharp total cost of the first central ascent, including a rank-four exception. Half-weight repair and a cubic preparation explain why scalar repair alone does not unlock ascent. Restricted envelopes, finite free covers and cyclic cohomology provide the structural inputs; the resulting general constructions also give nonmetabelian class-c models in dimension $O(2^{c/2})$.

Keywords. Quadratic map; three-Engel Lie algebra; restricted square; polynomial reconstruction; Lie homology; central extension.

Contents

1 Introduction **3**

2 The polar Lie algebra **5**

3 Solvability and nilpotence **7**

4 Global nilpotence of the polar algebra **8**

5 Central reconstruction and cyclic Lie cohomology **9**

6 The free two-point model **11**

7 Canonical structure beyond the metabelian range **12**

8 General growth and nonmetabelian dimension compression **16**

9 Exact free scalars and the universal quotient **19**

10 An ordinary Lie description of every quadratic solution **22**

11 The exact cost of repairing the derived square **25**

12 Scalar obstructions and central ascent **26**

13 A sharp first ascent over the compact bottoms **28**

14 Further questions **30**

A Proof of global polar nilpotence **30**

B Proofs for the canonical general structure **33**

C Proofs for central reconstruction **40**

D Fixed quadratic rigidity and the complete free-quotient obstruction **42**

E Global stability and the universal restricted derived quotient **43**

F Proofs for general growth and finite universal objects **50**

G The current family and its binary trace restriction **57**

H Proofs of the ordinary Lie reconstruction **61**

I Proofs for class-four obstructions and central covers **64**

J Proofs of exact repair and the sharp first ascent **67**

K Exact inputs and reproducibility **72**

1 Introduction

Let V be a finite binary vector space. The equation

$$f(f(x)) = f(x) + x \tag{1}$$

makes each nonzero orbit the three nonzero points of a binary plane. We study maps of binary algebraic degree at most two. The central question is how these orbit planes interact with the given addition, and which familiar algebraic structures describe all such interactions.

The answer is an ordinary Lie category with an explicit identity. Put $k = \mathbb{F}_4$, with $\omega^2 + \omega = 1$, and let $\mathcal{E}$ consist of finite-dimensional ordinary k-Lie algebras satisfying

$$[y, x, x, x] = 0, \qquad \mathcal{R}(a, b, c) = 0,$$

where, for left-normed brackets,

$$q(a, b) = [a, b, b, a], \qquad \mathcal{R}(a, b, c) = q([a, b], c) + q([b, c], a) + q([c, a], b) + [[a, b, c], [b, c, a]].$$

Theorem 10.1 gives mutual changes of addition and bracket between this category and all finite quadratic solutions of (1). Every binary-linear intertwiner is retained. Centers, lower-central terms, class, point rank and actual automorphism groups are preserved. The new ordinary operations differ from the original binary polar operations, and retain the entire underlying set.

There are two strong degree consequences. Every quadratic solution has a conjugator $\eta f = \omega \eta$ with

$$\deg_{\mathbb{F}_2} \eta \leq 2, \qquad \deg_{\mathbb{F}_2} \eta^{-1} \leq 2.$$

These bounds are independent of dimension and nilpotency class. On the entire ordinary three-Engel category the inverse construction still defines binary addition and an orbit rule of degree at most three. Its third additive difference is exactly $\mathcal{R}$. Thus the sextic identity specifies precisely the quadratic locus inside this larger encoding, and a 21-dimensional ordinary example shows that the cubic bound is sharp.

1.1 From polar structure to the ordinary correspondence

The original polar form $B(x, y) = f(x + y) + f(x) + f(y)$ is a Lie bracket. We first prove its global nilpotence (Theorem 4.1). Special biderivations constrain possible simple factors; Premet's Cartan-subalgebra theorem [11, Theorem 3(ii)] closes the binary-centroid endpoint. The explicit free two-point object then gives identities in every finite solution. A reduced restricted structure and a character argument prove quadratic stability of the distinguished biderivation.

The proof order matters. A preliminary class bound first constructs finite free covers. Cyclic cohomology and the fixed quadratic sector then force exact derived scalars on those covers. The resulting whole-solution restricted envelope allows the change $I+P$; its operations descend as fixed terms to arbitrary quotients. Relative PBW and the Hopf formula identify the exact ordinary image. None of these later steps is used to prove the preceding global nilpotence theorem. The detailed proofs specify the finite coefficient inputs and their roles.

The point rank is the least number of generators under binary addition and f; it equals $d(L_f) = \dim_k L_f / L'_f$. General dimension applications compare polar class with ordinary three-Engel class growth and give nonmetabelian class-c models in binary dimension $O(2^{c/2})$. We write $m(c)$ for the least dimension among all solutions of polar class c, and $\mu(c)$ for the corresponding metabelian minimum, determined separately in [20].

1.2 The homological obstruction and central covers

The restricted envelope can have a central source kernel N_f. The ordinary correspondence gives a precise explanation: $N_f = \operatorname{im} \lambda_{H_f}$ for a natural Frobenius-semilinear map from ordinary second Lie homology to the center (Theorem 10.4). Its quotient is the universal central reflection admitting a specified derived restricted square. The whole homology group need not vanish.

The obstruction also determines an exact enlargement cost. Theorem 11.1 constructs a minimum central stem repair $A_H \to H$ with $N(A_H) = 0$, preserving class and head rank. Its ordinary dimension cost is $\dim_k N(H)$, and its extension isomorphism class over H is unique. Every partial reduction of the obstruction is attained at its exact corresponding cost. For a bracket-length grading, repair coordinates occur at half the obstruction weight. They therefore do not unlock an ascent from a centrally rigid bottom.

In class four, the obstruction is controlled by head relations and their decomposable bivectors. We give the exact four-generator Plücker criterion for existence of an obstructed lift, and compact indecomposable families at all even ranks $2r \geq 4$ with obstruction dimension $\binom{r}{2}$. Removing that obstruction by a central cover costs exactly $\binom{r}{2}$ dimensions. Both the compact base S_r and a minimum cover still resist a class-raising central three-Engel extension. Theorem 13.1 goes further: among same-head class-five epimorphisms $G \to S_r$ with $N(G) = 0$, the total added ordinary dimension is exactly $\binom{r}{2} + 3$ for $r \geq 3$, and is five at $r = 2$. The lower bounds allow arbitrarily long central towers. Three arrows suffice; doing the minimum repair alone first requires four in total. An explicit cubic preparation attains the repair-plus-ascent bound $n(H) + r_{\mathrm{eff}}$. These are relative minima over the specified bottoms, rather than a formula for the global $m(5)$.

1.3 Context, conventions and organization

Complete mappings carry a classical order-three parastrophic action [9]; its strict fixed objects satisfy (1). Related three-cycle constructions are studied in [7, 15, 8]. Quaternary Mendelsohn quasigroups and their central descriptions occur in [16]. The present question keeps the original binary addition and its degree filtration throughout the structural analysis.

Restricted squares and quadratic maps have the classical $I + P$ interface [10, Section 9]. Restricted and ordinary extension comparisons, semilinear changes of a full restricted map, and reduction of a fixed nonabelian kernel to its center are classical [5, Theorems 3.1–3.3 and 4.1, p. 575]. Our repair problem instead prescribes a square only on the derived algebra, including its commutator values and square-zero condition. The distinction concerns the structures being repaired; the ordinary central-cover and cohomology methods retain their classical attribution. Bahturin–Olshanskii [1, Theorem 2.4] give a general recovery theorem for nilpotent polynomial quasigroups, and related Lie/loop changes of operations occur in [4, Section 5]. Here the explicit operations identify the precise three-Engel image, control both binary degrees uniformly, and identify the exact cubic defect. The Bogomolov quotient, decomposable-bivector geometry and Galois classification of underlying Lie algebras have the established sources [18, 19, 17]; their specified scalar-kernel applications are developed below.

The metabelian extremal and population theory is developed in the companion [20]; ordinary central-depth stability and optimal centers are treated in [21]. These are distinct questions arising from the unified research manuscript [22]. Each central structural input used here is proved within this paper.

Unless otherwise stated, dimensions are binary, brackets are alternating and left-normed, $L^{(j+1)} = [L^{(j)}, L^{(j)}]$, and $\gamma_{j+1}(L) = [\gamma_j(L), L]$. A nonzero abelian algebra has class one. The main correspondence and degree conclusions are in Section 10, their applications in Sections 11–13. The earlier sections establish the polar and restricted inputs; the appendices

give the longer proofs.

2 The polar Lie algebra

Proposition 2.1. *Every solution of* (1) *satisfies*

$$f^3 = \mathrm{id}, \qquad f(0) = 0, \qquad f(x + f(x)) = x.$$

Its only fixed point is zero. Every nonzero orbit spans a binary plane, and these planes partition $V \setminus \{0\}$. *In particular* $\dim V$ *is even.*

Proof. Evaluate (1) at $f(x)$. It gives $f^3(x) = f^2(x) + f(x) = x$, so f is bijective. If $a = f(0)$, then $f(a) = a$; applying f^3 at zero gives $a = 0$. A fixed point in (1) must be zero. The remaining identity follows from $x + f(x) = f^2(x)$. The nonzero orbits have length three and sum zero, and $3 \mid 2^{\dim V} - 1$ implies the parity assertion. □

Theorem 2.2. *Let* f *be a quadratic solution. Its polar form* $[x, y] = f(x + y) + f(x) + f(y)$ *is a Lie bracket. It satisfies*

$$[x, f(x)] = 0, \tag{2}$$

$$f([x, y]) = [f(x), f(y)] + [x, y] + [f(x) + f(y), [x, y]], \tag{3}$$

$$[[a, b], [c, d]] + [[a, c], [b, d]] + [[a, d], [b, c]] = 0. \tag{4}$$

Proof. The polar is alternating and bilinear because f is quadratic and $f(0) = 0$. Expand $f(x + f(x)) = x$ by polarization. The terms $f(x) + f^2(x)$ sum to x, leaving (2). Take the third additive difference at zero of $x \mapsto [x, f(x)]$. The terms involving individual values of f cancel, leaving

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

This is Jacobi.

Next polarize $f^2 = f + \mathrm{id}$. Since $f(x + y) = f(x) + f(y) + [x, y]$, the expansion of the left side gives (3).

For (4), introduce Boolean variables $t_1, \ldots, t_4$ and vectors $x_1, \ldots, x_4$. The exact expansion is

$$f\Big(\sum_i t_i x_i\Big) = \sum_i t_i f(x_i) + \sum_{i<j} t_i t_j [x_i, x_j].$$

In applying f again, the coefficient of $t_1 t_2 t_3 t_4$ can only come from a polar cross term between two disjoint pair summands. There are three such pairings, giving the left side of (4). The composition has degree at most two, so this coefficient vanishes. All coefficient comparisons take place in the Boolean function algebra, with $t_i^2 = t_i$. □

The two polarization steps have distinct roles. Binary degree two gives an alternating bilinear form; the commuting identity (2) forces its Jacobi identity. The fourth polarization then supplies (4), an additional restriction on the resulting Lie algebra. The polynomial in (4) is the associative standard polynomial of degree four expressed as a Lie element in characteristic two [2, Theorem 9]. Its forced occurrence here gives the explicit exclusions in the ordinary reconstruction in Section 10.

Proposition 2.3. *All terms of the derived, lower central, and upper central series of the polar Lie algebra* $L = (V, [\ ,\ ])$ *are* f*-invariant. The center is* f*-invariant and* f *is linear on it. Every nonzero factor of any of these series has even binary dimension.*

Proof. Let I be an f-invariant ideal. In (3), take $x \in I$. Each term on the right belongs to $[I, L]$, which is an ideal. Quadratic expansion extends invariance from bracket generators to their span: the extra polar terms also lie in that ideal. The same argument with $x, y \in I$ proves invariance of $[I, I]$, using that it is an ideal of L. Induction starts with $I = L$.

Polarization of (2) gives

$$[x, f(y)] + [y, f(x)] + [x, [x, y]] + [y, [x, y]] = 0. \tag{5}$$

If $x \in Z(L)$, this says $[y, f(x)] = 0$ for every y. Thus $Z(L)$ is invariant; its polar is zero, so f is linear there. Pass successively to quotients by invariant central ideals to obtain the upper central assertion. Every series factor inherits a solution of (1), so Proposition 2.1 proves its parity. □

Lemma 2.4 (Ideal descent)**.** *For every Lie ideal I of L,*

$$\{f(x) + I : x \in I\} \subseteq Z(L/I).$$

Every perfect ideal is f-invariant. Every ideal I for which $Z(L/I) = 0$ is also f-invariant.

Proof. For $x \in I$, all terms in (5) except $[y, f(x)]$ belong to I. This proves the inclusion. The map $\phi : I \to L/I$, $x \mapsto f(x) + I$, is binary-linear because its polar takes values in I. Equation (3) gives $\phi([x, y]) = [\phi(x), \phi(y)]$ for $x, y \in I$. Its image is central, so $\phi([I, I]) = 0$. This proves the perfect-ideal assertion. The centerless-quotient assertion follows immediately from the inclusion. Bijectivity and finiteness turn each invariant inclusion into equality. □

Corollary 2.5. *If a quadratic solution has a nonsolvable polar algebra, then there is a quadratic solution whose polar algebra is nonabelian simple, of even binary dimension no larger than the original dimension.*

Proof. The stabilized derived ideal P is nonzero and perfect, hence invariant. Choose a maximal proper Lie ideal M of P. The quotient P/M is nonabelian simple and centerless. Apply Lemma 2.4 inside P to see that M is invariant. Restriction and quotient therefore give the required quadratic solution. □

Lemma 2.6. *A nonabelian nilpotent polar Lie algebra has binary abelianization dimension at least four.*

Proof. The abelianization is a nonzero even-dimensional space with the linear action induced by f. If its dimension were two, any nonzero coset $x + L'$ and its image $f(x) + L'$ would form a binary basis. The subalgebra H generated by $x, f(x)$ is abelian by (2). Moreover $L = H + L'$. Inductively, $L = H + \gamma_j(L)$ implies $L = H + \gamma_{j+1}(L)$, because $[H + \gamma_j, H + \gamma_j] \subseteq H + \gamma_{j+1}$. Nilpotency then gives $L = H$, a contradiction. □

Proposition 2.7 (Canonical graded scalars)**.** *For any quadratic solution, the operators induced by f make $\mathrm{gr}_\gamma L$ canonically an $\mathbb{F}_4$-Lie algebra. This assertion concerns the lower-central graded algebra and requires neither metabelianity nor nilpotency.*

Proof. Write F_i for the induced operator on γ_i/γ_{i+1}. Proposition 2.3 and $[\gamma_i, \gamma_i] \subseteq \gamma_{i+1}$ show that F_i is linear and $F_i^2 + F_i = \mathrm{id}$. For $x \in \gamma_i$, $y \in \gamma_j$, reduce (5) modulo γ_{i+j+1} to get $[F_i\bar{x}, \bar{y}] = [\bar{x}, F_j\bar{y}]$. The double brackets have depths at least $i + j + 1$. Equation (3) on the same quotient gives

$$F_{i+j}[\bar{x}, \bar{y}] = [F_i\bar{x}, F_j\bar{y}] + [\bar{x}, \bar{y}].$$

Apply the first identity to $(\bar{x}, F_j\bar{y})$, then use $F_j^2 = F_j + \mathrm{id}$. The right side is $[\bar{x}, F_j\bar{y}]$, proving $[F_i\bar{x}, \bar{y}] = [\bar{x}, F_j\bar{y}] = F_{i+j}[\bar{x}, \bar{y}]$. Thus the graded bracket is $\mathbb{F}_4$-bilinear. □

The induced scalars are intrinsic. A lift to the original space need not make its polar bracket $\mathbb{F}_4$-bilinear; that stronger condition is the subject of the metabelian compatibility problem in [20]. Theorem 4.1 shows that this filtration exhausts the whole algebra for every finite solution.

Example 2.8. Let $\mathbb{F}_{16}$ contain $\mathbb{F}_4$, let $\eta \in \mathbb{F}_4 \setminus \mathbb{F}_2$, and write $\chi(x) = 1$ off $\mathbb{F}_4$ and $\chi(x) = 0$ on $\mathbb{F}_4$. The switching construction of [7, Theorem 22], with seed $x \mapsto \eta x$, direction $\gamma = 1$, and Boolean selector χ, gives the following map; it is also included in [15, Corollary 6]:

$$G(x) = \eta x + \chi(x) = \eta x + \mathrm{Tr}_{\mathbb{F}_{16}/\mathbb{F}_4}(x)^3.$$

The trace is binary-linear, and its cube is binary-quadratic. Both $\mathbb{F}_4$ and its complement are invariant under multiplication by η and under translation by one. Hence G is a permutation with $G^3 = \mathrm{id}$ and only zero fixed. The same holds for $G + \mathrm{id}$. Nevertheless

$$G^2(x) + G(x) + x = \eta\chi(x),$$

so G does not satisfy (1). This separates the equation studied here from simultaneous triple-cycle complete permutation behavior even within binary degree two.

3 Solvability and nilpotence

Assume in this section that the polar Lie algebra is metabelian. Then $W = L'$ is abelian, $f|_W$ is linear, and the induced map on $U = V/W$ is linear. Both operators satisfy $T^2 + T = \mathrm{id}$. We give U and W the $\mathbb{F}_4$-structures in which these operators are multiplication by ω.

Lemma 3.1 (The operator pencil)**.** *In any binary-linear splitting of $V \to U$, write*

$$f(x, z) = (\omega x, \omega z + L_x z + q(x)). \tag{6}$$

The binary-linear pencil $x \mapsto L_x$ satisfies

$$L_x\omega = \omega L_x, \quad L_x^3 = 0, \quad L_{\omega x} = \omega L_x + L_x^2, \quad L_x L_y = L_y L_x, \quad L_x^2 L_y = 0.$$

Proof. Put $J = \omega\,\mathrm{id}_W$ and $M_x = J + L_x$. Both M_x and $\mathrm{id} + M_x$ are invertible: they are the restrictions on coset differences of the permutations f and $f + \mathrm{id} = f^2$. Substitution in (1) gives

$$M_{\omega x} M_x = \mathrm{id} + M_x.$$

For $A = M_x$, the three matrices along the ω-orbit are $A, \mathrm{id} + A^{-1}, (\mathrm{id} + A)^{-1}$. Their sum is J, because the three L-terms sum to zero. Thus J is a rational function of A, and in particular commutes with it. Clearing the denominators gives

$$JA(\mathrm{id} + A) = A^3 + A + \mathrm{id}\,.$$

Since $J^2 + J = \mathrm{id}$ and $J^3 = \mathrm{id}$, this is $(A + J)^3 = 0$. It follows that L_x commutes with J and is cube zero. Expanding $(J + L_x)^{-1}$ now gives $L_{\omega x} = JL_x + L_x^2$. The additivity of the left side forces the L_x's to commute.

For $P = L_x, Q = L_y$, the identity $(P + Q)^3 = 0$ yields $P^2Q = PQ^2 =: T$. Apply the same identity to $P' = JP + P^2 = L_{\omega x}$ and Q. It gives $(J^2 + J)T = P^2Q^2$, hence $T = P^2Q^2 = PT$. As P is nilpotent, this implies $T = 0$. □

Corollary 3.2. *Every finite metabelian polar algebra is nilpotent.*

Proof. The pencil operators in Lemma 3.1 commute and are nilpotent. Finitely many of them span the pencil, so the associative ideal they generate is nilpotent. Its successive common kernels give a strictly triangular flag on W. For every x, the adjoint on L/W is zero and its action on W is one of these nilpotent operators. Engel's theorem proves the assertion. □

We now drop the metabelian hypothesis.

Theorem 3.3. *A finite-dimensional polar Lie algebra is solvable if and only if it is nilpotent. Its solvable radical is invariant under f and is nilpotent.*

Proof. Induct on derived length. The abelian case is immediate. Put $W = L'$, $W' = L''$. The restricted solution on W has nilpotent polar algebra by induction. The quotient L/W' is metabelian, hence nilpotent by Corollary 3.2. For fixed $x \in L$, the derivation $D = \operatorname{ad} x|_W$ is therefore nilpotent on W/W'.

We use the following elementary derivation fact. If N is nilpotent and D is nilpotent on N/N', then it is nilpotent on N. Each factor $\gamma_i(N)/\gamma_{i+1}(N)$ is a quotient of $(N/N')^{\otimes i}$. The induced derivation is the sum of the i commuting operators acting in individual tensor positions. If $D^s = 0$ on N/N', the sum has power $i(s-1)+1$ equal to zero, by the pigeonhole principle on its expanded products. The finite lower central filtration proves the assertion on N, in every characteristic.

Apply this fact to $N = W$. Since $\operatorname{ad} x$ maps L into W, it is nilpotent on L. Engel's theorem now gives nilpotency of L. The converse is immediate. If R is the solvable radical, L/R has zero center. Lemma 2.4 makes R invariant, and the result just proved applies to the restricted solution. □

4 Global nilpotence of the polar algebra

The polar identities do more than equip a quadratic orbit system with a Lie bracket: they exclude a nonsolvable polar algebra altogether. Thus nilpotence is a consequence of the orbit equation, rather than an additional hypothesis on the solutions considered below.

Theorem 4.1 (Global polar nilpotence)**.** *Let V be a finite-dimensional binary vector space and let $f : V \to V$ have binary algebraic degree at most two and satisfy $f \circ f = f + \mathrm{id}$. Its polar Lie algebra L_f is nilpotent.*

The proof combines identities specific to the functional equation with classical finite-dimensional Lie theory. Here is the structural step that makes the combination possible. Put $B(x, y) = [x, y]$ and, in the ordinary universal enveloping algebra of L_f, define

$$\begin{aligned} R_x = x^2 + f(x), \quad D(x,y) = [R_x, y] = [f(x), y] + [x, [x, y]], \\ G(x,y) = [R_x, R_y], \qquad S(x,y) = G(x,y) + B(x,y) + D(x,y). \end{aligned} \tag{7}$$

Although f is generally nonlinear, R is binary-linear. The three maps D, G, S take values in L'_f and are alternating biderivations. Moreover,

$$G([x, y], z) = [G(x, y), z]. \tag{8}$$

Consequently, on a perfect centerless polar algebra, G factors through the bracket by an element of the centroid.

Proof of Theorem 4.1: roadmap. By Corollary 2.5, a nonsolvable polar algebra would have a nonabelian simple polar subquotient. On such a simple algebra the finite centroid is a field K, and (8) gives $G = \kappa B$ for some $\kappa \in K$. The biderivation identities impose polynomial

constraints on $\operatorname{ad} x$. If $|K| \geq 8$, they give $(\operatorname{ad} x)^4 = 0$; if $K = \mathbb{F}_4$, they give $(\operatorname{ad} x)^5 = 0$. Engel's theorem excludes both cases.

The remaining case $K = \mathbb{F}_2$ requires a different argument. A square-zero associative product built from D first shows that there are no nonzero ad-nilpotent elements, and then eliminates the nilpotent part of every adjoint operator. The resulting identity

$$(\operatorname{ad} x)^4 + (\operatorname{ad} x)^2 + \operatorname{ad} x = 0$$

has square-free annihilating polynomial. Every adjoint is therefore semisimple, and Premet's theorem over perfect C_1-fields forces the algebra to be commutative [11, Theorem 3(ii)]. This excludes the last simple case. The polar algebra is consequently solvable, and Theorem 3.3 makes it nilpotent. Appendix A supplies all enveloping-algebra, centroid, and square-zero-product details. □

The use of Premet's result concerns ordinary finite-dimensional Lie algebras; no restricted structure on the original polar algebra is assumed. The proof of Theorem 4.1 also precedes, and does not use, the finite two-generator model or the universal Engel identities established in the next section. This order separates the global nilpotence argument from the subsequent structure theory.

5 Central reconstruction and cyclic Lie cohomology

Central extensions provide a route from the metabelian correspondence to solutions with nonabelian derived ideal. The Lie bracket records the polar part of an extension, while the functional equation also constrains its quadratic refinement. We determine both constraints and retain the linear maps between the resulting extensions.

Throughout this section, $g : V \to V$ is a quadratic solution, $n = \dim_{\mathbb{F}_2} V$, $B = \operatorname{pol} g$, and $W = L'_g$. Initially the coefficient line is $A = \mathbb{F}_4$, with fixed scalar $\omega^2 + \omega + 1 = 0$. The notation $H^2_{\mathrm{CE}}(L_g, A)$ always means ordinary *binary* Lie cohomology with trivial coefficients; its scalar structure comes from A. No nilpotency or metabelian hypothesis is imposed on g here. The constructions tensor with finite $\mathbb{F}_4$-coefficient spaces when these are needed below.

5.1 Quadratic lifts and their two obstructions

Let $\mathcal{Q}_{\leq j}(V, A)$ be the zero-constant functions of binary degree at most j, and put $\mathcal{L}(V, A) = \operatorname{Hom}_{\mathbb{F}_2}(V, A)$. Composition acts on the full function space by $Tq = q \circ g$. Define

$$\Delta_g = T + \omega^2 \operatorname{id}, \qquad \delta = T + \omega \operatorname{id}, \qquad N = \operatorname{id} + T + T^2. \tag{9}$$

The operator Δ_g measures the functional defect, whereas δ describes linear changes of section.

Proposition 5.1 (One-step central lifts)**.** *In a binary-linear section, a quadratic solution with marked central kernel* $(A, \omega \operatorname{id})$ *and quotient* g *is precisely*

$$F_q(x, a) = (g(x), \omega a + q(x)), \qquad q \in \mathcal{Q}_{\leq 2}(V, A), \quad \Delta_g q = 0. \tag{10}$$

Its marked extension classes form the vector space

$$H_g(A) = \frac{\ker(\Delta_g|_{\mathcal{Q}_{\leq 2}(V,A)})}{\delta \mathcal{L}(V, A)}. \tag{11}$$

Polarization induces a natural injection $H_g(A) \hookrightarrow H^2_{\mathrm{CE}}(L_g, A)$.

Proof. Centrality makes the mixed polar term with a vanish, giving the displayed form. Substitution into $F_q^2 = F_q + \mathrm{id}$ gives $q \circ g = \omega^2 q$. A section change by linear t replaces q by $q + \delta t$, whose polar changes by the CE coboundary tB. If $\operatorname{pol} q = tB$, then $r = q + \delta t$ is linear and satisfies $Tr = \omega^2 r$. Thus $\delta r = r$, and $q = \delta(t + r)$. This proves the injection. □

For a CE cocycle α, choose a quadratic refinement q with $\operatorname{pol} q = \alpha$. Such a refinement exists on every binary basis. The first obstruction is

$$\Theta[\alpha] = \Delta_g q \bmod \mathcal{Q}_{\le 2}(V, A) \quad \text{in } \mathcal{Q}_{\le 4}(V, A)/\mathcal{Q}_{\le 2}(V, A), \qquad K_g = \ker \Theta. \tag{12}$$

This is well defined: a refinement change is linear, and a CE coboundary is refined by δt, with $\Delta_g \delta t = 0$. Thus K_g consists of the ordinary cohomology classes whose quadratic refinements remain quadratic after composition with g. For $[\alpha] \in K_g$, define the remaining obstruction

$$\Psi[\alpha] = [\operatorname{pol}(\Delta_g q)]. \tag{13}$$

Theorem 5.2 (Complete central lifting criterion)**.** *The map $\Psi : K_g \to K_g$ is well defined. An ordinary central Lie extension class $[\alpha]$ admits a quadratic lift with the prescribed coefficient scalar if and only if*

$$\Theta[\alpha] = 0, \qquad \Psi[\alpha] = 0.$$

The two tests distinguish high-degree leakage from an incorrect quadratic weight. Their sufficiency includes the freedom to change the linear part of a refinement. For example, when $g = \omega \,\mathrm{id}$ on $\mathbb{F}_4^2$, the cocycle $\alpha((a, b), (c, d)) = (ad + bc)^2$ has $\alpha(x, gx) = 0$ and satisfies the four-variable obstruction. Its refinement $q = a^2 b^2$ has $\Delta_g q = q$, whose nonzero polar cannot be removed by a linear adjustment. This class fails the second test. The full correction argument is given in Appendix C.

5.2 The cyclic action and the fixed quadratic sector

Theorem 5.3 (A canonical cyclic action on quadratic-stable cohomology)**.** *On K_g, the formula*

$$\tau[\alpha] = [\operatorname{pol}(q \circ g)] \tag{14}$$

defines a natural $\mathbb{F}_4$-linear automorphism with $\tau^3 = \mathrm{id}$. Moreover

$$\Psi = \tau + \omega^2 \,\mathrm{id}, \qquad H_g(A) = (K_g)_{\omega^2}, \qquad \Pi = \mathrm{id} + \omega\tau + \omega^2 \tau^2 \tag{15}$$

is the projector onto the admissible extension space.

Here g need not be additive. The action is constructed through quadratic refinements and their CE classes, on the specified domain K_g. The projector selects a cohomology class; a choice of actual lift still amounts to a choice of refinement and section. The construction commutes with pullback by every binary-linear solution homomorphism and with coefficient maps.

There is also a canonical part visible on all ordinary cohomology. Write

$$I_g(A) = \{r \in \mathcal{Q}_{\le 2}(V, A) : r \circ g = r\}.$$

Proposition 5.4 (The fixed-sector splitting)**.** *The map*

$$O_0 : H^2_{\mathrm{CE}}(L_g, A) \longrightarrow I_g(A), \qquad O_0[\alpha](x) = \alpha(x, g(x)) \tag{16}$$

has the canonical section $r \mapsto [\operatorname{pol} r]$. Its section image is exactly $(K_g)_1$, and

$$H^2_{\mathrm{CE}}(L_g, A) = \operatorname{pol}(I_g(A)) \oplus \ker O_0. \tag{17}$$

Proof. For any quadratic refinement q, $Nq(x) = \alpha(x, gx)$. Its third additive difference is $d_{\mathrm{CE}}\alpha$, so it is quadratic when α is a cocycle. It is invariant, and a coboundary contributes $tB(x, gx) = 0$. For invariant quadratic r, $Nr = r$; the same difference identity shows that its polar is CE-closed and $O_0[\mathrm{pol}\, r] = r$. Applying the norm to a refinement of any τ-fixed class proves the assertion about $(K_g)_1$. □

For the coefficient line put

$$b_2 = \dim_{\mathbb{F}_2} H^2_{\mathrm{CE}}(L_g, \mathbb{F}_2) = \dim_{\mathbb{F}_4} H^2_{\mathrm{CE}}(L_g, A), \quad i_g = \dim_{\mathbb{F}_4} I_g(A), \quad h_g = \dim_{\mathbb{F}_4} H_g(A).$$

Corollary 5.5 (A sharp universal lift-space bound)**.** *If $d = \dim_{\mathbb{F}_4}(L_g/L'_g)$ with its intrinsic scalar structure, then*

$$\dim K_g = i_g + 2h_g, \qquad b_2 = \operatorname{rank}\Theta + i_g + 2h_g, \qquad i_g \geq d^2. \tag{18}$$

Consequently

$$h_g \leq \left\lfloor \frac{b_2 - d^2}{2} \right\rfloor. \tag{19}$$

The last bound is attained for every d by $g = \omega\,\mathrm{id}$ on $\mathbb{F}_4^d$.

Proof. The cyclic action and its degree condition descend to $\mathbb{F}_2$. Coefficient Frobenius therefore exchanges its two nontrivial eigenspaces, giving equal dimensions h_g. The fixed-space dimension is i_g by Proposition 5.4. On the abelianization, the d^2 independent functions $u_i u_j^2$, for ordered pairs (i, j), are invariant binary quadratics; their pullbacks prove the lower bound for i_g. In the linear example, $\Theta = 0$, $i_g = d^2$, $b_2 = \binom{2d}{2}$, and $h_g = \binom{d}{2}$, proving equality. For an m-dimensional coefficient space, all the cohomological and invariant dimensions scale by m. □

The dimension law separates three contributions to failed lifting: high-degree leakage H^2_{CE}/K_g, the fixed quadratic weight, and the opposite nontrivial weight. The fixed part splits by (17). The other two parts have the natural exact sequence

$$0 \longrightarrow (K_g)_\omega \longrightarrow (\ker O_0)/H_g(A) \longrightarrow H^2_{\mathrm{CE}}(L_g, A)/K_g \longrightarrow 0. \tag{20}$$

6 The free two-point model

The following explicit map on $\mathbb{F}_4^8$, in the coordinate order (a, b, c, d, e, h, r, s), supplies the two-point identities used below. It is the final model in the central-cover construction of the unified preprint [22]. Its free property uses only the displayed formula, generation, and the final vanishing certificate.

$$\begin{aligned}
a' &= \omega a, & b' &= \omega b,\\
c' &= \omega c + ab,\\
d' &= \omega d + ac + a^2 b,\\
e' &= \omega e + bc + ab^2,\\
h' &= \omega h + ae + bd,\\
r' &= \omega r + A, & A &= ac^2 + a^2 e + ah + cd,\\
s' &= \omega s + B, & B &= bc^2 + b^2 d + bh + ce.
\end{aligned} \tag{21}$$

Each coordinate is binary-quadratic. The successive nonlinear terms q obey $qg = \omega^2 q$ on the preceding coordinates, by substitution; the final two expressions cancel in pairs or

Table 1: The complete elementary central-extension calculation for the sixteen-dimensional model. All ranks in the table are binary.

Unknowns	Equation rank	Kernel dimension	Section-image dimension	$\dim_k H_g(k)$
272	244	28	28	0

with coefficient $1 + \omega + \omega^2 = 0$. Thus the displayed rule is a solution. The points on the a, b axes and their orbit partners generate the model; bracket closure has binary dimensions $4, 8, 16$. Its lower-central dimensions are $(16, 12, 10, 6, 4, 0)$. The checker reconstructs the original coefficient rows and verifies the complementary kernel and section-image bases. The freeness criterion in Lemma 7.1 therefore applies. No classification of the smaller prefixes is required for this step.

7 Canonical structure beyond the metabelian range

Global nilpotence permits a uniform passage from the orbit equation to restricted Lie theory. The free two-generator solution supplies universal identities, and these identities give canonical scalars on a reduced derived ideal. We then prove quadratic stability of the special biderivation in every finite solution. On finite free covers this produces an exact scalar on the whole derived ideal; on arbitrary quotients it produces a universal restricted derived quotient, natural for every linear intertwiner. A reduced adjoint representation gives a complementary matrix realization with a kernel of class at most three and an abelian cokernel. The quotient structures below retain the original polar bracket.

7.1 The free two-generator solution and universal identities

For a solution (G, g), let G' denote its polar derived ideal. The quotient G/G' has its intrinsic $\mathbb{F}_4$-scalar, given by the map induced by g. For $k = \mathbb{F}_4$, write

$$H_g(k) = \frac{\{q : G \to k : \deg_2 q \le 2,\ q(0) = 0,\ qg = \omega^2 q\}}{\{tg + \omega t : t \in \mathrm{Hom}_{\mathbb{F}_2}(G, k)\}}. \tag{22}$$

This is the elementary central-extension space, whose general construction is given in Proposition 5.1. Theorem 4.1 has already established nilpotence for every finite target. Thus the following criterion applies to the entire finite class.

Lemma 7.1 (A freeness criterion from central extensions)**.** *Suppose a finite solution (G, g) is generated by marked points $\xi_1, \dots, \xi_s$, their classes form an $\mathbb{F}_4$-basis of G/G', and $H_g(k) = 0$. Every assignment of the marked points to points of a finite solution extends uniquely to a binary-linear solution homomorphism.*

The proof is a central splitting argument, given in Appendix B. The sixteen-dimensional model in (21) satisfies the hypotheses with marked points a, b: its full central-extension space vanishes, as certified by the final row of Table 1. It is therefore the free two-generator solution among finite solutions. The vanishing certificate can be used directly, before classifying the intermediate stages of its central-cover construction. The dependency is global nilpotence, followed by this central-splitting criterion and its finite certificate, followed by the identities below.

Retain D, G from (7). The free solution supplies identities valid in every finite quadratic solution.

Theorem 7.2 (Universal two-orbit identities)**.** *Every finite quadratic solution satisfies*

$$G(x,y) = [x,y] + D(x,y), \tag{23}$$
$$[x,[x,[x,y]]] = [[x,y],[x,[x,y]]], \tag{24}$$
$$(\operatorname{ad} x)^4 = 0. \tag{25}$$

The last bound is sharp: the fourteen-dimensional model has $(\operatorname{ad} a)^3 b \neq 0$*. One also has*

$$[x,[[x,y],[x,z]]] = 0. \tag{26}$$

The exact identity (24) explains why the fourth adjoint vanishes without making the third vanish. Its right side lies deeper in the lower central series, and retains information that would disappear in a homogeneous Engel approximation. Appendix B gives a sparse coordinate verification on the free solution and proves the propagation to arbitrary pairs. The three-variable identity (26) is derived directly from the unconditional biderivations; it does not require a three-generator classification.

7.2 A Lie pencil and a canonical restricted derived quotient

Theorem 7.3 (The polar Lie pencil)**.** *The special biderivation* D *is a Lie bracket and satisfies*

$$D([x,y],z) = [D(x,y),z]. \tag{27}$$

In particular it is an adjoint Lie two-cocycle, and $B + tD$ *is a Lie bracket after any extension of the binary coefficient field. Its derivations satisfy*

$$[D_x, D_y] = \operatorname{ad}_{[x,y]+D(x,y)}. \tag{28}$$

Put

$$W = L', \qquad C = W \cap Z(L), \qquad E = W/C. \tag{29}$$

Here the center is that of the whole polar algebra. Equation (27) makes the following prescription independent of the bracket expression:

$$J\left(\sum_i [x_i,y_i] + C\right) = \sum_i D(x_i,y_i) + C. \tag{30}$$

Theorem 7.4 (Canonical restricted structure)**.** *The operator* J *on* E *satisfies* $J^2 + J = \mathrm{id}$*, belongs to the centroid of its original Lie bracket, and commutes with every ambient adjoint action induced by* L*. Thus* J *gives* E *a canonical* $\mathbb{F}_4$*-Lie structure. The map* f *descends to* $\bar{f}$ *on* E*, and*

$$P = \bar{f} + J \tag{31}$$

is a canonical restricted square on this $\mathbb{F}_4$*-Lie algebra, with* $P^2 = 0$*. Explicitly,*

$$P(u+v) = P(u) + P(v) + [u,v], \qquad P(Ju) = J^2 P(u),$$
$$\operatorname{ad}_{P(u)} = (\operatorname{ad}_u)^2, \qquad P(P(u)) = 0.$$

These constructions are preserved by solution isomorphisms and by surjective solution homomorphisms.

This theorem produces scalars for the original quotient bracket, with no metabelian hypothesis. The central quotient is essential:

Proposition 7.5 (A whole-derived central obstruction)**.** *There is a quadratic solution of binary dimension* 22*, with metabelian polar algebra of class four, for which no binary-linear map* $J_W : W \to W$ *satisfies*

$$J_W[x, y] = D(x, y) \qquad (x, y \in L).$$

Its obstruction is supported on $W \cap Z(L)$*. The canonical structure of Theorem 7.4 on* E *remains valid.*

The explicit model and a two-cycle detecting the obstruction appear in Appendix B. This separates the intrinsic reduced-derived scalar from the auxiliary scalar structure furnished by the metabelian normal form.

7.3 Global quadratic stability

An alternating binary-bilinear cochain A is *quadratically stable* if $q_A \circ f$ has binary degree at most two for a zero-constant quadratic refinement q_A of A. The condition is independent of the refinement: a linear adjustment composes with f to a quadratic function.

Theorem 7.6 (Global stability of the special biderivation)**.** *For every finite quadratic solution, the special biderivation* D *is quadratically stable. Thus, for every quadratic refinement* q_D*,*

$$\deg_2(q_D \circ f) \leq 2. \tag{32}$$

Theorem 7.7 (The ambient cube annihilates the derived ideal)**.** *Every finite quadratic solution satisfies*

$$(\mathrm{ad}_z)^3(W) = 0 \qquad (z \in L). \tag{33}$$

The fourth-adjoint bound on the whole space remains sharp, as in Theorem 7.2.

The proof first transports the restricted structure of E through its faithful ambient adjoint action. This controls compositions with derived inputs. The remaining high-degree part lies on the intrinsic $\mathbb{F}_4$-abelianization; scalar-character projections and the cyclic norm eliminate it. The complete argument is given in Appendix E. Quadratic stability controls composition degree; the central CE class detected in Proposition 7.5 may still be nonzero. The same construction retains a canonical central square-height class, described in Proposition E.3; it records central data before passage to the universal derived quotient.

7.4 A standard restricted presentation with a low-class kernel

Use $k = \mathbb{F}_4$ and write $\omega = J$ on E. The canonical adjoint representation is

$$\rho : L \longrightarrow \mathrm{Der}_k(E), \qquad \rho(x)(w + C) = [x, w] + C. \tag{34}$$

Theorem 7.8 (Reduced adjoint representation)**.** *The map* ρ *is a binary-linear Lie homomorphism satisfying*

$$\rho(fx) = \rho(x)^2 + \omega\rho(x), \qquad \rho(x)^3 = 0.$$

Its kernel K *is an* f*-stable ideal with*

$$[K, W] \subseteq C, \qquad \gamma_3(K) \subseteq C, \qquad \gamma_4(K) = 0.$$

Every $\rho(x)$ *is a restricted derivation of* (E, P)*:*

$$\rho(x)(P(u)) = [u, \rho(x)u].$$

The cube in this statement is the associative matrix cube. It follows from Theorem 7.7; the Lie-adjoint powers in the scalar-envelope formula below concern a different operation.

Let $X = \rho(L)$. It is a binary subspace stable under $A \mapsto \omega A + A^2$. Passing to its scalar span gives the restricted structure that will carry the realization.

Theorem 7.9 (The scalar envelope)**.** *The space $N = kX = X + \omega X$ is the restricted k-Lie algebra generated by X in $\mathrm{Der}_{\mathrm{res}}(E, P)$. It is an ordinary three-Engel k-Lie algebra, and*

$$N' = X' = \mathrm{Inn}(E).$$

Define

$$\mathcal{M} = N'' \cap Z(N') = \mathrm{ad}_E\big(E' \cap Z_2(E)\big), \tag{35}$$

where $Z_2(E) = \{e \in E : [e, E] \subseteq Z(E)\}$. Then $\mathcal{M}$ is a restricted ideal, any two of its matrices have zero product, and

$$T^4 = 0 \qquad (T \in N). \tag{36}$$

More precisely, for $A, B \in X$,

$$(A + \omega B)^4 = (\mathrm{ad}_A)^3 B + \omega(\mathrm{ad}_B)^3 A. \tag{37}$$

The right side is zero in every finite solution.

Already $T \mapsto \omega T + T^2$ is a quadratic solution on all of N. The quotient $\mathcal{T} = N/\mathcal{M}$ is a further restricted Lie reduction, with square induced from matrices, and

$$F(\bar{T}) = \omega\bar{T} + \bar{T}^{[2]} \tag{38}$$

is a quadratic solution on its entire k-space.

Theorem 7.10 (Bounded-class-kernel realization)**.** *The binary-linear map*

$$q : L \longrightarrow \mathcal{T}, \qquad q(x) = \rho(x) + \mathcal{M}$$

intertwines f and F. Its kernel $I = \rho^{-1}(\mathcal{M})$ is an f-stable ideal with

$$\gamma_3(I) \subseteq C \subseteq Z(L), \qquad \gamma_4(I) = 0.$$

Its image $Y = q(L)$ is an F-stable binary Lie ideal containing $\mathcal{T}'$, and $kY = \mathcal{T}$. There are exact sequences of quadratic solutions

$$0 \longrightarrow I \longrightarrow L \longrightarrow Y \longrightarrow 0, \qquad 0 \longrightarrow Y \longrightarrow \mathcal{T} \longrightarrow \mathcal{T}/Y \longrightarrow 0,$$

where $\mathcal{T}/Y$ is abelian. In particular the kernel is metabelian of class at most three, uniformly in the dimension and class of L.

The realization places the interactions outside a class-three kernel inside a standard restricted presentation; the extra scalar directions contribute an abelian cokernel. Its map is onto Y, which need not be all of $\mathcal{T}$. It is canonical under isomorphisms and surjective solution maps, the maps that preserve the center quotients used in its construction.

Equation (37) and the full-derived cube identity make the scalar envelope itself a height-two restricted presentation. The further quotient retains the bounded-kernel realization above. Section 8 uses the envelope directly to control class growth and its comparison with ordinary three-Engel Lie algebras.

The same pencil also connects the orbit equation to Bockstein-closed quadratic group extensions: D and $G = B + D$ are the classical binding maps for f^2 and f, respectively. The restricted-square interpretation follows the framework of [10].

8 General growth and nonmetabelian dimension compression

The general structure gives quantitative control of every finite quadratic solution. The companion metabelian theory [20] supplies an exact comparison scale. There are two complementary conclusions. The functional equation bounds polar class quartically in the number of point generators and connects its optimal growth to the ordinary three-Engel problem, whereas an explicit nonmetabelian family reaches every class in dimension $O(2^{c/2})$. Together these give logarithmic maximal derived length and finite free objects for all finite targets.

Write $L = L_f$ and

$$d = d(L) = \dim_{\mathbb{F}_4}(L/L'),$$

using the intrinsic scalar induced by f on the abelianization. As defined in Section 1, the point rank is the minimum number of generators under binary addition and f. It equals $d(L)$ by Lemma F.5; ordinary binary Lie generation uses the corresponding orbit pairs. For $c \geq 2$, let $m(c)$ be the least binary dimension of a solution of polar class c, and let $\mu(c)$ be the corresponding metabelian minimum. The companion paper [20] determines $\mu(c)$ exactly.

8.1 A quartic bound on the general polar class

The class estimate separates a scalar Lie action from its remaining central module. The scalar action is controlled by the classical larger-field 3-Engel bound; the central module has a much shorter commuting action. For $d \geq 2$, put

$$q(d) = \left\lfloor \frac{(d+1)^4}{2} \right\rfloor, \qquad C_0(d) = q(d) + d + 2, \qquad C(d) = q(d) + d + 1. \tag{39}$$

Set $C_0(0) = C(0) = 0$ and $C_0(1) = C(1) = 1$.

Proposition 8.1 (A bound independent of higher-rank free objects)**.** *Every finite quadratic solution of point rank d has polar class at most $C_0(d)$.*

This first estimate uses the canonical action on $E = L'/(L' \cap Z(L))$, the full-derived cube identity, and paired point generation. It supplies finite free objects without assuming their exact whole-derived scalar in advance.

Theorem 8.2 (A uniform quartic class bound)**.** *Every finite quadratic solution of point rank d satisfies*

$$\operatorname{cl} L \leq C(d). \tag{40}$$

In particular the bound is $\lfloor (d+1)^4/2 \rfloor + d + 1$ for $d \geq 2$. Rank zero is the zero solution; a nonzero rank-one solution is the linear scalar plane and has class one.

Proof outline. Proposition 8.1 first gives the finite free solution F at point rank d. On $W = F'$, Theorem 9.1 supplies the exact scalar $J = \omega$, commuting with all ambient adjoints. The actual binary action image consists of cube-zero operators and is closed under $A \mapsto \omega A + A^2$. Its scalar envelope is a 3-Engel $\mathbb{F}_4$-Lie algebra with the same class as an ordinary d-generator subalgebra. The larger-field estimate of [14] bounds this class by $q(d)$.

After that many lower-central steps, the remaining term lies in $Z(W)$. Ambient actions on this space commute; the two companion cube identities give $a_i^3 = a_i^2 a_j = 0$. Consequently $d+1$ further actions kill it for $d \geq 2$. Every finite target of point rank at most d is a quotient of F. Appendix F proves both estimates in this order, including the independent construction of F. □

The quantitative 3-Engel input is the larger-field improvement after Theorem 1 of [14]. The application here uses the precise orbit-to-action interface and the separate central-module bound; no bound on arbitrary representation words is inferred from Lie class alone.

Proposition 8.3 (The action on an invariant ideal)**.** *Let I be an f-stable Lie ideal and let $M = I/I'$. The map $J = f_M$ gives M its intrinsic $\mathbb{F}_4$-scalar structure. For every $x \in L$, the induced operator $A_x = \mathrm{ad}_x|_M$ satisfies*

$$A_x J = J A_x, \qquad A_x^3 = 0, \qquad A_{f(x)} = \omega A_x + A_x^2. \tag{41}$$

The actual binary image is an ordinary 3-Engel Lie algebra.

8.2 The same-rank comparison with ordinary three-Engel algebras

For $d \geq 1$, let $\kappa(d)$ be the largest polar class of a finite quadratic solution of point rank at most d. Let $e(d)$ be the largest class of a finite-dimensional ordinary 3-Engel Lie algebra over $\mathbb{F}_4$, generated by at most d ordinary Lie generators. These maxima exist: the preceding bound controls κ, and the classical larger-field estimate controls e.

Proposition 8.4 (An Engel algebra realized without increasing point rank)**.** *Let $H \neq 0$ be a finite-dimensional ordinary 3-Engel $\mathbb{F}_4$-Lie algebra, generated by at most r elements and of class c. There is a finite binary quadratic solution of point rank at most r and polar class $c-1$. For an abelian input, the resulting solution may be zero.*

Theorem 8.5 (Same-rank class-growth comparison)**.** *For every $d \geq 1$,*

$$e(d) - 1 \leq \kappa(d) \leq e(d) + d + 1. \tag{42}$$

In particular $\kappa(d) = \Theta(e(d))$. For every $\alpha \geq 1$, the bound $\kappa(d) = O(d^\alpha)$ holds if and only if $e(d) = O(d^\alpha)$. If $e(d)/d \to \infty$, then $\kappa(d)/e(d) \to 1$.

For the upper comparison, retain the actual class of the scalar action in the preceding proof instead of replacing it by $q(d)$. For the reverse construction, take the restricted matrix envelope of $\mathrm{ad}\, H$. Classical 3-Engel operator identities make every fourth power zero. The standard map $A \mapsto \omega A + A^2$ then gives an orbit solution; the binary subsolution generated by the original adjoints retains the full scalar span and hence the class. Appendix F gives this construction and the rank-preserving argument. The restricted-square viewpoint is classical [10]; the comparison keeps track of the original point rank and polar class.

Thus the linear-growth question for the orbit category has the same difficulty as the ordinary $\mathbb{F}_4$ three-Engel class problem. Unbounded principal-ideal class already occurs in compatible orbit solutions, by the classical current construction discussed in Remark F.4. The comparison concerns growth and explicit realizations, rather than an equivalence of the two underlying categories.

8.3 An explicit family below the metabelian dimension scale

Let $k = \mathbb{F}_4$, and let A_m be the augmentation ideal of

$$k[t_1, \ldots, t_m]/(t_1^2, \ldots, t_m^2).$$

Thus $\dim_k A_m = 2^m - 1$, and every element of A_m has square zero. We take $A_0 = 0$.

Theorem 8.6 (The trace-restricted current family)**.** *On the binary vector space $k \oplus A_m^3$, define*

$$\begin{aligned} f_m(a,u,v,w) = \big(&\omega a,\ \omega u + av + (a + a^2)w + uv,\\ &\omega v + aw + uw,\ \omega w + vw\big). \end{aligned} \tag{43}$$

This is a quadratic solution of $f_m^2 = f_m + \mathrm{id}$, with

$$\dim_{\mathbb{F}_2} L_m = 6 \cdot 2^m - 4, \qquad d(L_m) = m + 1, \qquad \operatorname{cl} L_m = 2m + 1. \tag{44}$$

It is nonmetabelian for $m \geq 2$. For $m \geq 1$, its original binary polar algebra admits no restricted square.

Construction and proof outline. Use the classical three-dimensional algebra $S = \langle U, V, W \rangle_k$, with $[U, V] = U, [U, W] = V, [V, W] = W$, and the commuting derivations $\delta = \mathrm{ad}_U, \delta^2$. The restricted current presentation

$$\widehat{L}_m = (kx \oplus ky) \ltimes (S \otimes A_m)$$

has restricted square

$$P(a, b, u, v, w) = (0, a^2, av + bw + uv, aw + uw, vw), \qquad P^2 = 0. \tag{45}$$

Consequently $\widehat{f}_m = \omega\,\mathrm{id} + P$ is a quadratic solution. The binary graph $b = a + a^2$ is $\widehat{f}_m$-stable; its restriction is (43). A positive grading gives the exact class, and a direct derived-ideal calculation gives the intrinsic rank. The graph is binary-linear but is not closed under the ambient k-scalar or P. Appendix G proves all these assertions, including the nonrestrictability statement. □

The simple algebra and its derivations are classical [12, Proposition 3.3(1)]; the current multiplication is the one used in [13, Proposition 2.1]. The passage from restricted squares to quadratic maps is also classical [10, Lemma 9.5 and Example 9.6]. Here the binary invariant restriction and its exact polar class and point rank give the following dimension comparison.

Corollary 8.7 (Dimension compression in every class)**.** *Let $m(c)$ denote the general minimum from the definition in this section. Then*

$$\begin{aligned} m(2j+1) &\leq 6 \cdot 2^j - 4 \quad (j \geq 1), \\ m(2j) &\leq 6 \cdot 2^j - 6 \quad (j \geq 1). \end{aligned} \tag{46}$$

In particular $m(c) = O(2^{c/2})$, with $m(7) \leq 44$ and $m(8) \leq 90$. For every $c \geq 5$, the displayed constructions are nonmetabelian and have dimension strictly below $\mu(c) = 2^c + 2c - 6$.

Proof. Use L_j for odd class. For even class, quotient by its last lower-central term $kU_{[j]}$, an invariant central binary plane. The quotient has exact class $2j$ and loses two dimensions. The bracket $[U_{\{1\}}, V_{\{2\}}] = U_{\{1,2\}}$ survives for every resulting class at least five. The comparison with $\mu(c)$ follows immediately from $\lceil (c-1)/2 \rceil \leq c - 3$ for $c \geq 5$. □

The construction gives the general bound $m(c) = O(2^{c/2})$. Its odd-class model has center equal to its last lower-central plane, so every nontrivial solution quotient lowers its class (Proposition G.3). Determining the optimal general dimension scale is a separate problem in Section 14.

The same family has a quadratic linearizing permutation with a quadratic inverse (Proposition G.5). Thus unbounded polar class and derived length do not force large inverse degree. More generally, central lifts of an involutively linearized base have a quadratic linearizer with inverse degree at most three (Proposition G.6), with the involutivity hypothesis imposed at the base stage.

8.4 Logarithmic derived length and finite universal objects

Let $\ell_{\max}(s)$ be the maximum polar derived length among finite solutions generated by at most s points, with $L^{(0)} = L$, $L^{(j+1)} = [L^{(j)}, L^{(j)}]$.

Theorem 8.8 (Maximal derived length)**.** *For $s \geq 1$,*

$$\lceil \log_2(3s) \rceil - 1 \leq \ell_{\max}(s) \leq 4 \log_2(s+1) + 1. \tag{47}$$

In particular $\ell_{\max}(s) = \Theta(\log s)$. The same maximum is obtained at exact point rank s.

Proof outline. The upper bound follows from (40) and $L^{(j)} \subseteq \gamma_{2^j}(L)$. The family in Theorem 8.6 has

$$\mathrm{dl}(L_m) = \lceil \log_2(3m+3) \rceil - 1$$

by its exact support recurrence (Proposition G.4). Set $m = s-1$. Adding linear scalar factors raises a smaller point rank to s without changing nonzero derived length. Full details are in Appendix F. □

The logarithmic order controls the whole finite class, including its nonmetabelian members. It leaves both the optimal logarithmic coefficient and exact values of $\ell_{\max}(s)$ open. A further consequence of the class bound is a growing dimension cost: if $a(c) = \min\{d \geq 2 : c \leq C(d)\}$, then

$$m(c) \geq 2(c-1) + 2a(c) \geq 2c + 2(2c)^{1/4} - 6 \qquad (c \geq 2). \tag{48}$$

This bound follows from parity of the lower-central factors and provides a lower comparison to (46).

Theorem 8.9 (Finite free solutions and complete stem towers)**.** *For every $s \geq 0$, there is a finite solution $\mathfrak{F}_s^{\mathrm{fin}}$, generated by marked points $\xi_1, \ldots, \xi_s$, such that every assignment of these points to a finite solution extends uniquely to a binary-linear intertwiner. Every such target has nilpotent polar algebra by Theorem 4.1. Its abelianization is $\mathbb{F}_4^s$, and $H_{\mathfrak{F}_s^{\mathrm{fin}}}(\mathbb{F}_4) = 0$.*

Starting with the linear solution $\mathbb{F}_4^s$, iteratively take the full quadratic central cover with kernel $H_g(\mathbb{F}_4)^*$ whenever $H_g(\mathbb{F}_4) \neq 0$. Every step is stem, meaning that its kernel lies in the derived ideal. The process terminates after finitely many nontrivial steps, and its terminal marked solution is $\mathfrak{F}_s^{\mathrm{fin}}$.

The reason for finiteness is quantitative: paired Lie generation and Proposition 8.1 bound the dimension of every s-point-generated finite solution. The free object is obtained by a finite product of marked representatives. The central-cover description gives an independent recognition criterion: a finite s-generated object with marked abelianization basis is free exactly when its full quadratic extension space vanishes. The proofs, including all noninvertible arrows, are in Appendix F.

The universal property covers finite targets and, by passage to the subsolution generated by the marked images, locally finite targets. The stem-cover construction terminates precisely when its full extension space vanishes. Its quantitative dimension bound ensures termination; explicit free objects in higher point rank remain a computational problem.

9 Exact free scalars and the universal quotient

9.1 Exact restricted structure on finite free covers

Theorem 8.9 constructs a finite free solution $\mathfrak{F}_s^{\mathrm{fin}}$ at every finite point rank. Its admissible central quadratic extension space vanishes. Global stability now converts that projectivity into an exact scalar structure on its derived ideal.

Theorem 9.1 (Exact free-derived scalar and restricted square)**.** *Let (F, g) be a finite free solution and put $W_F = F'$. There is a unique binary-linear map $J_F : W_F \to W_F$ satisfying*

$$J_F B_F(x, y) = D_F(x, y) \qquad (x, y \in F). \tag{49}$$

It obeys $J_F^2 + J_F = \mathrm{id}$, is a centroid of the original bracket on W_F, and commutes with every ambient adjoint action of F. The map

$$P_F = g|_{W_F} + J_F \tag{50}$$

is a restricted square on this $\mathbb{F}_4$-Lie algebra, with $P_F^2 = 0$. In particular $g|_{W_F} = J_F + P_F$.

Every finite solution is a quotient of such an F. Under a quotient, central relations can fail to be stable under J_F. The next construction records exactly the additional relations needed to retain the restricted structure. The proof of Theorem 9.1, including the use of quadratic-stable cohomology, is in Appendix E.

9.2 A universal restricted derived quotient

Write $B, D : \Lambda^2_{\mathbb{F}_2} L \to W$ for the induced linear maps and fix $k = \mathbb{F}_4$, with $\omega^2 + \omega = 1$.

Definition 9.2 (The scalar-compatible derived quotient)**.** Define

$$\begin{gathered} Q_f = \frac{k \otimes_{\mathbb{F}_2} W}{\langle 1 \otimes D(c) + \omega \otimes B(c) : c \in \Lambda^2_{\mathbb{F}_2} L \rangle_k}, \\ j_f(w) = 1 \otimes w. \end{gathered} \tag{51}$$

Let $N_f = \ker j_f$.

Theorem 9.3 (Canonical restricted quotient and all linear arrows)**.** *The map $j_f : W \to Q_f$ is binary-linearly surjective, and $N_f \subseteq W \cap Z(L)$ is f-stable. The original bracket and $f|_W$ descend to Q_f, with*

$$\begin{gathered} [j_f(u), j_f(v)] = j_f([u, v]), \qquad f_Q(j_f(w)) = j_f(f(w)), \\ J_Q(j_f(B(c))) = j_f(D(c)), \end{gathered} \tag{52}$$

where J_Q is multiplication by ω. The map $P_Q = f_Q + J_Q$ is a restricted square on the original quotient bracket, and $P_Q^2 = 0$.

For every k-space T, a binary-linear map $h : W \to T$ with $hD = \omega hB$ factors uniquely through a k-linear map $Q_f \to T$. Thus N_f is the least kernel required by the scalar-factor relation. In particular there is a canonical surjective restricted k-Lie map $Q_f \to E$.

Every binary-linear solution homomorphism $T : (L, f) \to (L_1, g)$ induces a restricted k-Lie homomorphism

$$Q(T) : Q_f \longrightarrow Q_g, \qquad Q(T) j_f(w) = j_g(Tw). \tag{53}$$

These maps preserve identities and composition. Hence Q is a functor on all finite solutions and all their linear intertwiners.

Unlike the quotient by the full ambient center, this construction does not require an arrow to preserve that center. The tensor relation defines the map directly, while the free-cover description proves that its kernel is stable under the quadratic rule.

Theorem 9.4 (Exact free-presentation formula)**.** *Let $\pi : (F, g) \twoheadrightarrow (L, f)$ be any finite free presentation, and put $H = \ker \pi \cap F'$. Then*

$$Q_f \simeq F'/(H + J_F H), \qquad N_f \simeq (H + J_F H)/H. \tag{54}$$

The subspace $H + J_F H$ is the least J_F-stable subspace containing H; it is automatically stable under g and P_F, and is an ideal. Its image modulo H is central.

The formula shows that Q_f retains the central directions that are compatible with the scalar relation. On the example of Proposition 7.5, it removes the detected central obstruction. It is a derived quotient with an exact universal property; the ambient solution is still described by its quotient, action, and extension data. Proofs of both quotient theorems are in Appendix E.

Corollary 9.5 (Reversal of the orbit orientation)**.** *If a binary-linear map satisfies $Tf = g^2T$, then $j_f(w) \mapsto j_g(Tw)$ induces a Frobenius-semilinear Lie map $Q_f \to Q_g$ preserving the restricted square. Replacing f by f^2 leaves N_f unchanged, conjugates J_Q, and leaves P_Q unchanged. The zero map is the same underlying map for either orientation.*

9.3 A canonical restricted envelope of the whole solution

The canonical quotient Q_f also determines a restricted Lie algebra containing the entire solution modulo exactly the same central obstruction. This construction applies without a metabelian hypothesis and respects every binary-linear intertwiner. Put $k = \mathbb{F}_4$ and let B, D be as in Definition 9.2. Define

$$\mathcal{E}_f^+ = \frac{k \otimes_{\mathbb{F}_2} L}{\langle \omega \otimes B(c) + 1 \otimes D(c) : c \in \Lambda^2_{\mathbb{F}_2} L \rangle_k}, \qquad i_f(x) = 1 \otimes x.$$

Theorem 9.6 (The whole-solution tensor envelope)**.** *The space $\mathcal{E}_f^+$ carries a canonical restricted k-Lie structure with square P such that*

$$P(i_f x) = i_f(fx) + \omega i_f x, \qquad P^2(i_f x) = 0 \quad (x \in L).$$

Its derived ideal is $i_f(W) \simeq Q_f$, and $\ker i_f = N_f \subseteq W \cap Z(L)$. In particular, if $n = \dim_{\mathbb{F}_2} L$, $w = \dim_{\mathbb{F}_2} W$ and $n_0 = \dim_{\mathbb{F}_2} N_f$, then

$$\dim_k \mathcal{E}_f^+ = n - \frac{w + n_0}{2}.$$

Every binary-linear solution homomorphism induces a restricted k-Lie homomorphism between the corresponding envelopes. Hence $N_f = 0$ gives a faithful binary-linear realization of (L, f) inside a restricted k-Lie algebra, with $f = \omega I + P$ on its image.

Proof. Let R be the displayed relation space in $k \otimes L$. The identity $D([x, y], z) = [D(x, y), z]$ from Theorem 7.3 shows that R is a Lie ideal, so the scalar-extension bracket descends. Choose a binary complement V of W in L. As $R \subseteq k \otimes W$, the quotient is $(k \otimes V) \oplus Q_f$ as a k-space. Theorem 9.3 therefore gives $\ker i_f = N_f$, $i_f(W) = Q_f$, and the stated dimension. The derived ideal equals $i_f(W)$ because it is the scalar span of the bracket values and $i_f(W) = Q_f$ is already a k-space.

On Q_f use its restricted square P_Q. Choose a binary basis $v_1, \dots, v_s$ of V, prescribe $P(i_f v_j) = i_f(fv_j) + \omega i_f v_j$, and extend by $P(au) = a^2 P(u)$ and $P(u+v) = P(u) + P(v) + [u, v]$. These rules agree on the derived summand by the definition of P_Q. Both sides of the displayed formula for $P(i_f x)$ have the same polar form, so agreement on the chosen basis and on W proves it for every $x \in L$.

For source vectors, the defining relation and the definition of D give

$$[P(i_f x), i_f y] = i_f([fx, y]) + \omega i_f([x, y]) = i_f([x, [x, y]]).$$

Together with the restricted identity on Q_f, scalar extension and the Jacobi identity give $\mathrm{ad}_{P(u)} = (\mathrm{ad}_u)^2$ for all u in the envelope. Thus P is a restricted square. Finally, $f^2 x = x + fx$ and $[x, fx] = 0$ imply $P^2(i_f x) = 0$ by direct expansion. The tensor relation is preserved by every solution homomorphism, and the defining formula for P proves compatibility with the induced map. Since the source image spans the envelope over k, this also proves uniqueness and naturality of the square. □

The assertion $P^2(i_f x) = 0$ concerns the marked binary image. It does not assert that P^2 vanishes on the entire scalar envelope. The construction extends the quotient Q_f from the derived ideal to the whole solution while retaining the central obstruction N_f exactly.

10 An ordinary Lie description of every quadratic solution

The restricted core records the precise obstruction to a faithful restricted realization. There is also a description on the entire original set, including that central obstruction. Its two operations are explicit polynomials in the original polar bracket. This gives an ordinary Lie category for all solutions and a uniform bound on both degrees in a linearizing change of coordinates.

Throughout this section $k = \mathbb{F}_4$ and $\omega^2 + \omega = 1$. Brackets are left-normed. In an ordinary k-Lie algebra put

$$q(a, b) = [a, b, b, a], \tag{55}$$

$$\mathcal{R}(a, b, c) = q([a, b], c) + q([b, c], a) + q([c, a], b) + [[a, b, c], [b, c, a]]. \tag{56}$$

Let $\mathcal{E}$ be the category of finite-dimensional ordinary k-Lie algebras satisfying

$$[y, x, x, x] = 0, \qquad \mathcal{R}(a, b, c) = 0, \tag{57}$$

with all k-linear Lie homomorphisms. Let $\mathcal{Q}$ be the category of finite zero-constant binary quadratic solutions of (1), with all binary-linear intertwiners.

Theorem 10.1 (Ordinary Lie reconstruction)**.** *There is an equivalence $\mathcal{Q} \simeq \mathcal{E}$ given by mutual changes of operations on the same set. For $(V, +, f) \in \mathcal{Q}$, write $B(x, y) = f(x+y)+f(x)+f(y)$ and $c = B(x, y)$. Define*

$$x \diamond y = x + y + c + B(c, x) + B(c, y) + B(B(c, y), x), \tag{58}$$

$$\{x, y\} = c + B(c, x) + B(c, y), \qquad \omega * x = f(x). \tag{59}$$

All operations on the right of these formulas are original operations. Then $H_f = (V, \diamond, *, \{\ ,\ \})$ is an ordinary k-Lie algebra in $\mathcal{E}$. Conversely, for $H \in \mathcal{E}$, set

$$x \boxplus y = x + y + [x, y] + [x, y, x] + [x, y, y] + [x, y, y, x], \qquad f(x) = \omega x. \tag{60}$$

Here the right side uses ordinary Lie-vector addition. The operation $\boxplus$ makes H a binary vector space, and f is quadratic for this addition and satisfies $f^2 = f + \mathrm{id}$.

The two constructions are inverse, preserve every homomorphism and recover the actual automorphism groups. On the original set they preserve the center and every lower-central term. In particular

$$\dim_k H_f = \tfrac{1}{2} \dim_{\mathbb{F}_2} V, \quad \mathrm{cl}\, H_f = \mathrm{cl}\, L_f, \quad \dim_k H_f / H_f' = d(L_f).$$

The original polar Lie algebra L_f and the ordinary algebra H_f have different additions and brackets. The theorem characterizes the ordinary algebra obtained by (58)–(59); it does not identify the original bracket with an arbitrary k-bilinear bracket. This distinction lets the construction retain the central directions lost by the restricted-core quotient.

Corollary 10.2 (Biquadratic linearization)**.** *For every finite quadratic solution on n binary dimensions there is a bijection $\eta : V \to k^{n/2}$ such that*

$$\eta f = \omega \eta, \qquad \deg_{\mathbb{F}_2} \eta \le 2, \qquad \deg_{\mathbb{F}_2} \eta^{-1} \le 2.$$

For a nonlinear solution both bounds are sharp. The conjugator need not have the averaging form $f + J_0^2$ for an original binary-linear operator J_0.

Proof strategy. On a scalar-faithful free source, the restricted square P gives the coordinate change $I + P$. Expanding its transport of addition gives (58); transport of the bracket gives (59). These are terms in $+$ and f, so they descend through every finite free presentation, including sources with nonzero central kernel. A relative PBW construction and the Hopf formula identify the exact image as (57). Linear sections of the two vector-space quotients preserve the two-sided quadratic degree bound. Appendix H gives the construction and both round trips. □

The class- and rank-preserving ordinary description also gives the sharper comparison

$$e(d) - 1 \leq \kappa(d) \leq e(d).$$

The lower bound is Proposition 8.4; the upper bound follows directly by applying Theorem 10.1 to every point-rank-d solution. The earlier preliminary estimates were used to construct finite free covers before this ordinary description was available.

10.1 The sextic identity is exactly the degree obstruction

The same change of addition works on the entire ordinary three-Engel category. The extra identity in (57) has a concrete meaning: it is the third additive difference of the encoded rule.

Theorem 10.3 (Cubic encoding and its quadratic locus)**.** *Let H be any finite-dimensional ordinary three-Engel k-Lie algebra. Formula* (60) *defines a binary vector-space addition. For $F(x) = \omega x$ one has $F^2 = F +_{\boxplus} \mathrm{id}$ and*

$$\Delta_a \Delta_b \Delta_c F(0) = \mathcal{R}(a, b, c), \tag{61}$$

where differences use $\boxplus$. The right side is central, alternating, binary-trilinear, and factors through H/H' with Frobenius-semilinear k-weights in its three arguments. Thus $\deg_{\mathbb{F}_2} F \leq 3$, and F is quadratic exactly when $\mathcal{R} = 0$. The cubic bound is attained in binary dimension 42. The encoding is fully faithful on all ordinary Lie homomorphisms.

The proof reduces the associator and the third difference in three generators to the one-dimensional central space generated by $\mathcal{R}(a, b, c)$. A 21-dimensional ordinary Lie algebra determines the two coefficients: the associator coefficient is zero and the third-difference coefficient is one. Its complete sparse table and exact coefficient certificate are given in Appendix H. This finite detector fixes coefficients in a universal identity; it is not an inference from sampled inputs.

10.2 The scalar kernel as an ordinary homology image

Write $N_f = \ker j_f \subseteq L'_f$ for the canonical central kernel of Theorem 9.3. Through the same-set description it is also a central k-subspace of H_f. Its intrinsic ordinary Lie description uses Chevalley–Eilenberg homology.

Theorem 10.4 (Homology formula for the scalar kernel)**.** *For $H \in \mathcal{E}$ and a CE two-cycle $z = \sum_i a_i \wedge b_i$, so that $\sum_i [a_i, b_i] = 0$, put*

$$\lambda_H(z) = \sum_i q(a_i, b_i) + \sum_{i<j} [[a_i, b_i], [a_j, b_j]]. \tag{62}$$

This is well-defined and Frobenius-semilinear on cycles. It vanishes on boundaries and induces a natural map

$$\lambda_H : H_2^{\mathrm{CE}}(H; k)^{(1)} \longrightarrow Z(H).$$

For $H = H_f$, its image is exactly N_f, with the original and new central additions agreeing. The quotient $H/\operatorname{im}\lambda_H$ is the universal quotient admitting a derived restricted square p with

$$\begin{aligned} p(aw) = a^2 p(w), \quad p(w+z) = p(w) + p(z) + [w,z], \quad p^2 = 0, \\ [p(w), x] = [w, [w, x]], \qquad p([x,y]) = q(x,y), \qquad w, z \in H',\ x, y \in H. \end{aligned} \tag{63}$$

When it exists, this square is unique.

In particular scalar-faithfulness is an ordinary homology condition, $\lambda_H = 0$. The tensor kernel and the homology image are two descriptions of the same central reflection. The condition concerns this specified map, rather than vanishing of the entire second homology group.

10.3 Relation with the metabelian correspondence and classical methods

The metabelian construction in [20] uses the same cubic bracket but the simpler addition $x + y + C(x, y)$. The general addition has the additional central term $q_B(x, y) = B(B(B(x, y), y), x)$. On a metabelian target, the change $+ \mapsto + + q$ is an involutive self-equivalence: q is central, binary-biadditive, and zero on a derived input. Composing the earlier metabelian correspondence with this self-equivalence gives Theorem 10.1 on that subcategory. The earlier normal forms therefore retain their own coordinates, while dimensions, classes, isomorphism counts and automorphism groups are preserved.

The method uses ordinary PBW and the Hopf formula, as well as the restricted-square relation between $I + P$ and quadratic maps [10, Section 9]. The broader recovery of nilpotent polynomial quasigroups is treated in [1, Theorem 2.4]; related changes between loop and Lie operations occur in [4, Section 5]. Here the explicit mutual operations identify the precise characteristic-two image, retain every linear intertwiner, and give uniform binary degree two in both directions. Formula (61) also describes the larger three-Engel class beyond this quadratic locus.

10.4 Fixed polar data and the central scalar gauge

Because the new addition and bracket use only B, they give a fixed binary Lie algebra H_B when the original polar bracket is fixed. Its scalar choices are precisely the roots $J^2 + J + I = 0$ in its binary centroid. The original polar automorphisms are the binary Lie automorphisms of H_B; the automorphisms of a specified rule form the centralizer of its scalar root.

Corollary 10.5 (The labeled indecomposable fiber)**.** *Suppose the original polar algebra is nonabelian and indecomposable. Put $U = V/L'_f$, $Z = Z(L_f)$, $d = \dim_k U$, and $z = \dim_k Z$. Every quadratic rule with exactly this polar bracket is of one of the forms*

$$f + h \quad or \quad f^2 + h, \qquad h : U \to Z \text{ Frobenius-semilinear.}$$

There are exactly $2 \cdot 4^{dz}$ such labeled rules. Within either orientation the binary-linear central shears $I + h$ conjugate the rules. There are one or two conjugacy classes, according to whether an automorphism exchanges the two orientations.

Proof. Here $Z \subseteq H'_B$; otherwise a central vector outside the derived algebra splits off. In the centroid C, the ideal $D = \operatorname{Hom}_{\mathbb{F}_2}(U, Z)$ has square zero. Commutators of centroid elements kill H'_B and have central image, so C/D is commutative. Indecomposability makes its semisimple quotient a field. Each residue field containing k has two roots of $X^2 + X + 1$; these lift uniquely through a commutative nilpotent radical. Lifting to C leaves exactly $J + h$ with $Jh + hJ = h$, namely the k-antilinear h. The space has k-dimension dz, and $(I + h)J(I + h) = J + h$. The opposite orientation uses J^2. The change of operations preserves central sums and all shear maps, giving the displayed formulas in the original addition. □

The underlying-Lie-algebra classification up to independent Galois conjugation has the direct predecessor [17, Theorem 4.3]. The application here records its labeled central gauge for the original polar data.

11 The exact cost of repairing the derived square

The obstruction in Theorem 10.4 can be removed by enlarging the algebra rather than taking a quotient. Its dimension is exactly the cost of that enlargement. Throughout this and the next two sections, dimensions are over $k = \mathbb{F}_4$, $H \in \mathcal{E}$, and

$$N(H) = \operatorname{im} \lambda_H, \qquad n(H) = \dim_k N(H), \qquad d(H) = \dim_k H/H'.$$

A central extension is called stem if its kernel is contained in the derived algebra. A square always means the specified map (10.4), defined on the derived algebra.

Theorem 11.1 (Exact central repair)**.** *Every $H \in \mathcal{E}$ has a central stem extension $a_H : A_H \twoheadrightarrow H$ such that*

$$N(A_H) = 0, \qquad \dim_k \ker a_H = n(H), \qquad \operatorname{cl} A_H = \operatorname{cl} H, \qquad d(A_H) = d(H).$$

Every central extension $E \twoheadrightarrow H$ in $\mathcal{E}$ with $N(E) = 0$ has kernel dimension at least $n(H)$. All extensions attaining this minimum are isomorphic over the given bottom H, allowing the induced isomorphism of their kernels.

More generally, a central extension with kernel dimension z satisfies $n(E) \geq n(H) - z$. For each integer $0 \leq s \leq n(H)$ there is a central stem extension $E_s \twoheadrightarrow H$ of the same class and head rank, with

$$\dim_k \ker(E_s \to H) = s, \qquad n(E_s) = n(H) - s.$$

Thus the minimum refers to an extension isomorphism class, not to a unique map or a unique isomorphism. The construction uses a finite free cover with its specified square. Among its central relations, one kills precisely the largest subspace on which that square descends. The relations that remain map onto $N(H)$ by squaring. Appendix J proves existence, the sharp lower bound and the equality assertion.

Proposition 11.2 (Factorization over the minimum repair)**.** *Every epimorphism $G \twoheadrightarrow H$ in $\mathcal{E}$ with $N(G) = 0$ and $d(G) = d(H)$ factors through an epimorphism $G \twoheadrightarrow A_H$ over H. Its kernel need not be central. The factor map can depend on the chosen head lifts.*

The underlying methods are classical. Free central covers and their quotients are described, for example, in [6, Chapter 1]. Ordinary extension classes and restricted extension comparisons belong to the cohomological framework of [5, Theorems 3.1–3.3 and 4.1, p. 575]. In that framework a restricted map on the whole bottom is already given; here the square is prescribed only on H' by (10.4), including $p^2 = 0$. The statements above identify the minimum cost for this prescribed square. They do not assert that H has a restricted map on all of H. For a nonzero square on an abelian kernel, Hochschild's general comparison is used at the level of additive groups; the six-term k-linear sequence uses a strongly abelian kernel.

11.1 Half-weight repair and the remaining ascent cost

Suppose $H = \bigoplus_{j=1}^{c} H_j$ is generated by H_1, $[H_i, H_j] \subseteq H_{i+j}$, and $H_c \neq 0$. These are bracket-length gradings.

Theorem 11.3 (Half-weight repair and central rigidity)**.** *The minimum extension $A_H \to H$ admits a grading extending that of H. If $Z = \ker(A_H \to H)$, the maps $z \mapsto a_H(p(z))$ are Frobenius-semilinear isomorphisms*

$$Z_j \longrightarrow N(H)_{2j}, \qquad Z_j = 0 \quad \textit{unless } 2 \le j \le \lfloor c/2 \rfloor.$$

In particular $N(H)$ has only even weights. If every finite central extension of H in $\mathcal{E}$ has class c, the same is true of every finite central extension of A_H. Consequently a central extension $E \to H$ with $N(E) = 0$ which admits a further central extension of class $c+1$ must have

$$\dim_k \ker(E \to H) \ge n(H) + 1.$$

There is a quantitative version. Let $E \to H$ be a graded stem central extension of class c with $N(E) = 0$, let Z be its kernel, and suppose a scalar central cocycle α of pure weight $c+1$ raises its class. Define the effective pairing rank by

$$r_{\text{eff}} = \operatorname{rank}\big(Z_c \longrightarrow H_1^*,\ z \longmapsto \alpha(z, -)\big).$$

Then

$$\dim_k Z \ge n(H) + r_{\text{eff}}. \tag{64}$$

If H is centrally rigid as in Theorem 11.3, then $r_{\text{eff}} \ge 1$. The repair directions occur at half the obstruction weight, whereas the next ascent needs an effective direction at the highest weight. This explains why a cheapest repair can leave central rigidity intact. The next section supplies concrete bottoms, and Section 13 attains (64) after a minimal preparation.

In binary dimensions the full repair costs $\dim_{\mathbb{F}_2} N_f = 2n(H)$, and a partial repair of ordinary dimension s costs $2s$. If $m_0(c)$ is the minimum binary dimension at class c with $N_f = 0$, then

$$m(c) \le m_0(c) \le 2m(c).$$

The same inequalities hold for the corresponding minima restricted to nonmetabelian objects: a repair surjecting onto such an object is again nonmetabelian. Eliminating the scalar kernel therefore changes these dimension minima by at most a constant factor.

12 Scalar obstructions and central ascent

The ordinary description makes two different constraints visible. The homology image N measures the failure of a derived restricted square. A separate constraint controls whether a central extension can increase nilpotency class. The families below separate these phenomena at every even point rank at least four. Throughout the section all unqualified dimensions are over $k = \mathbb{F}_4$, and $N(H) = \operatorname{im} \lambda_H$.

12.1 Commuting relations and the class-four kernel

Let

$$\widetilde{B}_0(H) = H_2^{\mathrm{CE}}(H; k) / \operatorname{span}\{[a \wedge b] : [a, b] = 0\}$$

be the homological Bogomolov quotient. This is a classical quotient, studied via nonabelian exterior products in [18]; decomposable bivectors and their geometry are central in [19].

Proposition 12.1. *For every $H \in \mathcal{E}$, λ_H factors through a natural Frobenius-semilinear surjection $\widetilde{B}_0(H)^{(1)} \to N(H)$. If $\operatorname{cl} H \le 4$, let $U = H/H'$ and*

$$K = \ker\big(\Lambda^2 U \longrightarrow \gamma_2(H)/\gamma_3(H)\big).$$

For $z = \sum_i u_i \wedge v_i \in K$, choose head lifts and define

$$\lambda_2(z) = \sum_i [\widetilde{u}_i, \widetilde{v}_i, \widetilde{v}_i, \widetilde{u}_i] + \sum_{i<j} [[\widetilde{u}_i, \widetilde{v}_i], [\widetilde{u}_j, \widetilde{v}_j]]. \tag{65}$$

This is independent of lifts and has image $N(H) \subseteq \gamma_4(H)$. It kills every decomposable member of K. Consequently $N(H) = 0$ in class at most three, and also in class four when $\dim U \leq 3$.

For $\dim U = 4$, write P for the Plücker quadratic form on $\Lambda^2 U$ after choosing a volume form. If $N(H) \neq 0$, then $P|_K$ is a nonzero square of a linear form, $\dim K \leq 3$, and $\dim N(H) \leq 1$. Conversely, a prescribed four-generator class-two truncation admits a class-four three-Engel lift with nonzero N exactly when $P|_K$ is such a nonzero square. The condition concerns existence of a lift, rather than every lift. Proofs and the three normal forms for K are in Appendix I.

This identifies a particular Frobenius-semilinear image of commuting relations. It is finer than the dimension of the whole Bogomolov quotient: an anisotropic plane with Plücker form $x^2 + xy + \omega y^2$ has no nonzero decomposable vectors over k, but cannot support this scalar obstruction in class four.

12.2 Compact families at arbitrary rank

Put $m = \binom{r}{2}$, $r \geq 2$. Define a graded algebra S_r with basis

$$\begin{array}{c|l} 1 & A_i, B_i \quad (1 \leq i \leq r) \\ 2 & E_{ij}, U_{ij} \ (i < j), \quad V_j \ (2 \leq j \leq r) \\ 3 & R_{ij}, S_{ij}, T_{ij} \quad (i < j) \\ 4 & Z_{ij} \quad (i < j). \end{array}$$

The nonzero brackets, besides their reverses, are

$$\begin{aligned} [A_i, A_j] = [B_i, B_j] = E_{ij}, \qquad & [A_i, B_j] = U_{ij} \qquad (i < j), \\ [A_j, B_j] = V_j \qquad & \qquad (j \geq 2), \\ [A_i, E_{ij}] = [B_i, U_{ij}] = R_{ij}, \qquad & [A_j, E_{ij}] = [B_i, V_j] = S_{ij}, \\ [A_j, U_{ij}] = [A_i, V_j] = T_{ij}, \qquad & [A_i, S_{ij}] = [A_j, R_{ij}] = [B_i, T_{ij}] = Z_{ij}. \end{aligned} \tag{66}$$

All other brackets vanish. Define $A_r \twoheadrightarrow S_r$ by adjoining central degree-two vectors C_{ij} and replacing only

$$[A_i, A_j] = E_{ij} + C_{ij} \qquad (i < j). \tag{67}$$

Theorem 12.2 (Compact obstruction and its minimum repair)**.** *The algebra S_r is a class-four metabelian three-Engel algebra, indecomposable even after restriction to the binary field. It has*

$$\dim S_r = 3r^2 - 1, \quad d(S_r) = 2r, \quad N(S_r) = \gamma_4(S_r) = \operatorname{span}\{Z_{ij} : i < j\}.$$

Thus $\dim N(S_r) = m$. The class-four central cover A_r satisfies $N(A_r) = 0$ and $\dim \ker(A_r \to S_r) = m$. By Theorem 11.1, it is the unique minimum repair extension class over S_r.

The head relation kernel of S_r has basis

$$A_i \wedge A_j + B_i \wedge B_j, \quad B_i \wedge A_j \ (i < j), \quad A_1 \wedge B_1. \tag{68}$$

Only the matched relations contribute to N; their images are the independent Z_{ij}. The cover makes those matched relations independent central bracket directions. Its minimality and equality class are instances of the general repair theorem.

Theorem 12.3 (Compact nonmetabelian family)**.** *For every $r \geq 2$ there is a class-four, binary-indecomposable, nonmetabelian $T_r \in \mathcal{E}$ with*

$$d(T_r) = 2r, \qquad \dim T_r = 3r^2 + 33, \qquad \dim N(T_r) = \binom{r}{2}.$$

Consequently the associated quadratic orbit solutions have

$$\dim_{\mathbb{F}_2} V_r = 6r^2 + 66, \qquad \dim_{\mathbb{F}_2} N_{f_r} = r(r-1), \qquad \frac{\dim_{\mathbb{F}_2} N_{f_r}}{\dim_{\mathbb{F}_2} V_r} \longrightarrow \frac{1}{6}.$$

Here T_r is an actual fiber product. Let D be the explicit four-generator class-four three-Engel algebra of dimension 45 in the verification table, with head relation $[a,b]+[c,d]=0$ and a surjection $D \twoheadrightarrow S_2$. Set $T_r = S_r \times_{S_2} D$, using the first pair of indices. The shared head generates the entire fiber product. The kernel of $T_r \to S_r$ has dimension 34 and is not central. Appendix I establishes the dimension, generation, nonmetabelianity and binary indecomposability.

12.3 Removing the scalar obstruction does not unlock ascent

Theorem 12.4 (Central-extension rigidity)**.** *For every $r \geq 2$, every finite central extension of S_r or A_r inside the ordinary three-Engel category has class four and is metabelian. The extension need not respect the displayed grading.*

Thus the minimum central cost in Theorem 12.2 removes the scalar obstruction but leaves a second, independent barrier to raising class. This is a statement relative to the three-Engel identity, rather than capability among all ordinary Lie extensions. The next section determines the exact first-ascent cost after class-preserving preparation. It does not classify all central descendants.

The proof has a finite coefficient part and an all-rank part. If $Z^2_{\mathrm{CE},3E}(H)_j$ denotes homogeneous central cocycles of total weight j satisfying the full cubic Engel coefficients, then

$$Z^2_{\mathrm{CE},3E}(S_r)_4 = B^2(S_r)_4, \quad Z^2_{\mathrm{CE},3E}(A_r)_4 = B^2(A_r)_4, \quad Z^2_{\mathrm{CE},3E}(A_r)_5 = 0.$$

Twelve exact row certificates establish the cases $2 \leq r \leq 5$. Every relevant cochain component is contained in an ordered local subalgebra on at most five indices, and its primitive is fixed by $t(Z_{ij}) = \beta(A_i, S_{ij})$. The local primitives therefore glue. This proves the identities at every rank, not just at the ranks in the certificates. Weight five for S_r also vanishes because its matched cycles lift all of $\gamma_4 = N$ to central elements. The final filtration argument treats ungraded extensions; see Appendix I.

13 A sharp first ascent over the compact bottoms

For S_r, A_r of Section 12, put $m = \binom{r}{2}$. We now determine the total cost of reaching class five while keeping the head rank $2r$. Define

$$D_r = \min\{\dim_k G - \dim_k S_r : G \in \mathcal{E},\ N(G) = 0,\ \mathrm{cl}\, G = 5,\ G \twoheadrightarrow S_r,\ d(G) = 2r\}.$$

The epimorphism in this definition is unrestricted; in particular, no bound on the length of a central tower is imposed.

Theorem 13.1 (Sharp relative cost and arrow count)**.** *One has*

$$D_2 = 5, \qquad D_r = \binom{r}{2} + 3 \quad (r \geq 3).$$

Three central arrows are necessary and sufficient to reach such an endpoint from S_r. If the first arrow is a minimum repair $A_r \to S_r$, four arrows in total are necessary and sufficient, with the same minimum final dimension.

For $r \geq 3$ an attaining algebra G_r is binary-indecomposable and satisfies

$$\dim_k G_r = \frac{7r^2 - r + 4}{2}, \quad d(G_r) = 2r, \quad \operatorname{cl} G_r = 5, \quad N(G_r) = 0, \quad G_r'' = kZ \subseteq Z(G_r).$$

At $r = 2$ an attaining algebra has ordinary dimension 16*, head rank four, and the same class and nonmetabelian properties.*

This is a minimum over a specified bottom, not a formula for $m(5)$. The lower bound holds for endpoints of arbitrarily long towers. Its ingredients are Proposition 11.2, a two-arrow obstruction over A_r, and, at rank four, a complete finite exclusion over $\mathbb{F}_4$.

The attaining mechanism is a missing cubic face. For $r \geq 3$, adjoin P, Q, Z of weights $3, 4, 5$ to A_r, keeping its old table and adding only

$$\begin{aligned} [A_1, U_{23}] = [A_2, U_{13}] &= P, \\ [A_1, T_{23}] = [A_2, T_{13}] = [A_3, P] &= Q, \\ [B_3, Q] = [U_{13}, T_{23}] = [U_{23}, T_{13}] = [V_3, P] &= Z. \end{aligned} \tag{69}$$

This is an actual fiber product with a twelve-dimensional exterior-module central extension. The shared heads generate the whole added kernel. The construction proves the identities and head rank at every r, rather than extrapolating finite tables; see Appendix J.

Theorem 13.2 (Attained repair and ascent cost after preparation)**.** *For $r \geq 3$, let $\widetilde{C}_r$ be S_r with the central cubic P in the first line of* (69) *added. It has class four and $n(\widetilde{C}_r) = m$, and is centrally rigid inside $\mathcal{E}$. There are central arrows*

$$G_r \twoheadrightarrow B_r \twoheadrightarrow \widetilde{C}_r$$

with $N(B_r) = N(G_r) = 0$, kernel dimensions $1, m+1$ respectively, and effective pairing rank one. Thus

$$\dim_k \ker(B_r \to \widetilde{C}_r) = n(\widetilde{C}_r) + r_{\text{eff}} = m + 1.$$

A one-dimensional cubic preparation is minimal for this attainment. At $r = 2$, the corresponding minimum preparation has dimension two, followed again by kernels of dimensions $m + 1$ and one.

Preparation and repair can be combined without changing the minimum endpoint dimension:

	$r \geq 3$	$r = 2$
preparation first	$(1, m+1, 1)$	$(2, m+1, 1)$
repair and preparation together	$(m+1, 1, 1)$	$(m+2, 1, 1)$
minimum repair alone first	$(m, 1, 1, 1)$	$(m, 2, 1, 1)$.

Entries are successive central-kernel dimensions, read from the bottom to the endpoint. The ordinary three-Engel central-depth problem in general is the subject of [21]; here the extra requirements are the specified square, $N = 0$, and the relative dimension cost. No new result from that companion is needed in the proofs.

14 Further questions

The ordinary description reduces several orbit questions to precise Lie problems. The minimum dimensions at class c, including their restriction to nonmetabelian objects, lie between the general lower bound and $O(2^{c/2})$; polynomial-size models would change this scale. The same-rank comparison connects optimal class growth to the classical ordinary function $e(d)$.

The cubic encoding is fully faithful, but its image among all cubic orbit rules is not yet characterized. A corresponding intrinsic criterion would clarify the information carried by the sextic defect. For quadratic objects, the existence of biquadratic coordinates is settled; imposing the averaging form $f + J_0^2$ or a prescribed linear readout is a separate normalization problem.

Theorems 11.1 and 13.1 settle the specified scalar repair and the first ascent over the compact bottoms. Their equality towers are not classified. For the displayed optimal class-five endpoints, the sharp total cost of a further ascent remains open. Ordinary central extensions at class six must also be tested against $\mathcal{R} = 0$; the ordinary three-Engel condition alone does not ensure quadratic realizability. Increasing the head rank is another relative minimization problem.

A Proof of global polar nilpotence

Throughout this appendix, $L = L_f$, $B(x, y) = [x, y]$, and only the unconditional identities of Section 2 are assumed. In particular B is a Lie bracket, $[x, f(x)] = 0$, and $f^2(x) = f(x) + x$. No nilpotence hypothesis is used until the final invocation of Theorem 3.3.

A.1 Two biderivations in the enveloping algebra

For an alternating bilinear map $C : L \times L \to L$, write $C_x(y) = C(x, y)$. A biderivation means that every C_x is a derivation; in characteristic two an alternating bilinear map is symmetric, so this also gives the derivation rule in the first variable.

Lemma A.1. *The maps D, G, S of (7) are alternating binary-bilinear biderivations with values in L'. With $A = \operatorname{ad} x$ and $F = \operatorname{ad} f(x)$, one has*

$$S_x = F^2 + AF + A^3 + A^2. \tag{70}$$

They also satisfy

$$S(x, f(x)) = 0, \qquad \sum_{\text{cyc}} S([x, y], z) = 0. \tag{71}$$

Proof. The polarization of $D(x, y) = [f(x), y] + [x, [x, y]]$ in its first variable is zero: the extra terms are $[[x, z], y] + [x, [z, y]] + [z, [x, y]]$, which vanish by Jacobi. Thus D is bilinear. The commuting identity gives $D(x, x) = 0$, and $D_x = \operatorname{ad} f(x) + (\operatorname{ad} x)^2$ is a derivation in characteristic two. Hence D is an alternating biderivation.

In the ordinary enveloping algebra $U(L)$, $(x + y)^2 = x^2 + y^2 + [x, y]$. The polar term in $f(x + y)$ cancels this last summand, so $R_x = x^2 + f(x)$ is binary-linear. Consequently $G(x, y) = [R_x, R_y]$ and $S = G+B+D$ are alternating bilinear. The identities $[x^2, y] = [x, [x, y]]$ and $[x^2, y^2] = [x, [y, [x, y]]]$ give

$$\begin{aligned} G(x, y) = {} & [x, [y, [x, y]]] + [x, [x, f(y)]] \\ & + [y, [y, f(x)]] + [f(x), f(y)]. \end{aligned} \tag{72}$$

In particular these commutators lie in L', not merely in $U(L)$.

To calculate S_x, put $b = [x, y]$ and $c = [f(x), y]$. The polarized commuting identity, first at (x, y) and then at $(f(x), y)$, gives

$$[x, f(y)] = c + [x, b] + [y, b], \qquad [f(x), f(y)] = b + c + [f(x) + y, c].$$

Substitution in (72) gives

$$G(x, y) = b + c + [x + f(x), c] + [x, [x, b]],$$

and hence (70). Since $[x, f(x)] = 0$, the operators A, F commute. Thus S_x commutes with A, or

$$S(x, [x, y]) = [x, S(x, y)]. \tag{73}$$

The commuting elements $x, f(x)$ also make R_x and $R_{f(x)} = f(x)^2 + f(x) + x$ commute. Therefore $S(x, f(x)) = 0$. Its third additive polarization, using that the polar of f is B, is the second identity in (71). Polarizing (73) in x at $x + z$ yields

$$S(x, [z, y]) + S(z, [x, y]) = [x, S(z, y)] + [z, S(x, y)].$$

The cyclic identity identifies the left side with $S([x, z], y)$. This is the derivation rule. Hence S, and then $G = S + B + D$, are biderivations. □

Lemma A.2. *The biderivation G satisfies*

$$G([x, y], z) = [G(x, y), z].$$

If L is perfect and centerless, there is a centroid operator C such that $G(x, y) = C[x, y]$.

Proof. For every alternating biderivation Q in characteristic two, expansion of the derivation identities gives $\sum_{\text{cyc}} Q([x, y], z) = 0$. Another expansion gives

$$\sum_{\text{cyc}} [Q(x, y), z] = Q([x, y], z) + [Q(x, y), z].$$

Jacobi for R_x, R_y, R_z, together with $[R_x, w] = D(x, w)$, gives $\sum_{\text{cyc}} D(G(x, y), z) = 0$. Put $T_x = x + R_x$. Its mixed commutators cancel, so

$$[T_x, T_y] = B(x, y) + G(x, y), \qquad [w, T_z] = (B + D)(w, z).$$

Jacobi for the three T's therefore reads

$$0 = \sum_{\text{cyc}} (B + D)((B + G)(x, y), z).$$

The B-Jacobi sum, the biderivation sum $\sum D(B(x, y), z)$, and the preceding R-Jacobi sum vanish. The remaining sum is $\sum B(G(x, y), z) = 0$, which, by the displayed expansion, is the asserted identity. In particular no adjoint cocycle property of D has been assumed here.

When $L = L'$, define

$$C\left(\sum_i [x_i, y_i]\right) = \sum_i G(x_i, y_i).$$

If $\sum_i [x_i, y_i] = 0$, the asserted identity says that the proposed image commutes with every $z \in L$. Centerlessness makes that image zero, so C is well defined. The same identity gives $C[w, z] = [Cw, z] = [w, Cz]$; these are the centroid identities. □

A.2 Simple algebras with a larger centroid

Assume for a contradiction that L is a nonabelian simple polar algebra. Its centroid K is a finite field. By Lemma A.2, $G = \kappa B$ for some $\kappa \in K$. Every binary derivation d of L is K-linear. Indeed, $[d, c]$ is again in the centroid for $c \in K$, and $c \mapsto [d, c]$ is a derivation of the finite separable field K; such a derivation is zero. Thus every D_x is K-linear. Symmetry then makes D K-bilinear in both arguments.

Jacobi in $U(L)$, at R_x, R_y, z, gives

$$[D_x, D_y] = \operatorname{ad} G(x, y) = \kappa \operatorname{ad}[x, y]. \tag{74}$$

Choose $a \in K$ with $a^2 = \kappa$, and put $E(x, y) = D(x, y) + a[x, y]$. This is an alternating, commutative bilinear product whose left multiplications commute: the mixed commutators cancel by $[D_x, \operatorname{ad} y] = \operatorname{ad} D(x, y)$, and the two remaining terms cancel by (74). Commutativity and commuting left multiplications imply associativity, since $(x*y)*z = z*(x*y) = x*(z*y) = x*(y*z)$. Alternation therefore gives

$$E_x^2 = 0, \qquad D_x^2 = \kappa(\operatorname{ad} x)^2. \tag{75}$$

Write $A = \operatorname{ad} x$ and $H = D_x^2 + AD_x + A^2$. Substituting $F = D_x + A^2$ in (70), and using $G = S + B + D$, gives

$$G_x = H + A^4 + D_x + A. \tag{76}$$

Apply this identity at λx, and subtract λ times its value at x. Scalar bilinearity of D, G yields

$$(\lambda^2 + \lambda)H + (\lambda^4 + \lambda)A^4 = 0 \qquad (\lambda \in K). \tag{77}$$

If $|K| \geq 8$, a polynomial of degree at most four vanishing at all elements of K has zero coefficients, so $A^4 = 0$. If $K = \mathbb{F}_4$, choose $\lambda \notin \mathbb{F}_2$. Then $H = 0$, and (76) gives $D_x = A^4 + (\kappa + 1)A$. Substitution in $H = D_x^2 + AD_x + A^2 = 0$, using (75), gives $A^5 = 0$. Engel's theorem makes L nilpotent in both cases, contradicting nonabelian simplicity.

A.3 The binary-centroid case

It remains that $K = \mathbb{F}_2$. Now $a = \kappa \in \{0, 1\}$. For fixed x, set $N = E_x = D_x + \kappa A$. Equations (75) and (76) become

$$N^2 = 0, \quad [N, A] = 0, \quad N(A + I) = A^4 + A^2 + A. \tag{78}$$

Squaring the last equality yields

$$A^8 + A^4 + A^2 = 0. \tag{79}$$

First there are no nonzero ad-nilpotent elements of L. For if A is nilpotent, (79) gives $A^2(I + A^2 + A^6) = 0$. The second factor is invertible, so $A^2 = 0$. Equation (78) then gives $N = A$. For every y,

$$\operatorname{ad} E(x, y) = [\operatorname{ad} x, E_y] = [E_x, E_y] = 0.$$

Centerlessness gives $E_x = 0$, and hence $A = 0$, so $x = 0$.

The absence of ad-nilpotent elements is not yet a semisimplicity assertion. The next argument removes the square-zero product itself. Associativity gives $E_x E_y = E_{E(x,y)}$, so this product of derivations is again a derivation. Its Leibniz cross terms yield

$$[E_x u, E_y v] + [E_y u, E_x v] = 0.$$

Set $u = x$, and use $E_x x = 0$ and symmetry. It follows that $\operatorname{im} E_x$ is an abelian subalgebra for the original bracket. Fix $N = E_x$ and let $z = Ny$. Because $N^2 = 0$ and $\operatorname{im} N$ is abelian, the derivation rule gives

$$\operatorname{ad}_z(L) \subseteq \ker N, \qquad \operatorname{ad}_z(\ker N) \subseteq \operatorname{im} N, \qquad \operatorname{ad}_z(\operatorname{im} N) = 0.$$

For the first inclusion, $N[z, v] = [Nz, v] + [z, Nv] = 0$. For the second, if $Nv = 0$, then $[Ny, v] = N[y, v]$. The third is exactly the abelianness of $\operatorname{im} N$. Thus $(\operatorname{ad} z)^3 = 0$. The preceding paragraph forces $z = 0$, so $E_x = 0$ for every x.

Equation (78) now reduces to

$$(\operatorname{ad} x)^4 + (\operatorname{ad} x)^2 + \operatorname{ad} x = 0.$$

The polynomial $t^4 + t^2 + t$ has derivative one. Every adjoint operator is therefore semisimple over an algebraic closure: L is anisotropic. Premet's theorem says that a finite-dimensional anisotropic ordinary Lie algebra over a perfect C_1-field of positive characteristic is commutative [11, Theorem 3(ii)]. The finite field $\mathbb{F}_2$ is perfect and C_1, so this contradicts nonabelian simplicity. Equivalently, the displayed identity implies $(\operatorname{ad} x)^8 = \operatorname{ad} x$, and [11, Corollary 1] gives the same conclusion.

A.4 Conclusion

All possible finite centroids of a nonabelian simple polar algebra have been excluded. Corollary 2.5 now implies that every finite quadratic polar algebra is solvable. Theorem 3.3 then proves Theorem 4.1.

The specific interface with the classical results is the pair of biderivations in Lemma A.1 and the adjoint-closed map in Lemma A.2. Premet's theorem is applied only after semisimplicity has been proved, rather than inferred from the weaker absence of ad-nilpotent elements. No finite nilpotent model, Engel identity for the full class of solutions, or scalar extension of the nonlinear map f enters this argument.

B Proofs for the canonical general structure

B.1 Central splitting and the freeness criterion

We first prove Lemma 7.1 independently of any uniform dimension bound or free-object construction in higher rank. For a binary-linear intertwiner T, polarization shows that $T[x, y] = [Tx, Ty]$. The smallest subsolution containing marked points ξ_i is the Lie algebra generated by $\xi_i, g(\xi_i)$: the generating set is stable under g modulo its binary span, and invariance propagates through sums by quadratic polarization and through brackets by (3).

Proof of Lemma 7.1. By Theorem 4.1, every finite target has nilpotent polar algebra. Induct on the binary dimension of such a target (V, f). If its polar algebra is abelian, f is linear and is its intrinsic $\mathbb{F}_4$-scalar. The required map is the $\mathbb{F}_4$-linear map from the marked basis of G/G'.

Otherwise the target center Z is nonzero by nilpotence. Proposition 2.3 makes Z invariant, with linear scalar $f|_Z = \omega$. The smaller solution V/Z has, by induction, a map $h : G \to V/Z$ with the prescribed marked values. Its pullback

$$P_h = \{(x, v) \in G \oplus V : h(x) = v + Z\}$$

is a subsolution of the product and is a central extension of G by (Z, ω). To see its splitting criterion directly, choose a binary-linear section and write its map as

$$(x, z) \longmapsto (g(x), \omega z + q(x)).$$

The functional equation is exactly $qg = \omega^2 q$; changing the section by a linear $t : G \to Z$ changes q by $tg + \omega t$. Thus its class belongs to $H_g(k) \otimes_k Z = 0$. A linear solution section therefore exists. Project it to obtain an intertwining lift $S : G \to V$.

The differences between the prescribed target points and $S\xi_i$ lie in Z. A k-linear map $G/G' \to Z$ can assume exactly these differences on the marked basis. Its inflation t satisfies $tg = \omega t$, so centrality makes $S+t$ an intertwiner with the exact prescribed values. Uniqueness follows because those values determine the images of $\xi_i, g\xi_i$, which Lie-generate G. □

The freeness criterion uses the full elementary central-extension space (22). The sixteen-dimensional coordinate map (21) has $136 = 16 + \binom{16}{2}$ zero-constant binary monomials of degree at most two with coefficients in k. The equation $qF_{16} + \omega^2 q = 0$ therefore has 272 binary unknowns. The final certificate of Table 1 gives

$$\operatorname{rank} = 244, \qquad \dim_{\mathbb{F}_2} \ker = 28, \qquad \dim_{\mathbb{F}_2} \operatorname{im} \delta = 28.$$

It follows that $H_{F_{16}}(k) = 0$. Appendix K describes the exact row, kernel, and section-image checks. The sparse bracket table below also verifies that the points a, b, together with their images, Lie-generate the sixteen-dimensional space: successive bracket closure has binary dimensions $4, 8, 16$. Its abelianization is the two-dimensional k-space on a, b. The freeness criterion is therefore applicable. By Theorem 4.1, every finite target is nilpotent, so every prescribed pair in a finite solution receives a homomorphism from this free solution. Its image is precisely the subsolution generated by that pair.

B.2 Identities on the free two-generator solution and their propagation

We give the coordinate part of Theorem 7.2 in a form that can be checked without the intermediate-prefix classification. Use the k-coordinate axes $e_a, e_b, e_c, e_d, e_e, e_h, e_r, e_s$ of (21). For arbitrary $\alpha, \beta \in k$, the nonzero polar products between distinct axes are

(i,j)	$[\alpha e_i, \beta e_j]$
(a,b)	$\alpha\beta e_c + \alpha^2\beta e_d + \alpha\beta^2 e_e$
(a,c)	$\alpha\beta e_d + \alpha\beta^2 e_r$
(b,c)	$\alpha\beta e_e + \alpha\beta^2 e_s$
(a,e)	$\alpha\beta e_h + \alpha^2\beta e_r$
(b,d)	$\alpha\beta e_h + \alpha^2\beta e_s$
(a,h)	$\alpha\beta e_r$
(b,h)	$\alpha\beta e_s$
(c,d)	$\alpha\beta e_r$
(c,e)	$\alpha\beta e_s$.

(80)

All omitted products, including those on a single k-axis, are zero; reversing the inputs uses symmetry. The entries record the binary-bilinear bracket in the displayed k-coordinates.

Let the first two coordinates of x be a, b, and those of y be A, B, and put $z = aB + Ab$. Substitution of (80) in the definitions of D and (72) gives

$$D(x,y) = \omega[x,y] + z^2 e_h, \qquad G(x,y) = \omega^2[x,y] + z^2 e_h. \tag{81}$$

Here the scalar multiplication on the right is the displayed coordinate scalar. One economical way to verify both formulas is to use bilinearity of B, D, G: for each row of (80), the residuals $D+\omega B$ and $G+\omega^2 B$ are zero, except in row (a,b), where each equals $\alpha^2\beta^2 e_h$. The single-axis residuals are zero. These formulas give all binary coefficients and prove $G = B + D$ on the entire free solution.

For the nonhomogeneous identity, put $w = [x, y]$ and $v = [x, w]$. If C is the c-coordinate of y, the table gives

$$\begin{aligned} &w_c = z, \quad w_d = aC + Ac + a^2B + A^2b, \quad w_e = bC + Bc + aB^2 + Ab^2, \\ &v_c = 0, \quad v_d = az, \qquad\qquad v_e = bz, \qquad\qquad v_h = aw_e + bw_d. \end{aligned}$$

The third adjoint has only its final two coordinates, and they are

$$\begin{aligned} ([x, v])_r &= a^2bz + a^2w_e + abw_d + acz = az^2, \\ ([x, v])_s &= ab^2z + abw_e + b^2w_d + bcz = bz^2. \end{aligned}$$

The equalities follow by substituting w_d, w_e, z and cancelling paired terms. The only nonzero coordinates of $[w, v]$ are $w_c v_d = az^2$ and $w_c v_e = bz^2$, respectively. Hence $[x, [x, [x, y]]] = [[x, y], [x, [x, y]]]$ for all inputs in this model.

The table gives lower central dimensions $(16, 12, 10, 6, 4, 0)$, with final term $ke_r + ke_s$. The third adjoint just computed lies in that center, so the fourth adjoint is zero. At $x = e_a, y = e_b$ its third adjoint is $e_r \neq 0$. Every pair in an arbitrary finite solution is the image of this marked pair under the freeness criterion. All three two-variable identities therefore descend, proving (23)–(25), including sharpness.

It remains to prove the three-variable identity, without any three-generator finite input. By Lemma A.1, S_x, F^2, A^2 are derivations, where $A = \operatorname{ad} x$ and $F = \operatorname{ad} f(x)$. Equation (70) therefore makes $AD_x = AF + A^3$ a derivation. Its Leibniz cross term is

$$[[x, y], D(x, z)] + [D(x, y), [x, z]] = 0.$$

Substitute $D_x = F + A^2$. The terms involving F vanish by (4) at $(x, f(x), y, z)$, since $[x, f(x)] = 0$. The remaining terms are

$$[[x, y], [x, [x, z]]] + [[x, [x, y]], [x, z]] = [x, [[x, y], [x, z]]],$$

as required. This completes the proof of Theorem 7.2.

B.3 The Lie pencil and canonical scalars

Proof of Theorem 7.3. For every alternating biderivation, the cyclic sum $\sum D([x, y], z)$ is zero. Jacobi for R_x, R_y, R_z, and the identity $G = B + D$, consequently give

$$0 = \sum_{\text{cyc}} D(G(x, y), z) = \sum_{\text{cyc}} D(D(x, y), z).$$

This is Jacobi for D. Put $T_x = x + R_x$. Now $[T_x, T_y] = B(x, y) + G(x, y) = D(x, y)$, and Jacobi for the three T's gives $\sum (B + D)(D(x, y), z) = 0$. Subtract Jacobi for D to obtain $\sum [D(x, y), z] = 0$. For a biderivation the expansion used in Lemma A.2 identifies this with $D([x, y], z) = [D(x, y), z]$. Thus both brackets and the mixed Jacobi coefficient vanish, proving the pencil assertion after any scalar extension. Finally Jacobi at R_x, R_y, z gives $[D_x, D_y] = \operatorname{ad} G(x, y)$, which is (28). □

We next construct J. For any expressions $w = \sum_i [x_i, y_i]$ and $u = \sum_i D(x_i, y_i)$, (27) gives the ambient identity

$$D_w = \operatorname{ad}_u \quad \text{on all of } L. \tag{82}$$

If $w = 0$, then $u \in W \cap Z(L) = C$. If $w \in C$, center invariance gives $D_w = 0$, so again $u \in C$. This proves both independence of the bracket expression and descent to the quotient $E = W/C$.

Jacobi for D gives $[D_x, D_y] = D_{D(x,y)}$. Combine this with (28) and (82). Representing the successive J-images in W, the result is

$$\mathrm{ad}_{J^2([x,y]+C)} = \mathrm{ad}_{[x,y]+D(x,y)}\,.$$

Equality here means equality of ambient adjoints; its two representatives differ by C. Bracket values span W, so $J^2 + J = \mathrm{id}$ on E. If $w \in W$, $z \in L$, and u represents $J(w + C)$, then

$$J([w, z] + C) = D(w, z) + C = [u, z] + C.$$

This proves commutation with every ambient adjoint. Taking $z \in W$ and using symmetry proves both centroid identities.

B.4 The canonical restricted square

Both W and $Z(L)$ are invariant. For $c \in C$, $f(w + c) = f(w) + f(c)$, with $f(c) \in C$; hence $\bar{f}$ is well defined. Its polar is the quotient bracket, as is that of $P = \bar{f} + J$.

For $w \in W$, choose $u \in W$ representing $J(w + C)$. By (82), the vector $p(w) = f(w) + u$ represents $P(w + C)$, and

$$\mathrm{ad}_{p(w)} = (\mathrm{ad}_w)^2 \quad \text{on all of } L. \tag{83}$$

We prove scalar semilinearity in this ambient representation, where the kernel on W is exactly C. Set

$$A = \mathrm{ad}_w, \qquad F = \mathrm{ad}_{f(w)}, \qquad Q = D_w = F + A^2 = \mathrm{ad}_u\,.$$

The operators A, F commute. Applying (83) to u gives

$$\mathrm{ad}_{p(u)} = Q^2 = F^2 + A^4.$$

On the other hand $J^2 = I + J$ and (82) give, for $v \in W$, $\mathrm{ad}_{J^2(v+C)} = \mathrm{ad}_v + D_v$, with representatives understood. For $v = p(w) = f(w) + u$,

$$\mathrm{ad}_v = A^2, \qquad D_{f(w)} = F + A + F^2, \qquad D_u = A + Q.$$

Here the middle equality uses $f^2(w) = f(w) + w$, and the last uses $J^2 = I + J$. Linearity of D consequently gives

$$\mathrm{ad}_{J^2P(w+C)} = A^2 + (F + A + F^2) + (A + Q) = F^2.$$

The difference from $\mathrm{ad}_{p(u)}$ is $A^4 = 0$, by (25). Thus $P(J(w + C)) = J^2P(w + C)$. The other nontrivial scalar follows from $J^2 = I + J$, the polar rule, and $[v, Jv] = J[v, v] = 0$. Together with (83), these are exactly the characteristic-two restricted Lie axioms over k.

Finally $[v, P(v)] = 0$ follows from restrictedness. In expanding the equation for $\bar{f} = J + P$, therefore,

$$\begin{aligned}\bar{f}^{\,2}(v) &= J^2v + JP(v) + P(Jv) + P(P(v)) + [Jv, P(v)] \\ &= J^2v + P(v) + P(P(v)).\end{aligned}$$

But $\bar{f}^{\,2}(v) = \bar{f}(v) + v = J^2v + P(v)$, so $P^2 = 0$. Every operation is intrinsic. An isomorphism, or a surjection, of solutions preserves the global-center quotients and their bracket and D-values, hence preserves J, P. This proves Theorem 7.4 with its stated naturality.

The unreduced derived ideal also admits a restricted square, although the above proof does not canonically select it. Choose a binary basis w_i of W and representatives u_i for $J(w_i + C)$, prescribe $p(w_i) = f(w_i) + u_i$, and extend by the restricted addition rule. The identity (83) then holds on all of W. Central choices may change this square, so the canonical k-structure asserted in the theorem remains that on E.

B.5 The reduced adjoint representation

Proof of Theorem 7.8. The adjoint action preserves W, C, so it induces the Lie homomorphism ρ on E. Commutation with J, already proved, makes each operator k-linear. For $w \in W$, symmetry and (82) give

$$D(x,w) + C = D(w,x) + C = [J(w+C), x] = J\rho(x)(w+C).$$

Thus $D_x|_E = J\rho(x)$. Since $D_x = \mathrm{ad}_{f(x)} + (\mathrm{ad}_x)^2$, we obtain $\rho(fx) = \rho(x)^2 + \omega\rho(x)$. The fourth-power identity used in the scalar-envelope argument follows from (25). More precisely, Theorem 7.7 gives $(\mathrm{ad}_x)^3(W) = 0$, and hence the associative matrix identity $\rho(x)^3 = 0$ on E. Its proof in Appendix E uses only the canonical restricted structure and its faithful ambient adjoints, rather than this matrix realization. Matrix cubes here are distinct from the Lie-adjoint cubes appearing in (37).

The kernel K is a Lie ideal, and the polynomial rule makes it f-stable. Its defining condition is $[K, W] \subseteq C$. Since $[K, K] \subseteq W$, it follows that $\gamma_3(K) \subseteq C$ and $\gamma_4(K) = 0$.

To check restricted derivations, let $\delta = \rho(x)$ and $u = w + C \in E$. The polarized commuting identity gives

$$\begin{aligned}\delta(\bar{f}(u)) &= \rho(fx)u + \delta^2 u + [u, \delta u] \\ &= (\delta^2 + \omega\delta)u + \delta^2 u + [u, \delta u] = \omega\delta u + [u, \delta u].\end{aligned}$$

Subtract $\delta(Ju) = \omega\delta u$. The result is $\delta(Pu) = [u, \delta u]$, the restricted-derivation rule. These derivations form a restricted k-Lie subalgebra: the rule is preserved by scalar sums, commutators, and squares, either by the restricted identities or by direct application of the derivation rule. □

B.6 The scalar envelope and its square-zero layer

The matrix image $X = \rho(L)$ carries the solution $F(A) = \omega A + A^2$. Its own special biderivation is

$$D_X(A,B) = [F(A), B] + [A, [A, B]] = \omega[A, B], \tag{84}$$

because $\mathrm{ad}_{A^2} = (\mathrm{ad}_A)^2$ in the associative matrix algebra. In particular the binary derived space X' is already k-linear. The scalar span $N = kX = X + \omega X$ is bracket closed. For $A \in X$, $A^2 = F(A) + \omega A \in N$; expanding a square of a scalar sum then proves that N is square closed. Hence it is exactly the restricted closure of X. Further,

$$N' = kX' = X' = \rho(L') = \mathrm{Inn}(E), \qquad N'' = \mathrm{ad}_E(E').$$

The equality with inner derivations uses the surjection $W \to E$ and does not identify E with its inner derivation algebra.

We first check the ordinary three-Engel identity on the actual binary image. For $A, B \in X$, put

$$K_{21} = A^2B + ABA + BA^2, \quad K_{12} = AB^2 + BAB + B^2A,$$
$$Q = A^2B^2 + AB^2A + B^2A^2, \quad H = BAB^2 + B^2AB, \quad R = B^2AB^2.$$

The zero cube of $A + B$ gives $K_{21} = K_{12} =: K$. For each $\lambda \in \{\omega, \omega^2\}$, the square companion $\lambda B + B^2$ is an actual element of X. Expanding its sum with A gives $0 = K + Q + \lambda H + R$. Subtracting the two equations yields $H = 0$, namely $(\mathrm{ad}_B)^3 A = 0$.

For $A, B \in X$, now expand in the associative matrix algebra:

$$(A + \lambda B)^4 = C_1\lambda + C_2\lambda^2 + C_3\lambda^3.$$

The constant and fourth coefficients vanish since $A^4 = B^4 = 0$. The two outer coefficients are $C_1 = (\mathrm{ad}_A)^3 B$ and $C_3 = (\mathrm{ad}_B)^3 A$. At $\lambda = 1$, the fourth power is again zero, giving $C_2 = C_1 + C_3$. Substitution of $\lambda = \omega$ proves (37).

The ordinary three-Engel identity just proved makes both outer coefficients zero. Thus (37) vanishes for every A, B, and every element of $N = X + \omega X$ has fourth matrix power zero. The restricted fourth-power expansion of $U + \lambda V$, for $U, V \in N$, has degree at most three after removing its zero pure fourth powers. Its vanishing at all four scalars of k makes its linear coefficient $(\mathrm{ad}_U)^3 V$ zero. Hence N is an ordinary three-Engel k-Lie algebra.

For the further reduction $\mathcal{M} = N'' \cap Z(N')$, an inner derivation $\mathrm{ad}_E e$ centralizes $\mathrm{Inn}(E)$ precisely when $[e, E] \subseteq Z(E)$, that is, $e \in Z_2(E)$. Since $N'' = \mathrm{ad}_E(E')$, a representative may be chosen in $E' \cap Z_2(E)$; changing a representative adds an element of $Z(E) \subseteq Z_2(E)$. This proves (35). Every such matrix maps E into $Z(E)$ and kills $Z(E)$, so the product of any two is zero. Both N'' and $Z(N')$ are ideals of N, the latter by Jacobi and invariance of N'. Their intersection $\mathcal{M}$ is consequently an ideal, and its zero matrix square makes it a restricted ideal. This proves Theorem 7.9, including the optional square-zero reduction $\mathcal{M}$.

B.7 The completion and its kernel

All operations on $\mathcal{T} = N/\mathcal{M}$ are restricted Lie operations. In particular the square descends because $\mathcal{M}$ is a restricted ideal; an associative quotient is neither asserted nor needed. With $P_{\mathcal{T}}(T) = T^{[2]}$, its polarization is the quotient bracket and $P_{\mathcal{T}}^2 = 0$. The restricted rules give

$$F^2(T) = \omega^2 T + T^{[2]} + T^{[4]} = F(T) + T.$$

This proves (38). As a Lie quotient of N, the restricted algebra $\mathcal{T}$ is also ordinary three-Engel.

Proof of Theorem 7.10. The polynomial rule for ρ proves that $q(x) = \rho(x) + \mathcal{M}$ intertwines the quadratic maps. Its kernel I is an f-stable Lie ideal. Since $\mathcal{M}$ is abelian, $[I, I] \subseteq K \cap W$, where $K = \ker \rho$. For $w \in K \cap W$, the inner derivation $\rho(w)$ is zero, so $w + C \in Z(E)$. For $i \in I$, the operator $\rho(i) \in \mathcal{M}$ is inner and hence kills $Z(E)$. Consequently $[i, w] \in C$. Thus $\gamma_3(I) \subseteq C$ and $\gamma_4(I) = 0$. This uses both the abelian image and its inner-action property.

As $\mathcal{M} \subseteq X' \subseteq X$, the image of q is $Y = X/\mathcal{M}$, and

$$\mathcal{T}' = N'/\mathcal{M} = X'/\mathcal{M} \subseteq Y.$$

It follows that Y is a binary Lie ideal of $\mathcal{T}$. Intertwining gives its F-stability, and $N = kX$ gives $kY = \mathcal{T}$. The induced map on the binary quotient $\mathcal{T}/Y$ is well defined by the quadratic addition rule and the ideal property. Its bracket is zero, so it is linear and satisfies $F^2 = F + \mathrm{id}$. This proves the two exact sequences and the abelian-cokernel assertion. □

Naturality under surjections extends to the envelope and completion. A surjective solution homomorphism $S : L \to L_1$ induces a surjective restricted k-Lie map $S_E : E \to E_1$. The rule

$$\Phi\left(\sum_i \lambda_i \rho(x_i)\right) = \sum_i \lambda_i \rho_1(S x_i)$$

is well defined, since $S_E A = \Phi(A) S_E$ and surjectivity detects a zero target operator. The same intertwining proves that Φ preserves brackets and squares. It is surjective, so it sends $N'' \cap Z(N')$ into $N_1'' \cap Z(N_1')$, giving the required maps on the quotients. For a nonsurjective solution map even preservation of the original global center can fail; the argument supplies no such map on reduced-derived quotients.

The restricted ideal generated by the fourth powers of N is zero. Thus $\rho : L \to N$ already intertwines the quadratic rules f and $A \mapsto \omega A + A^2$. The quotient by $\mathcal{M}$ is a further square-zero-ideal reduction that records the canonical kernel and cokernel displayed above.

B.8 The central obstruction on the whole derived ideal

Proof of Proposition 7.5. On the binary space underlying k^{11}, use coordinates

$$(a, b, c, d, z, v, w, s_1, s_2, s_3, t).$$

Define $f = \omega\,\mathrm{id} + q$, where the first four coordinates of q are zero and

$$\begin{aligned} &q_z = ab + cd, && q_v = ad, && q_w = bd,\\ &q_1 = az + cv + a^2 b, && q_2 = bz + cw + ab^2, && q_3 = aw + bv,\\ &q_t = as_2 + bs_1 + cs_3. \end{aligned}$$

All have binary degree at most two, since squaring in k is binary-linear. We verify the equation on every input, by showing $qf = \omega^2 q$. The first three nonzero coordinates are immediate. For example,

$$q_1(fx) = \omega^2(az + cv) + \omega a(ab + cd) + \omega c(ad) + a^2 b = \omega^2 q_1(x).$$

The two acd-terms cancel and $\omega + 1 = \omega^2$. The same argument gives the formula for q_2; the extra terms for q_3 are twice ωabd. Finally

$$q_t(fx) = \omega^2 q_t(x) + \omega(aq_2 + bq_1 + cq_3),$$

where the parenthesis is zero: each of abz, acw, bcv, a^2b^2 occurs twice. Thus $f^2(x) = \omega^2 x + q(x) = f(x) + x$.

Use upper-case letters for the coordinates of a second vector. The complete nonzero polar components are

$$\begin{aligned} &B_z = aB + Ab + cD + Cd, && B_v = aD + Ad, && B_w = bD + Bd,\\ &B_1 = aZ + Az + cV + Cv + a^2 B + A^2 b,\\ &B_2 = bZ + Bz + cW + Cw + aB^2 + Ab^2,\\ &B_3 = aW + Aw + bV + Bv,\\ &B_t = aS_2 + As_2 + bS_1 + Bs_1 + cS_3 + Cs_3. \end{aligned}$$

The derived ideal is $W_0 = \langle z, v, w, s_1, s_2, s_3, t\rangle_k$, and its bracket is zero. Indeed the products $(c, d), (a, d), (b, d), (a, z), (b, z), (a, w), (a, s_2)$, with varying scalar, span the seven displayed axes. The same table gives

$$\gamma_3(L) = \langle s_1, s_2, s_3, t\rangle_k, \quad \gamma_4(L) = kt, \quad \gamma_5(L) = 0.$$

The center is exactly kt. Bracketing a central vector with s_1, s_2, s_3 first forces $b = a = c = 0$, and bracketing with a then forces $d = 0$. Within W_0, bracketing with a, b, c forces all six coordinates other than t to vanish. Thus $W_0 \cap Z(L) = kt$, as asserted.

The complete biderivation formula is

$$D(x, y) = \omega B(x, y) + (aB + Ab)^2 e_t. \tag{85}$$

Here ω denotes the displayed coordinate scalar, without any centroid assumption on the whole original polar bracket. To verify it by binary bilinearity, split the space into the top $U = \langle a, b, c, d\rangle_k$ and W_0. For top inputs, the lower polar table gives $D = \omega B$ except possibly in the t-component; that component reduces to

$$Aab^2 + Ba^2 b + a(aB^2 + Ab^2) + b(a^2 B + A^2 b) = a^2 B^2 + A^2 b^2.$$

For top $x = (a, b, c, d)$, its action on W_0 is

u	z	v	w	s_1	s_2	s_3	t
$[x, u]$	$as_1 + bs_2$	$cs_1 + bs_3$	$cs_2 + as_3$	bt	at	ct	$0.$

The square of this operator is zero by pair cancellation. Thus $D(x, y) = \omega B(x, y)$ for $x \in U, y \in W_0$. On $W_0 \times W_0$ both sides vanish, proving (85).

Let e_i be the coordinate unit vectors, and take the binary exterior two-chain

$$\xi = e_a \wedge e_b + e_c \wedge e_d + e_a \wedge e_z + e_b \wedge e_z.$$

The four bracket values are $z+s_1+s_2, z, s_1, s_2$, so $B(\xi) = 0$. Equation (85) gives $D(\xi) = e_t \neq 0$. A linear factor $J_W B = D$ would send this zero to a nonzero vector, a contradiction. □

The counterexample retains the auxiliary k-structure of the metabelian normal form. What fails is the stronger prescription $J_W B = D$ on the whole original derived ideal. After reduction modulo kt, its scalar from (30) is precisely the displayed ω. Equivalently, with W_0 regarded as a trivial Lie module, D is a two-cocycle not of the form $T \circ B$; the cycle ξ detects its nonzero class. The general factorization on E and its restricted square are therefore consistent with this obstruction.

C Proofs for central reconstruction

We prove the refinement and morphism assertions of Section 5. All functions have zero constant term. Cochain inputs and polynomial degrees are over $\mathbb{F}_2$; coefficient scalars are over $\mathbb{F}_4$.

C.1 Refinement identities and the complete criterion

Lemma C.1 (Norm and defect identities)**.** *For the operators in* (9)*,*

$$T^3 = \mathrm{id}, \quad N\mathcal{L} = 0, \quad \Delta_g \delta = N, \quad N\Delta_g = \omega N, \quad \Delta_g^2 = \Delta_g + N. \tag{86}$$

For any quadratic q, with $\alpha = \mathrm{pol}\, q$,

$$Nq(x) = \alpha(x, g(x)). \tag{87}$$

The third additive difference at zero of this function is

$$\alpha(a, B(b, c)) + \alpha(b, B(a, c)) + \alpha(c, B(a, b)) = d_{\mathrm{CE}}\alpha(a, b, c). \tag{88}$$

In particular Nq is quadratic whenever α is a CE cocycle.

Proof. The relation $g^2 x = gx + x$ gives $g^3 = \mathrm{id}$ and $T^2 t = Tt + t$ for linear t, proving the first two identities. Multiplying the polynomials in T, using $\omega^2 + \omega = 1$, proves the remaining three. Polarization gives $q(x) + q(gx) + q(x + gx) = \alpha(x, gx)$, proving (87).

The right side has degree at most three. In its third difference, terms involving the linear part of g cancel, while its quadratic part contributes exactly the three terms in (88). Vanishing of this difference removes all degree-three Boolean ANF coefficients. This proves the last assertion. □

Lemma C.2 (The quadratic defect is a CE cocycle)**.** *Let $[\alpha] \in K_g$, choose a quadratic refinement q, and put $p = \Delta_g q$. Then p is quadratic,* $\mathrm{pol}\, p$ *is a CE cocycle, and its class belongs to K_g. This class is independent of the refinement and CE representative.*

Proof. By definition p is quadratic. The norm identities give $Np = \omega Nq$, which is quadratic by Lemma C.1. Taking third differences therefore gives $d_{\mathrm{CE}} \operatorname{pol} p = 0$. Also $\Delta_g p = \Delta_g^2 q = p + Nq$ is quadratic, so the resulting class lies in K_g.

Changing q by linear ℓ changes $\operatorname{pol} p$ by ℓB, a CE coboundary. Changing α by tB can be realized by changing q by δt; its defect is unchanged, because $\Delta_g \delta t = Nt = 0$. These observations prove all choice independence, as well as the well-definedness of Θ. □

Proof of Theorem 5.2. Necessity follows from $\Delta_g q = 0$. Conversely, suppose $\Theta[\alpha] = \Psi[\alpha] = 0$, and put $p = \Delta_g q$. Its polar is a coboundary, so choose linear ℓ such that $\operatorname{pol} p = \ell B$. Replacing q by $q' = q + \ell$ makes $p' = \Delta_g q'$ have zero polar, since $\operatorname{pol}(\Delta_g \ell) = \ell B$. Hence p' is linear. The identities

$$0 = Np' = N\Delta_g q' = \omega N q', \qquad \Delta_g p' = \Delta_g^2 q' = p' + Nq' = p'$$

now give $\Delta_g(q' + p') = 0$. The adjustment $\ell + p'$ is linear, so the new eigen-refinement has exactly the prescribed polar cocycle α. Lemma C.2 supplies the remaining assertions about Ψ. □

The degree-four component of the first obstruction has the explicit fourth difference

$$\alpha(B(a,b), B(c,d)) + \alpha(B(a,c), B(b,d)) + \alpha(B(a,d), B(b,c)). \tag{89}$$

Indeed only the products of two disjoint quadratic parts of g contribute. Thus Θ includes the central lifting condition for (4), together with any degree-three leakage. Its vanishing still leaves the quadratic test Ψ.

C.2 The cyclic domain and the fixed splitting

Proof of Theorem 5.3. For $[\alpha] \in K_g$, the refinement Tq is quadratic, and $\operatorname{pol}(Tq) = \operatorname{pol}(\Delta_g q) + \omega^2 \alpha$ is CE-closed by Lemma C.2. Changing q by linear ℓ changes this cocycle by the coboundary ℓB. A CE representative change via δt changes Tq by $T\delta t = \omega^2 \delta t$, again with coboundary polar. This proves well-definedness.

The norm formula yields $T^2 q = q + Tq + Nq$, so all three iterates are quadratic. Their polars are CE-closed, since $N(T^j q) = Nq$ and (88) applies. Thus τ preserves K_g and $\tau^3 = \mathrm{id}$. Polarizing $\Delta_g q = Tq + \omega^2 q$ gives the formula for Ψ. The complete criterion identifies its kernel with $H_g(A)$. Finally the polynomial $1 + \omega X + \omega^2 X^2$ has values $0, 0, 1$ at $1, \omega, \omega^2$, respectively. The three roots are distinct, proving the projector assertion.

Pullback by a binary-linear solution homomorphism commutes with composition by g, quadratic degree bounds, polarization, and CE coboundaries. Coefficient maps commute with the same operations. This proves naturality without invertibility assumptions. □

The space K_g and its cyclic operator precede the choice of coefficient marking. One can define Θ using the high-degree part of Tq alone. To change a CE representative by tB, use the refinement change Tt, with $T^2 t = Tt + t$. All of these constructions are defined over $\mathbb{F}_2$. Extension to $\mathbb{F}_4$ then selects the ω^2-eigenspace for the prescribed scalar kernel.

Details of the fixed splitting and dimension law. For a CE cocycle α, Lemma C.1 makes $\alpha(x, gx)$ an invariant quadratic. A coboundary evaluates to $tB(x, gx) = 0$ by (2). For invariant quadratic r, $Nr = r$, and (88) proves that its polar is CE-closed. Hence $O_0[\operatorname{pol} r] = r$, proving the section assertion on all ordinary cohomology.

An invariant refinement represents a τ-fixed class. Conversely, for a fixed class $[\alpha] \in K_g$, its norm $r = Nq$ is invariant quadratic and

$$[\operatorname{pol} r] = (\mathrm{id} + \tau + \tau^2)[\alpha] = [\alpha].$$

Thus the section image is precisely $(K_g)_1$, establishing Proposition 5.4.

Binary descent of K_g and τ means that coefficient Frobenius interchanges its ω- and ω^2-eigenspaces. This proves $\dim K_g = i_g + 2h_g$. The independent reduced monomials $u_i u_j^2$ on the intrinsic abelianization prove $i_g \geq d^2$. Trivial coefficient extension gives $\dim_{\mathbb{F}_4} H^2(L_g, \mathbb{F}_4) = \dim_{\mathbb{F}_2} H^2(L_g, \mathbb{F}_2)$, so rank–nullity proves (18) and (19). For the linear example, every quadratic remains quadratic under composition. Its fixed monomials are $u_i u_j^2$; after linear section gauge, its ω^2-sector is spanned by the $\binom{d}{2}$ products $u_i u_j$, $i < j$. This supplies the sharpness values in the main text.

Finally O_0 restricted to K_g is its fixed-sector retraction. Hence $K_g \cap \ker O_0 = (K_g)_\omega \oplus H_g(A)$. The fixed splitting makes $\ker O_0 \to H^2_{\mathrm{CE}}/K_g$ onto, with this intersection as kernel. Quotienting by $H_g(A)$ proves (20). □

D Fixed quadratic rigidity and the complete free-quotient obstruction

Let $f : V \to V$ be a binary-quadratic solution of $f^2 = f + I$, and let $L = (V, B)$ be its polar Lie algebra. All functions below have zero constant term and values in $k = \mathbb{F}_4$. The coefficient space can more generally be any finite-dimensional k-space, by tensoring. Write $\gamma_i = \gamma_i(L)$.

Lemma D.1 (Fixed binary cocycles on a scalar Lie algebra)**.** *Let $\mathfrak{a}$ be a Lie algebra over k, with scalar operator $J = \omega\,\mathrm{id}$, and regard it as a binary Lie algebra. If an alternating binary CE two-cocycle $\eta : \mathfrak{a} \times \mathfrak{a} \to k$ satisfies*

$$\eta(Jx, Jy) = \eta(x, y),$$

then $\eta(\mathfrak{a}', \mathfrak{a}) = 0$.

Proof. Extend the input binary space and bracket to $\mathfrak{a}_{\mathrm{bin}} \otimes_{\mathbb{F}_2} k$, and extend η bilinearly to this space with values in k. The centroid operator J has the distinct eigenvalues ω, ω^2, so the extended Lie algebra is a direct sum of its two eigenspace ideals $\mathfrak{a}_+$ and $\mathfrak{a}_-$. They commute: applying the centroid to a cross bracket gives both distinct scalar multiples of that bracket. The fixed-form identity makes η vanish on each same-sign pair, since $\omega^2 \neq 1$ and $\omega \neq 1$. For $x, y \in \mathfrak{a}_+$, $z \in \mathfrak{a}_-$, its CE equation reduces to $\eta([x, y], z) = 0$. The analogous equation kills the other derived cross pair. Together with same-sign vanishing this gives the assertion on the extended algebra and hence on the original binary algebra. □

Theorem D.2 (First-integral rigidity in every nilpotency class)**.** *For every finite quadratic solution, L is nilpotent by Theorem 4.1. Every binary-quadratic function $r : V \to k$ satisfying $r \circ f = r$ factors through V/L'. If $d = \dim_k(V/L')$ with its intrinsic scalar structure, then*

$$I_f(k) = \{r \in \mathcal{Q}_{\leq 2}(V, k) : r \circ f = r\} \simeq I_{\bar{f}}(k), \qquad \dim_k I_f(k) = d^2.$$

Proof. Put $\alpha = \mathrm{pol}\, r$. The norm identity

$$r(x) + r(fx) + r(f^2 x) = \alpha(x, fx)$$

has left side $r(x)$, since r is invariant and three equals one in characteristic two. Taking its third additive difference proves $d_{\mathrm{CE}}\alpha = 0$.

If $\alpha = 0$, then r is linear; invariance and $f^2 x = fx + x$ force $r = 0$, so the claim holds. Otherwise nilpotency makes the following integer finite:

$$m = \max\{i + j : \alpha(\gamma_i, \gamma_j) \neq 0\}.$$

Assume $m \geq 3$. Because $2m > m$, the restriction $\alpha(\gamma_m, \gamma_m)$ vanishes. Thus $r|_{\gamma_m}$ is linear. The ideal γ_m is f-invariant. A linear invariant function on it vanishes: for $u \in \gamma_m$,

$$r(u) = r(f^2 u) = r(fu + u) = r(fu) + r(u) = 0.$$

For $x \in \gamma_i$, $y \in \gamma_j$ with $i + j = m$, polarizing $r \circ f = r$ gives

$$\alpha(x, y) = \alpha(fx, fy) + r(B(x, y)) + \alpha(fx + fy, B(x, y)). \tag{90}$$

Here $B(x, y) \in \gamma_m$, so the middle term vanishes. The last term vanishes too: its two pair depths are $i + m$ and $j + m$, both larger than m, and f preserves γ_i and γ_j. Consequently $\alpha(fx, fy) = \alpha(x, y)$ on these pairs.

The maximal-depth part of α defines an alternating binary two-form α_m on $\operatorname{gr}_\gamma L = \bigoplus_i \gamma_i/\gamma_{i+1}$: on factors of degrees i, j with $i + j = m$, use α, and on all other pairs use zero. It is well defined since replacing either input by the next lower-central term raises the pair depth above m. The CE equation for α on triples with total depth m is exactly the CE equation for α_m; on other homogeneous triples its terms are zero. Thus α_m is CE-closed.

The graded scalar theorem makes $\operatorname{gr}_\gamma L$ a k-Lie algebra under the operators induced by f. Equation (90) shows that α_m is fixed by these simultaneous scalars. By Lemma D.1, it vanishes if either input belongs to the derived algebra of $\operatorname{gr}_\gamma L$. The lower-central graded algebra is generated by degree one; in particular every component of degree at least two belongs to its derived algebra. When $m \geq 3$, every pair of positive degrees summing to m has such an input. This forces $\alpha_m = 0$, contradicting the definition of m.

It follows that $m \leq 2$. Therefore $\alpha(L', V) = 0$. The restriction $r|_{L'}$ is linear and invariant, so it is zero by the same argument above. Hence $r(x + w) = r(x)$ for every $w \in L'$, proving factorization.

On the abelianization, $\bar{f}$ is multiplication by ω. Its invariant quadratic functions have basis u_i^3, and $u_i^2 u_j, u_i u_j^2$ for $i < j$, of total dimension $d + 2\binom{d}{2} = d^2$.

□

Corollary D.3 (The fixed cohomological sector in every nilpotency class)**.** *For every finite quadratic solution, with coefficient line k,*

$$(K_f)_1 \simeq I_f(k) \simeq I_{\bar{f}}(k), \qquad \dim(K_f)_1 = d^2.$$

Consequently, writing $h_f = \dim_k H_f(k)$,

$$\dim K_f = d^2 + 2h_f, \qquad b_2 = \operatorname{rank}\Theta + d^2 + 2h_f, \qquad h_f = \frac{b_2 - \operatorname{rank}\Theta - d^2}{2}.$$

These identities determine the fixed sector and its contribution to the obstruction in arbitrary nilpotency class. They do not assign the general values of b_2 or $\operatorname{rank}\Theta$.

Proof. The canonical polarization section identifies invariant quadratics with $(K_f)_1$. Binary descent exchanges the two nontrivial cyclic eigenspaces, each of dimension h_f. Apply the theorem and rank–nullity for Θ. □

E Global stability and the universal restricted derived quotient

The proof of global stability uses the canonical restricted structure of $E = W/(W \cap Z(L))$, but no exact scalar factor on the whole derived ideal. We first obtain a low-class stability lemma directly from the two-orbit identities. Faithful ambient adjoints then control derived inputs in arbitrary class. A character projection on the abelianization and the cyclic norm complete the argument. Finite free objects enter only afterwards, to construct exact restricted covers and the universal quotient.

E.1 The exact cochain criterion and a low-class lemma

Let A be an alternating binary-bilinear cochain, with any finite binary coefficient space, and choose a zero-constant quadratic refinement q_A. Write

$$\mathcal{C}_A(x,y,z) = \Delta_x\Delta_y\Delta_z(q_A f)(0).$$

This is independent of the linear part of q_A. It vanishes identically exactly when $q_A f$ is quadratic. Direct polarization gives

$$\begin{aligned}\mathcal{C}_A(x,y,z) = {} & A(B(x,y),B(z,y)) + A(B(x,z),fy)\\ & + A(fx,B(z,y)) + A(fz,B(x,y))\\ & + A(B(x,z),B(x,y)+B(z,y)).\end{aligned} \tag{91}$$

The expression is a third additive difference, not an assumed trilinear form.

For $A = B, D$, adjoint closedness converts (91) into commutators. Substitute $fv = R_v + v^2$ in $U(L)$, where $R_v = v^2 + f(v)$ is linear. For $A = B$, the three R-terms have sum $\sum_{\mathrm{cyc}} D(B(x,y),z) = 0$: expanding an alternating biderivation gives six terms that cancel in pairs. For $A = D$, their sum vanishes by Jacobi for D, the identity $D_{D(x,y)} = \mathrm{ad}_{B(x,y)+D(x,y)}$, and adjoint closedness. Thus

$$0 = \mathcal{C}_B(x,y,z) = \sum_{\mathrm{cyc}} \big([x,[x,B(y,z)]] + [B(x,z),B(x,y)]\big), \tag{92}$$

$$\mathcal{C}_D(x,y,z) = \sum_{\mathrm{cyc}} \big([x,[x,D(y,z)]] + [D(x,z),B(x,y)]\big). \tag{93}$$

The first left side is zero because f itself refines B and $f^2 = f + \mathrm{id}$.

We shall also use the exact bracket correction

$$f([u,v]) + D(u,v) = K(u,v), \qquad K(u,v) = [u,[u,[u,v]]] + [v,[v,[u,v]]] + [u,[v,[u,v]]]. \tag{94}$$

Here is an enveloping-algebra derivation. The function $R_x^2 + x^2$ has polar D, since $[R_x,R_y] = B(x,y)+D(x,y)$. As $[x,fx] = 0$, it equals $x^4 + f(x)^2 + x^2$. The identity $f(x)^2 = R_{f(x)} + f^2(x)$ then gives

$$\mathrm{pol}(x \mapsto x^4)(u,v) = B(u,v)^2 + f(B(u,v)) + D(u,v).$$

The ordinary fourth-power expansion gives the same polar as $B(u,v)^2 + K(u,v)$, proving (94). By (24), for $u \in \gamma_a L$ and $v \in \gamma_b L$,

$$K(u,v) \in \gamma_{2a+2b} L. \tag{95}$$

Indeed the three terms have depths at least $3a+2b$, $2a+3b$, and $2a+2b$, respectively.

Lemma E.1 (A low-class stability bound)**.** *For every finite solution, $\mathcal{C}_D(L,L,L) \subseteq \gamma_8 L$. Consequently D is stable whenever $\mathrm{cl}(L) \le 7$, in particular on every subsolution generated by two marked points.*

Proof. Write the six summands of (92) as $t_i = [u_i, v_i]$. For the nested terms take $u_i = x$, $v_i = [x,[y,z]]$, of depths $(1,3)$; for the other terms take $u_i = [x,z]$, $v_i = [x,y]$, of depths $(2,2)$, with cyclic permutations. Each $t_i \in \gamma_4 L$ and $\sum_i t_i = 0$. Quadratic expansion gives

$$\sum_i f(t_i) = \sum_{i<j} [t_i, t_j].$$

Adjoint closedness and the biderivation rule identify $\sum_i D(u_i,v_i)$ with $\mathcal{C}_D$. Applying (94) therefore gives

$$\mathcal{C}_D = \sum_i K(u_i,v_i) + \sum_{i<j} [t_i,t_j] \in \gamma_8 L.$$

The two-point assertion uses the free sixteen-dimensional solution, whose class is five, as established in Appendix B. □

E.2 An order-three lift and its central square defect

Put $C = W \cap Z(L)$, $E = W/C$, and $j = f|_C$. Choose a binary-linear lift $J : W \to W$ of the canonical scalar on E, with $J^2 + J = \mathrm{id}$ and $J|_C = j$. Such a lift exists by splitting the binary vector sequence $0 \to C \to W \to E \to 0$ and transporting the two given $\mathbb{F}_4$-scalar structures. Equation (82) gives

$$[Jw, z] = D(w, z) \qquad (w \in W, z \in L). \tag{96}$$

Define

$$S = f|_W + J, \qquad r = S^2, \qquad a = D + JB. \tag{97}$$

The cochain a takes values in C, is an ordinary trivial-coefficient CE cocycle, and satisfies $a(x, fx) = 0$. The map S has polar $B|_{W^2}$ and satisfies $\mathrm{ad}_{S(w)} = (\mathrm{ad}_w)^2$ on all of L; it is a binary restricted square. It induces the canonical P on E, and $S(C) = 0$. In particular $r(W) \subseteq C$.

Lemma E.2. *The map $r : W \to C$ is quadratic, with polar $A = a|_{W \times W}$, and vanishes on C.*

Proof. The identities $[w, Jw] = D(w, w) = 0$ and ambient compatibility give $[Jw, Sw] = (\mathrm{ad}_w)^2(Jw) = 0$. Expansion of $f^2 = f + \mathrm{id}$ on W, with $f = J + S$, gives $S(Jw) + JS(w) + S^2(w) + S(w) = 0$. Polarization yields

$$\mathrm{pol}(S^2)(u, v) = B(Ju, Jv) + JB(u, v) + B(u, v).$$

This is bilinear, proving quadraticity. Express u, v as sums of original bracket values. Their J-images differ from the corresponding D-sums only by central terms. The identity

$$[D(x, y), D(z, t)] = [B(x, y), (B + D)(z, t)]$$

follows from adjoint closedness and $D_{D(z,t)} = \mathrm{ad}_{B(z,t)+D(z,t)}$. It gives $B(Ju, Jv) = B(u, v) + D(u, v)$ exactly. Substitution proves the asserted polar. Finally $S(C) = 0$ gives $r(C) = 0$. □

Proposition E.3 (The central square-height class)**.** *The function $r = S^2$ descends to $r_E : E \to C$, satisfies $r_E \bar{f} = j r_E$, and defines a class*

$$[r_E] \in H_{\bar{f}}(C, j^2) := \frac{\{q : E \to C : \deg_2 q \le 2,\ q(0) = 0,\ q\bar{f} = jq\}}{\{t\bar{f} + j^2 t : t \in \mathrm{Hom}_{\mathbb{F}_2}(E, C)\}}. \tag{98}$$

This class is independent of the order-three lift J. It is natural under solution isomorphisms and surjective solution maps, with pullback on the quotient and the induced map on center coefficients. The coefficient operator in this central-extension space is j^2, the reverse of the original center operator.

Proof. The cochain a annihilates a central input, and $r(C) = 0$, so the quadratic addition rule makes r constant on cosets of C. The functional equation gives $S(Jw) = J^2 S(w) + r(w)$ and hence $S(J^2 v) = JS(v) + r(v)$. Since $S(C) = 0$ and $r(Sw) = S^3 w = S(rw) = 0$, these identities give $r(Jw) = jr(w)$. Moreover $a(Jw, Sw) = [J^2 w, Sw] = 0$. Thus $r(fw) = r(Jw + Sw) = jr(w)$, proving the eigencondition.

For another lift $J' = J + t$, the map $t : W \to C$ vanishes on C. Expanding the two scalar polynomials gives $tJ = j^2 t$, and $S' = S + t$ gives $r' = r + tS = r + (tf + j^2 t)$. Both t and this gauge equation descend to E, proving independence.

For a surjective solution map $T : L \to L_1$, choose lifts J, J_1 and put $u = TJ + J_1 T : W \to C_1$. It vanishes on C, satisfies $uJ = j_1^2 u$, and obeys $TS = S_1 T + u$. Since $S_1(C_1) = 0$, it follows that $Tr = r_1 T + uS = r_1 T + (uf + j_1^2 u)$. This is precisely the asserted equality of the pushed-forward and pulled-back classes. □

The class retains a central square defect; its definition does not identify its vanishing with global stability of D, nor assert that a representative on E extends to an admissible refinement on all of L.

Whenever a refinement on all of L is needed, fix a binary-linear extension $\widetilde{J} : L \to W$ of J. Then $\widetilde{J}f$ refines JB and is stable, since $\widetilde{J}f^2 = \widetilde{J}(f + \mathrm{id})$ is quadratic. Choose a C-valued quadratic refinement q_a with $q_a|_W = r$; the matching polars make this possible by a linear adjustment. Put $h = q_a f$.

E.3 Faithful ambient coefficients and the cube on W

The map

$$\iota : E \longrightarrow \mathrm{End}_{\mathbb{F}_2}(L), \qquad \iota(w + C) = \mathrm{ad}_w$$

is injective and preserves brackets and restricted squares, by (83). For $z \in L$, put $d = \mathrm{ad}_z$ and let δ be its induced k-linear derivation on E. Ambient compatibility gives $dS(w) = [w, dw]$, so δ is a restricted derivation of (E, P).

Fix $u \in E$ and write $T = \iota(u)$, $v = Pu + \delta u$. Squaring twice in $\mathrm{End}_{\mathbb{F}_2}(L)$ gives

$$(d + T)^2 = d^2 + \iota(v), \qquad (d + T)^4 = \iota(Pv + \delta^2 v).$$

The fourth power is zero by (25), since $d + T$ is an actual ambient adjoint. Use $P^2 = 0$ and $\delta^2 Pu = [u, \delta^2 u]$ to obtain, by injectivity,

$$P\delta u + \delta^3 u + [u, \delta^2 u] + [Pu, \delta u] = 0.$$

For each $\lambda \in k$, a representative of λu in W again gives an actual ambient adjoint. Substitution gives

$$\lambda\delta^3 u + \lambda^2\big(P\delta u + [u, \delta^2 u]\big) + \lambda^3[Pu, \delta u] = 0.$$

This degree-three polynomial vanishes at all four field elements; therefore

$$\delta^3 u = 0, \qquad P\delta u = [u, \delta^2 u], \qquad [Pu, \delta u] = 0. \tag{99}$$

The coefficient extraction takes place in E, after injectivity of ι; it assumes no k-scalar multiplication on the whole ambient matrix algebra.

Proof of Theorem 7.7. The first identity in (99) says $d^3(W) \subseteq C$. Let $q = d^3|_W$. Successive applications of the restricted-derivation identity give

$$dSw = [w, dw], \qquad d^2Sw = [w, d^2 w], \qquad d^3Sw = [dw, d^2 w] + [w, d^3 w].$$

The last term vanishes by centrality, while (24) gives $[dw, d^2 w] = d^3 w$. Thus $qS = q$ pointwise. Iterating once gives $qS^2 = q$, but $S^2(W) \subseteq C$ and $q(C) = 0$. Hence $q = 0$. □

E.4 The exact coset formula

For $z \in L$, continue to write $d = \mathrm{ad}_z\,|_W$, and put

$$k_z(w) = dJw + Jdw = a(w, z) \in C.$$

One has $k_z(C) = 0$, $k_z J = j^2 k_z$, and

$$dSw = [w, dw], \qquad k_z(Sw) = A(w, dw), \qquad d_{f(z)} = dJ + d^2, \qquad k_{f(z)} = k_z d + j^2 k_z. \tag{100}$$

For the second identity, $D_z = dJ = d_{f(z)} + d^2$ is an S-restricted derivation, as are both summands. Apply it to Sw, use the first identity and $[w, Jdw] = D(w, dw)$, and rearrange.

The remaining identities follow by commutator expansion, using $dk_z = 0$. The exact lift relation also gives

$$A(Ju, v) = j^2 A(u, v) \qquad (u, v \in W). \tag{101}$$

Indeed both sides equal $B(u, v) + D(u, v) + JD(u, v)$.

Quadratic expansion of $h = q_a f$ now gives

$$h(w + z) + h(w) = r(dw) + A(fw, dw) + k_{f(z)}(fw) + k_{f(z)}(dw) + h(z).$$

Use $fw = Jw + Sw$, (100), and (101). The two $j^2 A(w, dw)$-terms cancel; the non-affine part in w is

$$F_d(w) = r(dw) + A(Sw, dw) + A(w, d^2 w). \tag{102}$$

We prove that this function is zero exactly. Put

$$U_0 = S(dw), \qquad V_0 = [Sw, dw], \qquad T_0 = [w, d^2 w], \qquad R_0 = S(d^2 w).$$

Equations (99) show that $U_0 + V_0 + T_0 \in C$ and $R_0 \in C$; for the latter apply $P\delta v = [v, \delta^2 v]$ at $v = \delta(w + C)$. The polar formula for S^2 is equivalent to

$$S([p, q]) = [Sp, Sq] + [Sp + Sq, [p, q]] + A(p, q).$$

Expand $S(U_0 + V_0 + T_0) = 0$ with this identity. The two $[U_0, V_0]$-terms cancel and $[U_0, T_0] + [V_0, T_0] = 0$ by centrality. Consequently

$$F_d(w) = [Sw, R_0] + [Sw + R_0, T_0] = [Sw, T_0].$$

On the other hand, $r(w) \in C$ implies $0 = d^2 r(w) = [Sw, d^2 Sw] = [Sw, T_0]$. This proves $F_d = 0$.

The remaining affine terms give the precise formula

$$h(w + z) + h(w) + h(z) = j^2 k_{f(z)}(w) + k_{f(z)}(dw) = j k_z(w) + k_z d^2 w.$$

Since $dk_z = 0$, one has $[J, d^3] = k_z d^2$. Theorem 7.7 therefore removes the last term:

$$h(w + z) + h(w) + h(z) = j\, a(w, z) \qquad (w \in W, z \in L). \tag{103}$$

In particular all third differences of h with a derived input vanish.

E.5 A character lemma and the cyclic-norm conclusion

For the coefficient space C, scalar multiplication by ω is the original center operator $j = f|_C$.

Lemma E.4 (Plane-wise quadratics in a nontrivial character)**.** *Let U, A be finite $k = \mathbb{F}_4$-spaces, and let $\phi : U \to A$ have binary degree at most four. Suppose its restriction to every k-subspace of dimension at most two is quadratic. If $\phi(\omega u) = \chi\phi(u)$, where $\chi = \omega$ or ω^2, then ϕ is quadratic.*

Proof. The three inputs of a third difference span at most a three-dimensional k-space, so it suffices to work in k^3. Coefficient functionals reduce to $A = k$, and output Frobenius interchanges the two characters. Take $\chi = \omega^2$. Use the function-polynomial basis with each coordinate exponent in $\{0, 1, 2, 3\}$. Its binary degree is the sum of the binary digit weights: $1, X, X^2, X^3$ have degrees $0, 1, 1, 2$, and tensor products give a basis adapted to the degree filtration. Modulo quadratics, restriction to the coordinate planes leaves only

$$aX^3YZ + bXY^3Z + cXYZ^3 + dXY^2Z^2 + eX^2YZ^2 + gX^2Y^2Z.$$

On $Z = \lambda X$, the forbidden cubic coefficients are $b\lambda$ and $d\lambda^2 + g\lambda$. Taking $\lambda = 1, \omega$ gives $b = d = g = 0$. On $Z = \lambda Y$ one obtains $a = e = 0$, and on $X = \lambda Y$ one obtains $c = 0$. □

Proof of Theorem 7.6. Equation (103) makes $\Delta_w h$ affine-linear for every $w \in W$. In an adapted binary basis no monomial of degree at least three can therefore contain a W-coordinate. Writing $\pi : L \to U = L/W$, we have

$$h = \phi\pi + q_0,$$

where $q_0 : L \to C$ is quadratic and $\phi : U \to C$ has degree at most four, all with zero constant. Every k-plane in U is covered by the subsolution generated by two marked lifts of its basis. Lemma E.1 makes D stable there, and $\widetilde{J}f$ makes JB stable there. Thus h, and hence ϕ, restricts to a quadratic function on that plane; use any binary-linear section of the covering map. No preservation of this subsolution by J is required.

Put $\phi_0 = \phi + \phi\omega + \phi\omega^2$, where scalars act on the input. This is the exact invariant-character projection. The other two character components are quadratic by Lemma E.4. Absorb them into q_0 and rename ϕ_0 as ϕ. Now $\phi(\omega u) = \phi(u)$ exactly.

Let $b_0 = \operatorname{pol} q_0$. Since $\phi\pi$ is unchanged by a W-shift, (103) gives $b_0(w, z) = j\, a(w, z)$. All bracket values lie in W, so

$$\sum_{\text{cyc}} b_0(B(x, y), z) = j \sum_{\text{cyc}} a(B(x, y), z) = 0.$$

Thus b_0 is an ordinary CE cocycle.

For a zero-constant quadratic q with polar b, the cyclic norm is $Nq(x) = q(x) + q(fx) + q(f^2x) = b(x, fx)$. As $a(x, fx) = 0$, we have $Nq_a = Nh = 0$. The quotient map satisfies $\pi f = \omega\pi$, so invariance gives $N(\phi\pi) = 3\phi\pi = \phi\pi$. Therefore

$$\phi\pi = Nq_0 = b_0(x, fx).$$

The right side has degree at most three, and its third additive difference is precisely the CE differential of b_0, which is zero. It is quadratic. A linear section of π shows that ϕ is quadratic, and hence so is $h = q_a f$. Finally $D = a + JB$, with stable refinement $\widetilde{J}f$ for the second summand. This proves stability of D, independently of the chosen refinement. □

E.6 Exact restricted structure on finite free covers

Proof of Theorem 9.1. By Theorem 8.9, a finite free solution (F, g) has $H_g(k) = 0$. The cyclic decomposition of Theorem 5.3 and coefficient Frobenius make both nontrivial eigenspaces of K_g zero. Its remaining fixed sector is detected injectively by O_0, by Proposition 5.4. Every scalar component of D_F is a CE cocycle, is stable by Theorem 7.6, and has zero checksum. Its class is therefore zero. For each $\mathbb{F}_2$-valued coordinate component of D_F, choose a k-valued primitive and compose it with a binary-linear projection $k \to \mathbb{F}_2$ taking 1 to 1. This gives an $\mathbb{F}_2$-valued primitive. Assembling these coordinate primitives gives a binary-linear map $T : F \to F'$ with $D_F = TB_F$. Set $J_F = T|_{F'}$. It is unique because bracket values span F'.

We give the remaining argument for any exact factor $JB = D$. Adjoint closedness implies

$$J[w, z] = D(w, z) = [Jw, z] \qquad (w \in W, z \in L),$$

so J is an ambient centroid on W. For $A = \operatorname{ad}_x$, $F_x = \operatorname{ad}_{f(x)}$, calculation on a bracket value gives

$$(J^2 + J + \mathrm{id})B(x, y) = (F_x^2 + AF_x + A^3 + A^2)y = 0.$$

The last equality is (70) with $G = B + D$. Thus $J^2 + J = \mathrm{id}$ on W.

Put $P = f|_W + J$. Its polar is the original bracket and its ambient adjoint is the square of the original adjoint. Expansion of the functional equation, using $[w, Jw] = [Jw, Pw] = 0$, gives

$$P^2(w) = P(Jw) + J^2P(w).$$

The right side is additive because its polar is $B(Ju, Jv) + J^2 B(u, v) = 0$. Moreover $P([u, v]) = K(u, v)$ by (94). For every f-closed Lie subalgebra H, this implies $P(\gamma_i H) \subseteq \gamma_{i+2} H$ for $i \geq 2$; the quadratic cross terms on sums have depth $2i \geq i + 2$. The two-point-generated subsolution containing x, y has class at most five. Hence $P^2([x, y]) \in \gamma_6 H = 0$. Additivity of P^2 and spanning by bracket values give $P^2 = 0$ on W. The preceding equation now gives $P(Jw) = J^2 P(w)$. Together with the polar and adjoint identities, this is exactly the restricted-square structure over k. □

The higher-rank finite free objects used here are obtained from the preliminary quartic bound of Proposition 8.1 through Theorem 8.9. That construction uses only the reduced-derived theory and its matrix action; the proof of global stability above uses only the independently certified two-point free object.

E.7 The tensor quotient and exact saturation formula

Proofs of Theorems 9.3 and 9.4. First use only the tensor presentation (51). Since B maps onto W, its relation shows that $j_f(W)$ is stable under ω, and hence is the whole Q_f. The map $k \otimes W \to E$, $\lambda \otimes w \mapsto \lambda(w + C)$, annihilates the relation space. Thus $N_f \subseteq C$. The original bracket descends, and (82) shows that scalar multiplication commutes with every ambient adjoint: for $w = B(c)$,

$$j_f([D(c), x]) = j_f(D(B(c), x)) = \omega j_f([B(c), x]).$$

The descended bracket is therefore k-bilinear. Scalar extension of a map h with $hD = \omega hB$ kills exactly the displayed relation generators, proving the stated universal property.

Now choose any finite free presentation $\pi : F \twoheadrightarrow L$. Use J_F, P_F from Theorem 9.1, and put $H = \ker \pi \cap F'$, $S_0 = H + J_F H$. The scalar polynomial makes S_0 the least J_F-stable subspace containing H. For $h \in H, x \in F$,

$$[J_F h, x] = D_F(h, x) = [g(h), x] + [h, [h, x]] \in H.$$

Thus $[S_0, F] \subseteq H$. The space H is g-stable, and restricted semilinearity gives

$$g(J_F h) = J_F h + J_F^2 g(h) \in S_0.$$

Quadratic expansion, with $[H, J_F H] \subseteq H$, proves $g(S_0) \subseteq S_0$. Hence S_0 is also P_F-stable and is an ideal. The quotient F'/S_0 is a restricted k-Lie algebra with square of height two and induced map $g = J_F + P_F$.

The composite $F' \to W \xrightarrow{j_f} Q_f$ sends $J_F w$ to $\omega j_f(\pi w)$: check this on bracket values, which span F'. It kills H and $J_F H$, so factors through F'/S_0. Conversely F'/S_0 is a scalar-compatible quotient of $W = F'/H$, and the tensor universal property gives the reverse map. The two maps agree on the generating images of W, hence are inverse. This proves (54), including f-stability of N_f and its precise central kernel. It also proves the restricted identities for $P_Q = f_Q + J_Q$. The map to E is now a restricted quotient map, since it agrees with the original projection on every $j_f(w)$.

For an arbitrary linear solution homomorphism T, scalar extension preserves the tensor relations because T preserves B, D. It therefore defines $Q(T)$ without a condition on the image of the global center. Its defining formula preserves brackets and f_Q, and its k-linearity then preserves P_Q. The same formula proves identity and composition laws. □

Proof of Corollary 9.5. If $Tf = g^2 T$, polarization and the definition of D give

$$TB_f = B_g(T \wedge T), \qquad TD_f = (D_g + B_g)(T \wedge T).$$

The Frobenius-semilinear map $\lambda \otimes w \mapsto \lambda^2 \otimes Tw$ sends a relation generator to

$$1 \otimes D_g(Tc) + (1+\omega^2) \otimes B_g(Tc) = 1 \otimes D_g(Tc) + \omega \otimes B_g(Tc).$$

Thus it descends. It preserves the restricted square because reversal changes f_Q, J_Q to f_Q^2, J_Q^2, whereas $f_Q^2 + J_Q^2 = f_Q + J_Q = P_Q$. Taking $T = \mathrm{id}$ proves the statements about N_f and the reversed scalar. Zero maps give the same underlying zero map for either orientation. □

F Proofs for general growth and finite universal objects

The quantitative argument has two stages. The reduced derived action first gives a quartic bound without higher-rank free objects. This constructs the finite free objects. Their exact whole-derived scalar then removes one central step. We keep that order explicit before giving the reverse Engel realization and the sharp ordinary centre-by-metabelian boundary.

F.1 Paired operators and scalar envelopes

Lemma F.1 (The scalar envelope of a paired binary action)**.** *Let $k = \mathbb{F}_4$, and let X be a binary Lie subalgebra of $\mathrm{End}_k(M)$ such that every $A \in X$ satisfies $A^3 = 0$ and $\omega A + A^2 \in X$. Then X is an ordinary 3-Engel Lie algebra. Its scalar span $N = kX$ is closed under matrix squares, every element of N has fourth power zero, and N is an ordinary 3-Engel k-Lie algebra.*

Proof. For $A, B \in X$, put

$$K_{21} = A^2B + ABA + BA^2, \qquad K_{12} = AB^2 + BAB + B^2A,$$
$$Q = A^2B^2 + AB^2A + B^2A^2, \qquad H = BAB^2 + B^2AB, \qquad R = B^2AB^2.$$

The cube of $A + B$ gives $K_{21} = K_{12} =: K$. The two iterates of the companion map are $\omega B + B^2$ and $\omega^2 B + B^2$, using $B^4 = 0$. For either $\lambda \in \{\omega, \omega^2\}$, expansion of the cube of the actual element $A + \lambda B + B^2$ gives

$$0 = K + Q + \lambda H + R.$$

Subtracting the two equations gives $H = 0$, which is $(\mathrm{ad}_B)^3 A = 0$. These substitutions all occur in X.

The scalar span is bracket closed and $A^2 = (\omega A + A^2) + \omega A \in N$ for every $A \in X$. The square expansion makes N square closed. Every element of N is $A + \omega B$ with $A, B \in X$. Write

$$(A + \lambda B)^4 = C_1\lambda + C_2\lambda^2 + C_3\lambda^3,$$

where the pure fourth powers are zero. Here $C_1 = (\mathrm{ad}_A)^3 B = 0$ and $C_3 = (\mathrm{ad}_B)^3 A = 0$. At $\lambda = 1$, the element is actual and its fourth power vanishes, so $C_2 = 0$. Hence every fourth power in N vanishes.

For arbitrary $U, V \in N$, remove the pure fourth powers from $(U + \lambda V)^4$. The remaining polynomial has degree at most three and vanishes at all four scalars of k. Its linear coefficient is $(\mathrm{ad}_U)^3 V$, proving the last assertion. The conclusion requires fourth powers in the scalar span to vanish; no assertion about all of its matrix cubes is needed. □

Lemma F.2 (Removing squared generators)**.** *For matrices $A_1, \ldots, A_d$ over a field of characteristic two, put $H = \mathrm{Lie}_k\langle A_i \rangle$ and $N = \mathrm{Lie}_k\langle A_i, A_i^2 \rangle$. Then $\gamma_j(N) = \gamma_j(H)$ for every $j \geq 2$. In particular their classes agree when $N \neq 0$ is nilpotent.*

Proof. The identities

$$[A_i^2, T] = [T, A_i, A_i], \qquad [A_i^2, A_j^2] = [A_i, A_j, A_j, A_i]$$

hold in every associative algebra of characteristic two. Expand a left-normed Lie word of length at least two in the displayed generators. Its initial bracket, if it contains a square, becomes an ordinary A_i-word of length at least two. Each later square entry acts by two successive adjoints of A_i. Thus every word of length at least $j \geq 2$ belongs to $\gamma_j(H)$. The reverse inclusion follows from $H \subseteq N$. If $H = 0$, all A_i and hence N are zero; otherwise the equality of lower-central terms also covers the abelian case. □

Lemma F.3 (The short commuting action)**.** *Let $d \geq 2$, and let $a_1, \ldots, a_d$ be commuting $k = \mathbb{F}_4$-linear operators. Suppose that every element of the binary span of $a_i, \omega a_i + a_i^2$ has zero cube. Then $a_i^3 = a_i^2 a_j = 0$ for all i, j, and every product of $d+1$ operators from that span is zero.*

Proof. For two of the actual operators a, b, the cube of $a + b$ gives $a^2 b = ab^2 =: M$. Hence $a^2 b^2 = ab^3 = 0$. The cube of $a + \omega b + b^2$ gives

$$0 = \omega a^2 b + a^2 b^2 + \omega^2 a b^2 = M.$$

The remaining monomials in the commuting generators are square-free monomials and single-generator squares. Their degrees are at most d and two, respectively. A product of $d + 1 \geq 3$ operators from the span expands into monomials of degree at least $d + 1$, all zero. □

F.2 The preliminary bound and the exact free-derived bound

Proof of Proposition 8.1. Put $W = L'$, $C = W \cap Z(L)$, and $E = W/C$. Use the intrinsic $k = \mathbb{F}_4$-structure on E from Theorem 7.4. Its actual action image $X = \{\rho_x : x \in L\}$, with $\rho_x = \mathrm{ad}_x|_E$, satisfies

$$\rho_x^3 = 0, \qquad \rho_{f(x)} = \omega \rho_x + \rho_x^2$$

by Theorems 7.7 and 7.8. Lemma F.1 makes $N = kX$ an ordinary 3-Engel k-Lie algebra. If $\xi_1, \ldots, \xi_d$ are point generators and $A_i = \rho_{\xi_i}$, paired generation (Lemma F.5) gives

$$X = \mathrm{Lie}_{\mathbb{F}_2}\langle A_i, \omega A_i + A_i^2 \rangle, \qquad N = \mathrm{Lie}_k\langle A_i, A_i^2 \rangle.$$

By Lemma F.2, N has the same class as the at most d-generator ordinary algebra $H = \mathrm{Lie}_k\langle A_i \rangle$. The larger-field improvement after Theorem 1 of [14] applies to H:

$$q_* := \mathrm{cl}\, N = \mathrm{cl}\, H \leq q(d).$$

This invokes a Lie-class estimate only.

Suppose $N \neq 0$, so $q_* \geq 1$. Then $\gamma_{q_*+1} L \subseteq \ker \rho \cap W$, and the image of this term in E lies in $Z(E)$. Ambient actions on $Z(E)$ commute: their commutator is the inner action of $[x, y] + C \in E$, which vanishes on $Z(E)$. They are the binary span of $a_i = A_i|_{Z(E)}$ and $\omega a_i + a_i^2$. For $d \geq 2$, Lemma F.3 therefore gives

$$\gamma_{q_*+d+2} L \subseteq C, \qquad \gamma_{q_*+d+3} L = 0.$$

Thus $\mathrm{cl}\, L \leq q(d) + d + 2$. If $N = 0$, then $[L, W] \subseteq C$ and $\gamma_4 L = 0$, which is within the same bound. Rank zero is zero by finite nilpotence; a one-point subsolution is the commuting scalar plane because $[x, f(x)] = 0$. This proves the two small-rank conventions.

This proof uses the canonical reduced-derived structure and the full-derived cube theorem, whose proofs use only the independently certified two-point free object. No higher-rank free object has entered the argument. □

The finite-product construction in Lemma F.6 and Theorem 8.9 now uses this preliminary bound. Its proof appears below and is independent of the sharper estimate that follows.

Proof of Theorem 8.2. For $d \geq 2$, take the finite free solution F on d marked points, constructed using C_0, and put $W = F'$. Theorem 9.1 supplies the exact scalar $J = \omega$ on W, commuting with all ambient actions. For $\rho_x = \operatorname{ad}_x|_W$, the exact factor identity and Theorem 7.7 give

$$\rho_{f(x)} = \omega\rho_x + \rho_x^2, \qquad \rho_x^3 = 0.$$

The preceding scalar-envelope argument now applies on all of W. Write q_* for the class of its scalar action algebra. It is the class of an at most d-generator ordinary k-Lie algebra, and hence $q_* \leq q(d)$.

If the action algebra is nonzero, then

$$\gamma_{q_*+1}F \subseteq \ker\rho \cap W = Z(W).$$

This center is a k-subspace and is invariant under every ambient adjoint. On it the actions commute because W acts trivially. Lemma F.3 gives $\gamma_{q_*+d+2}F = 0$, and therefore $\operatorname{cl} F \leq q(d) + d + 1$. If the action algebra is zero, W is central and $\operatorname{cl} F \leq 2$. Every finite target of point rank at most d is a quotient of F, so the same bound holds throughout the category. The rank-zero and rank-one cases were handled in the preliminary proof. □

F.3 The action on an invariant ideal

Proof of Proposition 8.3. The ideal I' is f-stable by the bracket-image identity and quadratic expansion. Thus $J = f_M$ is binary-linear and $J^2 + J = \mathrm{id}$. Put $A = \operatorname{ad}_x|_M$ and $F = \operatorname{ad}_{f(x)}|_M$. Polarized commuting and bracket-image identities give

$$F = AJ + A^2, \qquad JA = FJ + A + FA. \tag{104}$$

The discarded terms are $[y,[x,y]]$ and $[f(y),[x,y]]$ for $y \in I$; both lie in I'. Set $T = JA + AJ$. Substitution gives $(\mathrm{id} + A)T = A^3$. Since $A^4 = 0$ by Theorem 7.2, this gives $T = A^3$. Commutation with J is an idempotent derivation:

$$[J,[J,A]] = J^2A + AJ^2 = T.$$

But $[J,T] = [J,A^3] = TA^2 + ATA + A^2T = A^5 = 0$. Hence $T = 0$, $A^3 = 0$, and $F = JA + A^2$. Lemma F.1 proves the ordinary 3-Engel assertion for the actual binary image. □

F.4 Reverse Engel realization and the same-rank comparison

Proof of Proposition 8.4. Let $k = \mathbb{F}_4$, put $X = \operatorname{ad} H \subseteq \operatorname{End}_k(H)$, and let $A_1, \dots, A_r$ be the adjoints of chosen generators of H. Every actual $A \in X$ has zero matrix cube. The classical operator identities of [14, Lemmas 1–2 and equations (3)–(4)] give, for $A, B \in X$,

$$\begin{gathered} B^2AB^2 = B^2A^2B^2 = 0, \qquad [A,B]^2 = [A^2,B^2], \\ [A,B]A[A,B] = 0. \end{gathered} \tag{105}$$

The last equality is the commutator sandwich in the proof of Lemma 2 of that source with A, B exchanged. These identities apply to the genuine adjoint representation of the ordinary 3-Engel algebra H.

Set $N = X + \operatorname{span}_k\{B^2 : B \in X\}$. Because k is perfect and $B^2 + C^2 = (B+C)^2 + [B,C]$, every element of N is $A + B^2$ with $A, B \in X$. The square-commutator identities make N bracket closed, with $N' = X'$, and

$$(A + B^2)^2 = A^2 + [A,B^2] \in N.$$

To verify the height-two condition, put $T = [A, B^2]$ and $C = [A, B]$. Equations (105) give

$$T^2 = A(B^2AB^2) + AB^4A + B^2A^2B^2 + (B^2AB^2)A = 0,$$
$$(A + B^2)^4 = [A^2, T] = [C^2, A].$$

Partial polarization of the cube in the k-linear space X gives $C^2A + CAC + AC^2 = 0$. The sandwich identity gives $CAC = 0$, so $[C^2, A] = 0$. Thus every fourth power in N is zero.

The standard restricted square $P(Z) = Z^2$ on N therefore satisfies $P^2 = 0$. On the underlying binary space define

$$f_N(Z) = \omega Z + Z^2.$$

Its polar is the original commutator bracket and

$$f_N^2(Z) = \omega^2 Z + Z^2 + Z^4 = f_N(Z) + Z.$$

This is the restricted-square construction of quadratic maps [10, Lemma 9.5 and Example 9.6], here with the displayed ω-coefficient and height-two condition.

Squares of Lie words of length at least two belong to X', by the commutator-square identity. Hence

$$N = X + \mathrm{span}_k\{A_1^2, \dots, A_r^2\} = \mathrm{Lie}_k\langle A_i, A_i^2\rangle.$$

Lemma F.2 gives $\gamma_j N = \gamma_j X$ for $j \geq 2$, so their classes agree when $X \neq 0$.

Now take only the binary subsolution Y generated by the r points A_i inside (N, f_N). Its point rank is at most r. The scalar span kY is Lie closed and contains $y^2 = f_N(y) + \omega y$ for every $y \in Y$. The square polarization formula makes kY square closed. It therefore contains the restricted envelope of the A_i, which is N, and $kY = N$. Bilinearity gives

$$\gamma_j N = k\gamma_j Y \qquad (j \geq 1).$$

The class of Y is thus that of $X = H/Z(H)$, namely $c - 1$ when H has class $c \geq 2$. For an abelian input take $Y = 0$. The scalar-span argument preserves point rank without identifying the intrinsic scalar of a binary abelianization with the ambient scalar on N. □

Related restricted-power calculations occur in [3, Section 5.5.4, p. 78]: on a support tensor algebra over an infinite field of characteristic two, a canonical restricted fourth-power identity is equivalent to the linearized 3-Engel identity. The construction above uses the adjoint algebra of an arbitrary finite-dimensional 3-Engel $\mathbb{F}_4$-algebra. The binary subsolution Y keeps its point-generator rank within the ordinary generator rank of the input algebra.

Proof of Theorem 8.5. The reverse construction gives $e(d) - 1 \leq \kappa(d)$. For the upper bound use the finite free solution at point rank d and repeat the whole-derived proof of Theorem 8.2. Its ordinary action algebra is generated by at most d elements over k, so its class is at most $e(d)$. The remaining commuting center action costs at most $d + 1$ further steps. Rank one is checked separately, with $e(1) = \kappa(1) = 1$.

For completeness, $e(d) \geq d$. Put $m = d - 1$ and let $R = k[t_1, \dots, t_m]/(t_i^2)$. The standard metabelian algebra $k^m \ltimes R$, with commuting generators y_i acting on the abelian ideal R by multiplication by t_i, is generated by $y_1, \dots, y_m, 1$. It is 3-Engel: on R each adjoint squares to zero, and on the quotient its first image lies in R. Its class is $m + 1 = d$, since $\gamma_j = (t_1, \dots, t_m)^{j-1}R$ for $j \geq 2$. For $d = 1$, use the one-dimensional abelian algebra. The two-sided comparison now gives $\kappa = \Theta(e)$ and the stated equivalence of polynomial upper bounds. Dividing it by $e(d)$ proves the final limit when $e(d)/d \to \infty$. □

Remark F.4 (Unbounded principal-ideal class). The current example in [13, Proposition 2.1] already obstructs a uniform principal-ideal argument within the orbit category. For $m \geq 1$, let $A_m = (t_1, \dots, t_m) \subset k[t_i]/(t_i^2)$, and on $kb \oplus A_m^3$ take the inherited restricted square

$$P(b_0, u, v, w) = (0, b_0 w + uv, uw, vw), \qquad f = \omega \,\mathrm{id} + P.$$

This is the $a = 0$ restricted subalgebra of the current presentation in Appendix G; it has $P^2 = 0$. Writing $A_{\geq j}$ for monomials with support size at least j, the smallest solution ideal containing b is $I = kb \oplus A_{\geq 1}U \oplus A_{\geq 2}V \oplus A_{\geq 3}W$. For $j \geq 2$, its lower-central terms are

$$\gamma_j I = A_{\geq 2j-1}U \oplus A_{\geq 2j}V \oplus A_{\geq 2j+1}W.$$

The current products add disjoint supports, giving both inclusions by partitioning any support above the indicated threshold. Thus $\operatorname{cl} I = \lfloor (m+1)/2 \rfloor$, which is unbounded. Here an ideal generated inside the ambient algebra is distinct from the one-point subsolution, which is only a scalar plane.

F.5 Point generation, derived length, and dimension

Lemma F.5 (Point generation and point rank)**.** *The subsolution generated by $x_1, \dots, x_s$ is*

$$\mathrm{Lie}_{\mathbb{F}_2}\langle x_1, f(x_1), \dots, x_s, f(x_s)\rangle. \tag{106}$$

This identity does not require finite dimension. In a finite solution, the minimum number of point generators is $d(L) = \dim_{\mathbb{F}_4} L/L'$.

Proof. Let H be the right-hand side of (106). The span of its displayed generators is f-stable by $f^2(x) = f(x) + x$. For Lie words use recursively the bracket-image identity

$$f([u, v]) = [f(u), f(v)] + [u, v] + [f(u) + f(v), [u, v]].$$

If the f-values of shorter words belong to H, so does the f-value of their bracket. The quadratic expansion

$$f\Big(\sum_i u_i\Big) = \sum_i f(u_i) + \sum_{i<j} [u_i, u_j]$$

then proves $f(H) \subseteq H$. Conversely any subsolution containing the points contains their orbit partners and their polar brackets, hence contains H.

The images of s point generators span the abelianization over its intrinsic $\mathbb{F}_4$, giving $d(L) \leq s$. Conversely lift an $\mathbb{F}_4$-basis of L/L'. The corresponding paired elements form a binary basis modulo L'. Their Lie subalgebra H satisfies $L = H + L'$. Induction on the lower-central series gives $L = H + \gamma_j(L)$ for every j: replace the entries in $[L, L]$ by $H + \gamma_j(L)$. Finite nilpotence gives $H = L$. Apply the first assertion. □

Proof of Theorem 8.8. By Lemma F.5, an object counted by $\ell_{\max}(s)$ has $d \leq s$. Theorem 8.2 gives $\operatorname{cl} L \leq C(s)$. The standard commutator inclusion $L^{(j)} \subseteq \gamma_{2^j}(L)$ implies

$$\operatorname{dl}(L) \leq \lfloor \log_2 \operatorname{cl} L \rfloor + 1.$$

For $s \geq 1$, $C(s) \leq (s+1)^4$. Hence the right-hand side is at most $4 \log_2(s+1) + 1$. The preliminary bound already makes the possible derived lengths a nonempty bounded set of integers, so their maximum exists before the finite free-object construction.

For the lower bound, take L_{s-1} from Theorem 8.6. It has exact point rank s, and Proposition G.4 gives the lower endpoint in (47). The case $s = 1$ is the nonzero linear scalar solution. Finally direct sum with linear scalar planes raises any smaller rank to s while preserving a positive derived length. Since $\ell_{\max}(s) \geq 1$, the maximum at exact rank equals $\ell_{\max}(s)$. □

Proof of (48). For a class-$c \geq 2$ solution, the first lower-central factor has binary dimension $2d$; all remaining $c-1$ nonzero factors have positive even dimension by Proposition 2.7. Thus

$$\dim_{\mathbb{F}_2} L \geq 2d + 2(c-1).$$

Lemma 2.6 gives $d \geq 2$, and the class bound gives $d \geq a(c)$. The explicit quartic bound satisfies

$$c \leq C(d) \leq \frac{(d+1)^4}{2} + d + 1 \leq \frac{(d+2)^4}{2}.$$

Thus $d \geq (2c)^{1/4} - 2$, proving the second inequality. In particular $m(c) \geq 2c + \Omega(c^{1/4})$; the integer inverse bound through $a(c)$ retains the sharper constants. □

F.6 Finite free objects and central stem towers

We use s for point rank and $k = \mathbb{F}_4$ for the coefficient field. All arrows below are binary-linear intertwiners.

Lemma F.6 (A finite bound at fixed point rank)**.** *Put*

$$B(s) = \sum_{j=1}^{C_0(s)} (2s)^j \qquad (s \geq 1).$$

Every finite solution generated by at most s marked points has binary dimension at most $B(s)$. There are finitely many marked isomorphism types of such solutions.

Proof. The paired generation lemma supplies at most $2s$ binary Lie generators, and Proposition 8.1 gives class at most $C_0(s)$. Jacobi expands every bracket word into left-normed words, and only lengths at most $C_0(s)$ are needed. There are at most $(2s)^j$ such words of length j. This proves the dimension bound. On each standard binary vector space of dimension at most $B(s)$ there are finitely many quadratic maps and finitely many ordered s-tuples. Restricting to solutions and generating tuples preserves finiteness. □

Existence and universal property in Theorem 8.9. Choose one representative of each marked isomorphism type in Lemma F.6, and form their finite direct product. Let $\mathfrak{F}_s^{\mathrm{fin}}$ be its subsolution generated by the diagonal marked points. It is finite, and its dimension is at most $B(s)$ by the lemma.

For any desired point values in a finite target, their generated subsolution is marked-isomorphic to one chosen factor. Projection to that factor, followed by its isomorphism and inclusion into the target, gives the required map. A binary-linear intertwiner preserves polar brackets and the f-images of generators, so Lemma F.5 proves uniqueness. The marked classes span the abelianization over k. Mapping to the linear solution k^s with its standard basis proves that they are independent, hence a basis. For $s = 0$, take the zero solution. □

Lemma F.7 (Generation in a stem extension)**.** *If a surjective morphism of finite solutions $E \to G$ has kernel $A \subseteq E'$, arbitrary lifts of point generators of G generate E.*

Proof. The map on abelianizations is an isomorphism. The subalgebra generated by the lifted points and their orbit partners therefore satisfies $E = H + E'$. The nilpotent generation argument in Lemma F.5 gives $E = H$. □

Proposition F.8 (The full quadratic cover is stem)**.** *For a finite solution g, put $A = H_g(k)^*$. A central quadratic extension representing*

$$\mathrm{id} \in H_g(k) \otimes_k H_g(k)^*$$

has its entire kernel A in the derived ideal. It is nontrivial exactly when $H_g(k) \neq 0$.

Proof. The central-lift construction of Proposition 5.1 tensors with coefficient spaces, so the displayed class has a representative. Every nonzero k-linear functional λ on A pushes this extension out to a nonzero scalar quadratic extension class. Its ordinary binary CE class is nonzero by the polarization injection in that proposition.

Suppose $A \cap E' \neq A$. This intersection is a k-subspace: it is f-stable, and $f|_A = \omega\,\mathrm{id}$. Choose a nonzero k-linear $\lambda : A \to k$ vanishing on $A \cap E'$. It extends to a binary-linear functional on E vanishing on E'. In a section $E = V \oplus A$, write this functional as $t(x) + \lambda(a)$. Its vanishing on brackets says $\lambda\alpha(x, y) = t([x, y])$, exhibiting the scalar polar cocycle as a CE coboundary. This contradicts the injection. Hence $A \subseteq E'$. □

Proposition F.9 (Recognition by vanishing quadratic extensions)**.** *Let F be a finite solution generated by marked points whose classes form a k-basis of F/F'. Then F has the finite-target free property on those points if and only if $H_F(k) = 0$.*

Proof. Assume $H_F(k) = 0$. We construct the map to a finite target E, with arbitrary desired marked values y_i, by induction on its binary dimension. If E is abelian, the map exists through F/F', by k-linearity. Otherwise global nilpotence gives a nonzero full center $Z(E)$, which is f-stable and on which f is its intrinsic scalar ω. The induction hypothesis supplies a map $h : F \to E/Z(E)$ with the required marked cosets.

Pull back the central extension $E \to E/Z(E)$ along h. Its coefficient space is $Z(E)$, and its quadratic extension space is $H_F(k) \otimes_k Z(E) = 0$. It therefore admits a binary-linear solution section. Composition gives a lift $S : F \to E$ of h. The discrepancies $z_i = y_i + S\xi_i$ lie in $Z(E)$. There is a unique k-linear $t : F/F' \to Z(E)$ taking the marked basis to the z_i. Inflate it to F. Since its values are central,

$$f_E(Sx + tx) = S(f_F x) + \omega t(x) = (S + t)(f_F x).$$

Thus $S + t$ takes the exact desired marked values. Uniqueness follows from paired Lie generation.

Conversely, given a scalar central extension of a finite free F, choose lifts of its marked points. The universal property gives a solution homomorphism from F into the extension. Its composite with the quotient fixes every generator, hence is the identity. Every scalar extension therefore splits, so $H_F(k) = 0$. □

Termination assertion in Theorem 8.9. For $s = 0$, the starting solution is zero, its extension space is zero, and the procedure terminates immediately. Assume $s \geq 1$. At each nontrivial full-cover step, Propositions F.8 and Lemma F.7 preserve the marked abelianization k^s and point generation. All stages have dimension at most $B(s)$. Each nonzero central coefficient space adds at least two binary dimensions, so there are at most

$$\left\lfloor \frac{B(s) - 2s}{2} \right\rfloor$$

nontrivial steps. At termination the full extension space vanishes; otherwise another nontrivial full cover would exist. Recognition by Proposition F.9 identifies the terminal marked object with $\mathfrak{F}_s^{\mathrm{fin}}$. □

More generally a fixed-rank finite stem tower with $s \geq 1$, starting in E_0, has at most $\lfloor (B(s) - \dim_{\mathbb{F}_2} E_0)/2 \rfloor$ nontrivial stages. Nonstem extensions can enlarge the abelianization and are not subject to this bound. The universal property extends to locally finite infinite targets because finitely many marked points then generate a finite subsolution. Extending it to arbitrary infinite targets would require an additional local-finiteness or residual argument, which is not supplied here.

Finally $\mathrm{dl}(\mathfrak{F}_s^{\mathrm{fin}}) = \ell_{\max}(s)$: every finite solution generated by at most s points is its quotient, and the free object itself has point rank s. The construction by products, the nilpotent generation argument and the central projectivity criterion are classical mechanisms. Their finite scope here comes from the independent preliminary quartic bound and the specific quadratic extension space; none requires a computation of a named terminal stage.

G The current family and its binary trace restriction

G.1 A classical restricted presentation

Put $k = \mathbb{F}_4$, and write

$$R_m = k[t_1, \dots, t_m]/(t_1^2, \dots, t_m^2), \qquad A_m = (t_1, \dots, t_m) \subseteq R_m.$$

For nonempty $B \subseteq [m]$, let $t_B = \prod_{i \in B} t_i$. These monomials form a basis of A_m. Their product is $t_{B \cup C}$ for disjoint supports and zero otherwise. In particular every element of A_m has square zero.

The classical three-dimensional simple algebra $S = W(1;2)^{(1)}$ has basis U, V, W and table

$$[U, V] = U, \qquad [U, W] = V, \qquad [V, W] = W.$$

The inner derivation $\delta = \mathrm{ad}_U$ sends $U \mapsto 0, V \mapsto U, W \mapsto V$. In characteristic two its square is again a derivation. The commuting pair δ, δ^2 spans a restricted derivation algebra with $\delta^{[2]} = \delta^2$ and $(\delta^2)^{[2]} = 0$. These derivations occur in the classical description of $\operatorname{Der} S$ in [12, Proposition 3.3(1)].

Let x, y act on $S \otimes A_m$ by δ, δ^2, respectively, and put $[x, y] = 0$. In $\widehat{L}_m = (kx \oplus ky) \ltimes (S \otimes A_m)$, write $U_B = U \otimes t_B$, and similarly for V_B, W_B. Besides the current products on disjoint supports, the nonzero external products are

$$[x, V_B] = U_B, \qquad [x, W_B] = V_B, \qquad [y, W_B] = U_B. \tag{107}$$

The semidirect-product construction proves Jacobi. Its current table is the disjoint-support table of [13, Proposition 2.1]; adjoining δ supplies a square root of that source's external δ^2-action.

Lemma G.1 (Restricted square and exact height)**.** *The map P in* (45) *is a restricted square on $\widehat{L}_m$, and $P^2 = 0$. Thus $\widehat{f}_m = \omega\,\mathrm{id} + P$ satisfies $\widehat{f}_m^2 = \widehat{f}_m + \mathrm{id}$.*

Proof. Polarization of P gives exactly the displayed Lie table, and $P(\lambda X) = \lambda^2 P(X)$. On its monomial basis, $P(x) = y$, with all other basis squares zero. The table gives $(\mathrm{ad}_x)^2 = \mathrm{ad}_y$. Every other basis adjoint squares to zero: any repeated nonempty current support makes the two-step product zero, including on the external vectors. The discrepancy $\mathrm{ad}_{P(X)} + (\mathrm{ad}_X)^2$ is additive and Frobenius-semilinear, so its vanishing on this basis proves the restricted adjoint identity.

Put

$$u' = av + bw + uv, \qquad v' = aw + uw, \qquad w' = vw.$$

The second iterate is

$$P^2(a, b, u, v, w) = (0, 0, a^2 vw + u'v', u'w', v'w') = 0.$$

Indeed $u'v' = a^2 vw$ after the two $auvw$ terms cancel, and the other products contain u^2, v^2, or w^2. The restricted adjoint identity also gives $[X, P(X)] = 0$. Consequently

$$\begin{aligned}(\omega\,\mathrm{id} + P)^2(X) &= \omega^2 X + (\omega + \omega^2)P(X) + P^2(X) + [\omega X, P(X)] \\ &= \omega^2 X + P(X) = (\omega\,\mathrm{id} + P)(X) + X.\end{aligned}$$

This calculation uses classical restricted-square identities; the broader quadratic-map correspondence is already present in [10, Lemma 9.5 and Example 9.6]. □

Lemma G.2 (Exact class of the ambient algebra)**.** *For $m \geq 1$, the class of $\widehat{L}_m$ is $2m+1$, and $\gamma_{2m+1}(\widehat{L}_m) = kU_{[m]}$.*

Proof. The weights

$$\operatorname{wt} x = 1, \quad \operatorname{wt} y = 2, \quad \operatorname{wt} U_B = 2|B|+1, \quad \operatorname{wt} V_B = 2|B|, \quad \operatorname{wt} W_B = 2|B|-1$$

give a positive grading: every nonzero bracket adds weights. The maximum is $2m+1$, so $\gamma_{2m+2}(\widehat{L}_m) = 0$. The left-normed word

$$W_{\{1\}}, x, x, W_{\{2\}}, x, \ldots, W_{\{m\}}, x$$

has length $2m+1$ and value $U_{[m]}$. The first two x-actions give $U_{\{1\}}$, and each later $W_{\{i\}}, x$ pair enlarges its support by i. Scalar multiples in the first current input give the full k-line. That line is the only homogeneous component of weight $2m+1$, which proves the last assertion as well. □

The ambient Lie algebra is in fact 3-Engel by a direct calculation. For $X = (a, b, u, v, w)$, set $h = a + u$. Its action on the current ideal and its square are

$$T_X = \begin{pmatrix} v & h & b \\ w & 0 & h \\ 0 & w & v \end{pmatrix}, \qquad T_X^2 = \begin{pmatrix} hw & hv + bw & a^2 \\ vw & 0 & hv + bw \\ 0 & vw & hw \end{pmatrix}.$$

Since $h^2 = a^2$ and $u^2 = v^2 = w^2 = 0$, multiplication gives $T_X^3 = 0$. The image of an external vector $(A, B, 0, 0, 0)$ is the current vector $A(v, w, 0) + B(w, 0, 0)$, which T_X^2 kills. Hence $(\operatorname{ad}_X)^3 = 0$ on all of $\widehat{L}_m$. This separate verification also applies to its binary Lie subalgebras.

G.2 The invariant binary trace graph

Proof of Theorem 8.6. On the external plane the map $\widehat{f}_m$ is

$$(a, b) \longmapsto (\omega a, \omega b + a^2).$$

If $b = a + a^2$, its second coordinate is $\omega a + \omega^2 a^2 = \omega a + (\omega a)^2$. Thus the binary-linear graph

$$L_m = \{(a, a + a^2, u, v, w) : a \in k,\ u, v, w \in A_m\}$$

is $\widehat{f}_m$-stable. Restriction gives exactly (43); the functional equation and binary degree at most two are inherited. The term $a^2 w$ remains binary quadratic because $a \mapsto a^2$ is binary-linear. The graph has binary dimension $2 + 6(2^m - 1) = 6 \cdot 2^m - 4$.

For $m \geq 1$, it contains x and the full current ideal. The class-attaining word in Lemma G.2 therefore lies in it, and the ambient upper bound applies. It follows that

$$\operatorname{cl} L_m = 2m + 1, \qquad \gamma_{2m+1}(L_m) = kU_{[m]}.$$

For $m = 0$, the graph is the single linear scalar block, with class one.

For $m \geq 1$, direct calculation gives

$$L'_m = \operatorname{span}_k\{U_B, V_B : B \neq \varnothing\} \oplus \operatorname{span}_k\{W_B : |B| \geq 2\}. \tag{108}$$

The external products with x supply all scalar multiples of U_B, V_B, since the current inputs have arbitrary coefficients in k. Disjoint current products supply all W_B on supports of size at least two. The reverse inclusion follows from the table: no bracket has external component or singleton W component. On the quotient, the induced f_m is multiplication by ω on $a, W_{\{1\}}, \dots, W_{\{m\}}$. Thus the intrinsic abelianization rank is $m+1$. For $m \geq 2$, both $V_{\{1\}}$ and $U_{\{2\}}$ lie in L'_m, and their bracket is the nonzero $U_{\{1,2\}}$.

Finally the original binary algebra is not restrictable for $m \geq 1$. Indeed

$$(\mathrm{ad}_x)^2 V_{\{i\}} = 0, \qquad (\mathrm{ad}_x)^2 W_{\{i\}} = U_{\{i\}}.$$

If $(\mathrm{ad}_x)^2 = \mathrm{ad}_z$, write the external coordinates of z as $(a, a+a^2)$. The singleton $U_{\{i\}}$ coefficient of $[z, V_{\{i\}}]$ is a, because current-current products strictly increase support degree or vanish. It must be zero, so $a = 0$, and the trace relation gives $b = 0$. But then $[z, W_{\{i\}}]$ has no singleton $U_{\{i\}}$ component, a contradiction. A restricted square would make every squared adjoint inner, so none exists on this binary Lie algebra. This does not conflict with the restricted structure on its larger parent. □

Proposition G.3 (Center and relative quotient minimality)**.** *For $m \geq 1$, both $\widehat{L}_m$ and L_m have center exactly $kU_{[m]}$. No nontrivial solution quotient of L_m retains class $2m+1$.*

Proof. For a central vector, bracketing first with x forces $v = w = 0$. Bracketing with $V_{[m]}$ forces $a = 0$, since positive-support current products with $t_{[m]}$ vanish. In the parent, bracketing with $W_{[m]}$ also forces $b = 0$; in the trace graph this already follows from $a = 0$. The remaining u must annihilate A_m, whose annihilator in A_m is $kt_{[m]}$. To see this, multiplication by t_i kills precisely the monomial terms already containing i; the other supports remain distinct. Simultaneous annihilation by all t_i leaves only the full support.

Every nonzero ideal in a finite nilpotent Lie algebra meets its center: take the last nonzero iterated ambient-adjoint image of that ideal. For an f_m-stable ideal, its intersection with the center is also f_m-stable. The center is a single binary irreducible scalar plane, so that intersection is the whole center. It is also the last lower-central term. Any nonzero quotient kernel therefore kills that term and lowers the class. □

Details for Corollary 8.7. For odd $c = 2j+1$, use L_j. For even $c = 2j$, quotient L_j by $kU_{[j]} = \gamma_{2j+1}(L_j)$. This is an f_j-stable central plane, since f_j acts there as ω. The quotient has class exactly $2j$: a nonzero lower-central series in a nilpotent Lie algebra cannot have two consecutive equal terms. Its dimension decreases by two, giving (46).

For every resulting class $c \geq 5$, the derived bracket $U_{\{1,2\}} = [U_{\{1\}}, V_{\{2\}}]$ survives. It has ambient grading weight five, whereas every element of $\gamma_6(\widehat{L}_j)$ has weight at least six. In the even construction the quotient term is $\gamma_{c+1}(L_j) \subseteq \gamma_6(\widehat{L}_j)$, so it cannot kill this bracket. The odd construction has no quotient. Thus all advertised comparisons for $c \geq 5$ are non-metabelian.

With $j = \lceil (c-1)/2 \rceil$, both dimensions are at most $6 \cdot 2^j - 4$. For $c \geq 5$, $j \leq c-3$, giving

$$6 \cdot 2^j - 4 \leq \tfrac{3}{4}\, 2^c - 4 < 2^c + 2c - 6 = \mu(c).$$

The even bound is two dimensions smaller than this common bound. □

G.3 Exact derived length

Proposition G.4 (Derived-support recurrence)**.** *For every $m \geq 0$,*

$$\mathrm{dl}(L_m) = \lceil \log_2(3m+3) \rceil - 1.$$

Proof. For $m \geq 1$, equation (108) gives the support thresholds $(a_1, b_1, c_1) = (1, 1, 2)$ for the three current letters in L'_m. The product of the spans of monomials of degree at least $p, q \geq 1$ is exactly the span of those of degree at least $p+q$. Indeed a support of size at least $p+q$ can be partitioned into disjoint parts meeting the two thresholds. Thus the table gives the exact recurrence

$$(a_{r+1}, b_{r+1}, c_{r+1}) = (a_r + b_r, a_r + c_r, b_r + c_r) \qquad (r \geq 1).$$

Induction gives

$$(a_r, b_r, c_r) = \begin{cases} ((2^{r+1}-1)/3, (2^{r+1}-1)/3, (2^{r+1}+2)/3), & r \text{ odd}, \\ ((2^{r+1}-2)/3, (2^{r+1}+1)/3, (2^{r+1}+1)/3), & r \text{ even}. \end{cases}$$

The minimum threshold is $\lfloor 2^{r+1}/3 \rfloor$. Consequently $L_m^{(r)} = 0$ first when $2^{r+1} \geq 3(m+1)$, which gives the stated formula. For $m = 0$, the solution is nonzero abelian and both sides equal one. □

G.4 Low-degree linearization and a conditional extension theorem

Proposition G.5 (Quadratic inverse for the entire family)**.** *For the displayed binary-linear scalar $J = \omega \,\mathrm{id}$ on $k \oplus A_m^3$, the map $H = f_m + J^2$ is a quadratic permutation with quadratic inverse, and $Hf_m = JH$. Explicitly,*

$$H(a, u, v, w) = (a, u + av + (a + a^2)w + uv, v + aw + uw, w + vw),$$
$$H^{-1}(a, U, V, W) = (a, U + aV + aW + UV, V + aW + UW, W + VW).$$

Proof. In the full restricted presentation, $\widehat{H} = \mathrm{id} + P$ is an involution, because $P^2 = 0$ and $[X, P(X)] = 0$. It maps the external graph $b = a + a^2$ to the linear diagonal $b = a$. Restricting it in these two directions gives precisely the displayed formulas; the diagonal is stable under ambient multiplication by ω. Alternatively, their compositions cancel directly using only $u^2 = v^2 = w^2 = 0$. The general identity $(f + J^2)f = J(f + J^2)$, for $f^2 = f + \mathrm{id}$ and $J^2 + J = \mathrm{id}$, proves the intertwining assertion. □

Proposition G.6 (Central lifts of an involutively linearized base)**.** *Let g be a binary quadratic solution, and suppose a binary-linear J_0, with $J_0^2 + J_0 = \mathrm{id}$, makes $H_0 = g + J_0^2$ an involution. For any finite k-space A and quadratic $q : V \to A$ with $qg = \omega^2 q$, the central lift*

$$F_q(x, a) = (g(x), \omega a + q(x))$$

has a quadratic linearizing permutation

$$H(x, a) = (H_0(x), a + q(x))$$

whose inverse has binary degree at most three. In particular this applies to every central lift of a height-two restricted base $g = \omega \,\mathrm{id} + P$, $P^2 = 0$.

Proof. Write $\alpha = \mathrm{pol}\, q$. The inverse is

$$H^{-1}(y, b) = (H_0(y), b + qH_0(y)).$$

Quadratic expansion and the eigenfunction equation give

$$qH_0(y) = \omega^2 q(y) + q(J_0^2 y) + \alpha(g(y), J_0^2 y).$$

The first two terms have binary degree at most two and the last has degree at most three. Moreover $H_0 g = J_0 H_0$, and $qg + q = \omega q$, so $HF_q = (J_0 \oplus \omega \,\mathrm{id})H$. For a restricted base, $H_0 = \mathrm{id} + P$ is involutive by the calculation in Proposition G.5. □

The preceding proposition controls one central-extension level, with any number of simultaneous central coordinates. It does not identify a noncentral class-three kernel with a central extension, or establish the same inverse bound for arbitrary successive lifts. The trace family supplies a separate infinite collection with quadratic inverse; its class and derived-length formulas concern the original polar bracket, which the nonlinear change of coordinates does not preserve.

H Proofs of the ordinary Lie reconstruction

H.1 The scalar-faithful source

We first work with $N_f = 0$. In the whole restricted envelope of Theorem 9.6, identify V with its faithful binary image X, and write P for the restricted square. The existing stability theorem gives

$$E' = X', \quad P(E') \subseteq E', \quad P^2(E') = 0, \quad P^2(X) = 0, \quad f|_X = (\omega I + P)|_X.$$

The image of P on the abelian quotient E/E' is semilinear. Since X scalar-spans E, $P^2(E) \subseteq E'$, and hence $P^4(E) = 0$. For $K = I + P$, restricted powers of a single element commute, so $K^2 = I + P^2$ and $K^4 = I$.

The set $U = K(X)$ is a k-linear Lie subalgebra. Indeed K permutes E' and maps each E'-coset bijectively to a coset. On E/E' it is binary-linear; therefore U is binary-linear. For $x \in X$, $P^2x = 0$ gives $K(\omega x + Px) = \omega Kx$, proving scalar closure. All brackets of elements of U lie in $E' \subseteq U$. Both restrictions $K : X \to U$ and $K : U \to X$ are quadratic and inverse.

Expanding the pullback of addition gives

$$K(Kx + Ky) = x + y + [x, y] + [x, y, x] + [x, y, y] + [x, y, y, x],$$

where these brackets are the original polar brackets. For the bracket, put $c = [x, y]$ and $A = [Kx, Ky] = c + [c, x] + [c, y] + [c, y, x]$ in the envelope. The two-point solution has class at most five. The restricted square of its depth-three and depth-four terms is zero; the global bracket correction gives

$$Pc = [c, y, x] + [c, [c, x]] + [c, [c, y]].$$

The two depth-five cross terms cancel the corresponding terms in $P(A)$, leaving $P(A) = [c, y, x]$. Thus $K(A) = c + [c, x] + [c, y]$. This proves (58)–(59). Expanding in the reverse direction gives (60).

The ordinary algebra U is three-Engel: for $u \in U$, $(\operatorname{ad} u)^3 = 0$ follows by expansion from the two-point identities. Its derived square $p = P|_{U'}$ satisfies (10.4). In particular $p([a, b]) = [a, b, b, a]$. Applying p to the Jacobi relation $[a, b, c] + [b, c, a] + [c, a, b] = 0$ gives precisely $\mathcal{R} = 0$.

For $j \geq 2$, the restricted correction sends the original γ_j into γ_{j+2}. The leading bracket is unchanged, and K permutes each lower-central subset. Induction down the nilpotent filtration therefore gives $\gamma_j(U) = \gamma_j(L_f)$ as subsets of the envelope. Centrality is also preserved: all correction brackets vanish with a central input, and the inverse formulas prove the converse.

H.2 Descent retains the entire central kernel

Every finite solution has a finite free presentation $\pi : F \twoheadrightarrow V$; the exact free-derived scalar theorem gives $N_F = 0$. The operations (58) and (59) are terms in $+$ and f. Hence π preserves them, and their group, scalar, Lie and three-Engel identities descend to V. The same applies to $\mathcal{R} = 0$ and the inverse operation (60). No central direction is removed. Every original

linear intertwiner preserves the new terms; every new ordinary Lie homomorphism preserves the inverse terms and ω. This proves fullness as well as faithfulness on the image.

For the degree assertion, let $X = \ker \pi$ and let $\eta_F : F \to U_F$ be the preceding biquadratic map. Its image $I = \eta_F(X)$ is an ordinary Lie ideal. The fibers of π are cosets for both additions, since π is a homomorphism for each addition with the same kernel set. Thus η_F induces a bijection $\bar{\eta} : V \to U_F/I$. Choose a linear section s of the original binary quotient and an ordinary binary-linear section t of the new vector quotient. With quotient maps π and q,

$$\bar{\eta} = q\eta_F s, \qquad \bar{\eta}^{-1} = \pi\eta_F^{-1} t.$$

Both degrees are at most two. A conjugator or its inverse of degree one would force f itself to be linear, proving sharpness in the nonlinear case. These choices establish existence; they do not prescribe a particular averaging conjugator.

Lower-central subsets, their dimensions, and the center descend through the same quotients. On the center the two additions agree. Mutual term generation proves the assertion about point rank. These arguments use the established global nilpotence, finite-free and exact-scalar theorems; they do not replace their earlier proofs.

H.3 Relative PBW realization

Suppose an ordinary three-Engel H has a square p as in (10.4). Choose a basis $w_1, \dots, w_s$ of H' and a complement basis $x_1, \dots, x_d$. In the ordinary associative enveloping algebra, the elements $z_i = w_i^2 - p(w_i)$ are central: their commutator with h is $[w_i, [w_i, h]] - [p(w_i), h] = 0$. PBW shows that the enveloping algebra is free over $k[z_1, \dots, z_s]$ with basis

$$w_1^{\epsilon_1} \cdots w_s^{\epsilon_s} x_1^{a_1} \cdots x_d^{a_d}, \qquad \epsilon_i \in \{0, 1\}, \quad a_i \geq 0.$$

Indeed multiplication by $\prod z_i^{b_i}$ has distinct leading PBW exponents $2b_i + \epsilon_i$; its other terms have smaller filtration degree. Triangular induction proves spanning and independence. Consequently quotienting by the z_i retains H faithfully.

Its restricted closure is $H \oplus \operatorname{span}\{x_i^{2^j} : j \geq 1\}$. The fourth powers x_i^4 are central by the Engel identity. Factoring out their restricted span gives a finite restricted algebra

$$E(H) = H \oplus \operatorname{span}\{s_1, \dots, s_d\}, \qquad s_i = x_i^2.$$

Here $[s_i, h] = [h, x_i, x_i]$, $[s_i, s_j] = q(x_i, x_j)$, $P(x_i) = s_i$, $P(s_i) = 0$, and $P|_{H'} = p$. The fourth-power cross term on $a, b \in H$ is

$$[Pa, [a, b]] + [Pb, [a, b]] + P([a, b]) + [Pa, Pb] = 0 :$$

the first two terms are third adjoints, and the last two are both $q(a, b)$. Thus P^2 is additive on H and vanishes there.

Now $K = I + P$ sends H bijectively to a binary-linear subspace X, and $(\omega I + P)K(u) = K(\omega u)$. On X the map $\omega I + P$ is quadratic and squares to itself plus the identity. The transported ordinary core is exactly H. PBW also gives the universal property of $E(H)$, independent of the chosen complement. Uniqueness of p follows because bracket values span H' and its polar and semilinear rules prescribe its value on every sum.

H.4 The exact sextic image and the homology map

For an arbitrary ordinary three-Engel H, form its finite first restricted envelope $T = H \oplus \operatorname{span}\{e_i^2\}$ by killing central fourth and higher powers of a basis in the PBW restricted closure. PBW retains H. Put

$$r(a, b) = P([a, b]) + q(a, b), \qquad V_r = \operatorname{span}_k r(H, H).$$

This r is central, alternating and Frobenius-semilinear in both arguments. To check centrality, write $A = \operatorname{ad} a$, $B = \operatorname{ad} b$. The full cubic Engel coefficients give $A^2B + ABA + BA^2 = AB^2 + BAB + B^2A = 0$, and hence $(AB + BA)^2 = \operatorname{ad} q(a, b)$. This is also $\operatorname{ad} P([a, b])$. For additivity the cross term is

$$[[a, b], [a', b]] + [a, b, b, a'] + [a', b, b, a] = 0,$$

another repeated Engel coefficient combined with Jacobi.

The fourth-power cross term on H is $r(a, b)$, so $P^2(H) \subseteq V_r$. Also V_r is square-stable. If $T_0 = [a, b, b]$ and $z = [a, b, b, a, a]$, the universal three-Engel identity $[z, H] = 0$ gives $q(T_0, a) = [z, T_0] = 0$ and $P(q(a, b)) = r(T_0, a) \in V_r$. Therefore $P(r(a, b)) = P^2([a, b]) + P(q(a, b)) \in V_r$. These identities are certified as ordinary free-Lie polynomial consequences in the accompanying universal identity DAGs, whose leaves are full cubic Engel coefficients; their replay checks Lie membership of every substituted polynomial. Thus T/V_r is restricted, and $H/(H \cap V_r)$ has the unique admissible derived square.

The square-copy component of $r(a, b)$ is $[a, b]^{(1)}$. Consequently $H \cap V_r$ is exactly the image on CE cycles of (62). This also proves independence of wedge decomposition. The polar cross term of the sum of restricted squares accounts for the second sum there. On a boundary one obtains

$$\lambda_H\big(\partial(a \wedge b \wedge c)\big) = \mathcal{R}(a, b, c). \tag{109}$$

To prove sufficiency of $\mathcal{R} = 0$, let F be the relatively free ordinary three-Engel algebra on d generators with class at most C, where C is a classical uniform bound for finite d-generator three-Engel targets [14]. This is finite and maps onto any target under consideration. The Hopf formula, for an ordinary free presentation $F = \mathfrak{L}/I$, is $H_2(F; k) = (I \cap [\mathfrak{L}, \mathfrak{L}])/[\mathfrak{L}, I]$. The Engel relators have cycle representatives $[y, x, x] \wedge x$, whose λ-value is zero. Their polarized coefficients also have zero value. The additional class-C relators have weight $C + 1$; the corresponding λ-values have weight $2(C + 1) > C$ and vanish. The defining relators generate this Hopf quotient, so

$$F \cap V_r = \operatorname{span}_k\{\mathcal{R}(a, b, c) : a, b, c \in F\}.$$

Thus $F/(\mathcal{R})$ has an admissible square and a PBW realization. Every H with $\mathcal{R} = 0$ is its ordinary Lie quotient. The binary operation (60) is a Lie polynomial; an ordinary ideal and its cosets are congruence classes for it. Hence the quadratic solution descends to H, and both mutual term identities descend. This finishes Theorem 10.1.

If $\mathcal{R} = 0$, equation (109) makes λ a homology map. Every map into an algebra admitting the specified square kills $H \cap V_r$, and the quotient itself admits that square. It is therefore the universal such quotient. The canonical source quotient $f \to f/N_f$ has the same property under the established equivalence. Their kernels agree, proving Theorem 10.4.

H.5 Cubic encoding: a one-dimensional universal defect

The universal Engel identities also show that $\mathcal{R}$ is central, alternating, Frobenius-semilinear in each slot, and zero on a derived input. Work in a finite relatively free three-generator three-Engel algebra, with a class bound covering finite three-generator targets. Its $\mathcal{R}$-ideal is thus spanned by the single value $\mathcal{R}(a, b, c)$. By the exact-image theorem, both the associator of (60) and its prescribed third-difference defect vanish modulo that line. They are respectively $\rho\mathcal{R}(a, b, c)$ and $\sigma\mathcal{R}(a, b, c)$ for scalars $\rho, \sigma \in k$.

Here is a finite detector fixing those scalars. In dimension 21, use basis $e_1, \ldots, e_{21}$ with the following nonzero brackets. An entry $(i, j; v)$ means $[e_i, e_j] = v$; omitted brackets are zero.

(i,j)	Value	(i,j)	Value
$(1,2)$	e_4	$(1,3)$	e_6
$(1,5)$	e_{13}	$(1,6)$	e_7
$(1,9)$	e_{14}	$(1,10)$	e_5+e_8
$(1,11)$	e_{16}	$(1,12)$	e_{15}
$(1,15)$	e_{19}	$(1,18)$	e_{20}
$(1,20)$	e_{21}	$(2,3)$	e_{10}
$(2,5)$	e_{16}	$(2,6)$	e_8
$(2,7)$	e_{13}	$(2,8)$	e_{16}
$(2,9)$	e_{17}	$(2,10)$	e_{11}
$(2,12)$	e_{18}	$(2,14)$	e_{19}
$(2,17)$	e_{20}	$(2,19)$	e_{21}
$(3,4)$	e_5	$(3,5)$	$e_{15}+e_{17}$
$(3,6)$	e_9	$(3,7)$	e_{14}
$(3,8)$	e_{15}	$(3,10)$	e_{12}
$(3,11)$	e_{18}	$(3,13)$	e_{19}
$(3,16)$	e_{20}	$(4,6)$	e_{13}
$(4,9)$	e_{19}	$(4,10)$	e_{16}
$(4,12)$	e_{20}	$(4,15)$	e_{21}
$(4,17)$	e_{21}	$(5,8)$	e_{21}
$(6,8)$	e_{19}	$(6,10)$	$e_{15}+e_{17}$
$(6,16)$	e_{21}	$(7,11)$	e_{21}
$(8,10)$	e_{20}	$(10,13)$	e_{21}

Jacobi on all basis triples and all 37191 full cubic Engel coefficients vanish. For $a=e_1, b=e_2, c=e_3$ one computes $\mathcal{R}(a,b,c)=e_{21}$, zero associator, and third difference e_{21}. Therefore $\rho=0$ and $\sigma=1$ universally. The table is independently checked by the supplied coefficient checker; relative freeness of this particular example is not needed.

Symmetry, $x \boxplus x = 0$, and the zero identity now make $\boxplus$ an elementary abelian group operation. Brackets of scalar multiples of one point vanish, so $F^2 = F +_{\boxplus} \mathrm{id}$. The trilinearity and centrality of $\mathcal{R}$ imply vanishing of every fourth difference, proving the cubic degree bound and its sharpness. On every pair of inputs the generated ordinary subalgebra has $\mathcal{R}=0$. The inverse terms therefore recover ordinary addition and bracket even for a globally cubic encoding. This proves fullness on all arrows and completes Theorem 10.3.

I Proofs for class-four obstructions and central covers

I.1 Reduction to head relations and the Plücker criterion

The cycle $a \wedge b$ with $[a,b]=0$ has λ-value zero, which proves the Bogomolov factorization. In class at most four, use the complete formula (65), including its cross sum. For a head relation in K, the total bracket lies in γ_3. Changing a head lift changes only words of weight at least five. The partial reflection in Appendix H shows that the displayed value is independent of wedge decomposition and is Frobenius-semilinear on K: its polar is the bracket of the two total bracket sums, of weight at least six. The cross sum in the formula is retained; it can be nonzero in a nonmetabelian algebra.

Projecting any CE cycle to $\Lambda^2 U$ gives a member of K. Terms with one derived input have zero λ-value by weight. Conversely a lifted head relation can be completed to a CE cycle by adding wedges with one derived input, because its bracket sum lies in γ_3. This proves that the image is exactly $N(H)$. A decomposable member has $[a,b] \in \gamma_3$, and hence $q(a,b) \in \gamma_5 = 0$. Every bivector in at most three dimensions is decomposable, giving the stated vanishing boundary.

For four-dimensional U, the decomposable cone is $P = c_{12}c_{34} + c_{13}c_{24} + c_{14}c_{23} = 0$. Extend

scalars to an algebraic closure. The Engel identity extends, since its individual coordinate degrees are less than $|k| = 4$; the semilinear map has the same rank after extension. Its kernel contains the zero cone of $P|_K$. A nonzero homogeneous quadric has its zero cone in a proper hyperplane only when it is a square of a linear form. Indeed two distinct linear factors have zero hyperplanes spanning the whole space; an irreducible reduced quadric cannot be contained in a hyperplane. Thus nonzero λ_2 requires a nonzero square, and then its rank is at most one. The polar of P is nondegenerate alternating on $\Lambda^2 U$, so such a K is totally isotropic and has dimension at most three.

For sufficiency, write $K = kw + L$ with $P(w) \neq 0$ and $P|_L = 0$. Darboux normalization gives $w = a \wedge b + c \wedge d$. Its stabilizer is $\mathrm{Sp}_4(k)$. A one-dimensional L is normalized to $k(a \wedge c)$; a two-dimensional decomposable plane has a common factor, and orthogonality gives $L = \mathrm{span}(a \wedge c, b \wedge c)$. Hence the three cases are

$$K_1 = \mathrm{span}(ab + cd), \quad K_2 = K_1 + \mathrm{span}(ac), \quad K_3 = K_2 + \mathrm{span}(bc),$$

where juxtaposition denotes a wedge in this display. The free class-four three-Engel quotient by any K_i maps onto the following eleven-dimensional algebra J:

$$\begin{gathered}
[a,b] = [c,d] = e, \quad [a,d] = u, \quad [b,d] = v, \\
[a,e] = [c,u] = r, \quad [b,e] = [c,v] = s, \quad [a,v] = [b,u] = t, \\
[a,s] = [b,r] = [c,t] = z.
\end{gathered}$$

All omitted brackets vanish. The derived algebra is abelian; its head actions commute and square to zero. The four head Jacobi identities cancel in pairs, so this is three-Engel. Its head relation space is K_3 and $\lambda_2(ab + cd) = z \neq 0$. Naturality proves sufficiency for all three cases. This J is S_2, not the full free quotient by K_3.

The anisotropic example in the main text follows from $v_1 = ab + cd$, $v_2 = ab + ac + \omega bd$: $P(xv_1 + yv_2) = x^2 + xy + \omega y^2$. Its polar is nonzero, so the scalar image vanishes even though its rational decomposable subspace is zero. The quotient $H \to H/N$ need not preserve nonzero commutators: in J, $z = [a,s]$ is killed. Thus it is not being identified with a classical commutativity-preserving cover.

I.2 The compact families and their central cost

Place a copy of J at every pair $i < j$. In the product, choose shared heads sending A_i, A_j, B_i, B_j to a, b, c, d in that factor and other heads to zero. The generated subalgebra has precisely (66): every E, U, R, S, T, Z has its single pair coordinate, whereas V_j is the sum of its v-coordinates over $i < j$. These vectors are independent and have dimensions $2r, 2m + r - 1, 3m, m$. This proves existence and the dimension formula. Its head kernel is (68), whose matched values are exactly the independent Z_{ij}. The other basis relations are decomposable, proving $N(S_r) = \gamma_4(S_r)$.

The central cochain defining (67) factors through the head and is CE-closed. It preserves the Engel identity and class. The new head kernel consists only of the decomposable relations in (68), so $N(A_r) = 0$. Theorem 11.1 now proves minimality and uniqueness of this repair over S_r.

To prove indecomposability, first consider the four-dimensional head of J. Its bracket radical is zero. A commuting $2 + 2$ head decomposition would put four independent mixed wedges in its three-dimensional relation kernel. A $1+3$ decomposition would make that entire kernel decomposable, contrary to $ab + cd$. Commuting head images cannot overlap, since an overlap would lie in the radical. In a decomposition of S_r into commuting ideals, projection to each J factor therefore assigns its entire head to one ideal. Pairs sharing an index make the same assignment; connectivity makes it uniform. The other ideal has zero head and is

zero by nilpotence. There is no central head vector, so $Z(S_r) \subseteq S_r'$. For a binary centroid idempotent, its commutator with ω has central image and kills the derived algebra. Its head action is therefore k-linear, and the preceding argument still applies. This proves binary indecomposability.

For the nonmetabelian factor D, literal free associative word elimination constructs a table with homogeneous dimensions $(4, 5, 16, 20)$ and head relation $[a, b] + [c, d] = 0$. The supplied finite table is checked directly for Jacobi and all three-Engel coefficients, for generation by its four head vectors, and for its surjection onto J. These properties suffice here; relative freeness of D is not required. Nonmetabelianity also has a short word certificate: the functional with support

$$caba + cbaa + dacc + dcac$$

kills the full Engel coefficients and the degree-four consequences $[[ab + cd, u], v]$, but evaluates to one on $[[a, b], [a, c]]$. It therefore survives in D. The presentation gives a surjection $D \twoheadrightarrow J$.

The subalgebra of S_r on A_1, A_2, B_1, B_2 is exactly J. Every element of $\ker(D \to J)$ is represented by a word vanishing in this subalgebra. The shared heads in $S_r \times_J D$ therefore generate that entire kernel and the whole fiber product. The dimension is $\dim S_r + 45 - 11 = 3r^2 + 33$. Its head kernel removes exactly the two independent decomposable relations A_1B_1 and B_1A_2 from (68). Thus $\dim N \leq m$; projection to the pair factors gives the reverse inequality. Projection to D gives nonmetabelianity, while projection to S_r transfers the head rigidity and absence of a central head. The same centroid argument proves binary indecomposability.

I.3 Finite coefficient certificates and all-rank gluing

For a homogeneous central cochain β of weight j, the equations are the ordinary CE equations and every coefficient of

$$\beta([y, x, x], x) = 0. \tag{110}$$

For a basis index y and an unordered triple (a, b, c), the latter coefficient is

$$\sum_{(i,j,k) \in \mathrm{Perm}(a,b,c)} \beta([e_y, e_i, e_j], e_k) = 0,$$

where distinct permutations are counted once. No diagonal-only test is used. Table coefficients lie in $\mathbb{F}_2$, so binary row rank is also rank over k. The certificate dimensions are

r	$\dim S_r$	$\dim A_r$	$\dim Z^2(S_r)_4$	$\dim Z^2(A_r)_4$	$\dim Z^2(A_r)_5$
2	11	12	1	1	0
3	26	29	3	3	0
4	47	53	6	6	0
5	74	84	10	10	0

Here Z^2 includes (110). Each weight-four kernel is exactly the independent coboundaries from γ_4. For example the largest three systems have $(576, 566)$, $(861, 851)$ and $(1120, 1120)$ as (number of variables, equation rank). Twelve certificates list independent original equation rows and a complementary coboundary basis. An independent pivot convention replays all 3903 selected equations and checks that rank plus kernel dimension equals the full variable count. This is the finite, computer-assisted part of Theorem 12.4.

For the general-rank argument, every basis vector has at most two index labels; the vector V_j additionally uses the anchor 1. For each ordered index subset I containing 1, its head-generated subalgebra is precisely $S_{|I|}$ or $A_{|I|}$ after relabeling. This follows from the displayed table even though V_j has additional coordinates in the product realization. Every cochain component of weight four or five involves at most four labels; adjoining 1 gives at most five.

Therefore a weight-five cocycle on A_r restricts to zero on every such subalgebra, and is zero. For weight four, prescribe $t(Z_{ij}) = \beta(A_i, S_{ij})$. Every local primitive has exactly these values, so they agree on overlaps. Subtract the global coboundary δt; all local restrictions vanish, and hence the entire cochain vanishes. This proves the weight-four assertion for S_r and A_r at every rank.

For S_r, a pure weight-five extension preserves all matched quadratic cycles. Their λ-values lift a basis of γ_4 to central vectors. Thus the lifted fourth term is central, and no weight-five value can be generated. This proves the remaining weight-five vanishing without an additional rank extrapolation.

Finally an arbitrary central extension of a class-four algebra has class at most five. Choose a linear section adapted to the lower-central filtration. Only total cochain weights at most five can occur. Its CE and full Engel equations separate by total weight; equivalently use scalar dilation of the graded base and the full coefficient identities. Each individual coordinate degree is less than four, so these coefficient identities hold over k. The weight-five part is zero. Remove the weight-four coboundary. The remaining cochain vanishes on pairs of derived inputs, whose total weight is at least four. The base is metabelian and the kernel central, so the extension is metabelian as well. Surjection onto the base forces class exactly four. This proves the ungraded statement and completes Theorem 12.4.

J Proofs of exact repair and the sharp first ascent

J.1 The minimum repair and its equality class

Take a finite free $\mathcal{E}$-object F on a minimal head tuple of H, with kernel $I \subseteq F'$. The finite free construction and Theorem 10.4 give $N(F) = 0$ and its specified derived square p. The ideal $[I, F]$ is square-stable: on its spanning brackets,

$$p([i, f]) = [i, f, f, i] \in [I, F],$$

and the polar terms of sums also lie in $[I, F]$. Consequently

$$C = F/[I, F], \qquad K = I/[I, F] \subseteq Z(C) \cap C'$$

has the descended square and $N(C) = 0$. On K the square is additive and Frobenius-semilinear, with central values. Set

$$K_0 = \{k \in K : p(k) \in K\}.$$

Since $p^2 = 0$, this is the largest square-stable subspace of K. Thus $A = C/K_0$ still has the specified square.

The semilinear map

$$\rho : K \longrightarrow H, \qquad k \longmapsto p(k) \bmod K \tag{111}$$

has image exactly $N(H)$. Indeed, lift a CE cycle $z = \sum_i a_i \wedge b_i$ from H to C and put $k_z = \sum_i [\widetilde{a}_i, \widetilde{b}_i] \in K$. Expanding the square of this sum gives $\rho(k_z) = \lambda_H(z)$. Conversely each element of $K \subseteq C'$ is such a bracket sum and defines a cycle downstairs. The kernel of (111) is K_0, so $\dim \ker(A \to H) = n(H)$.

The square respects the lower-central filtration:

$$p(\gamma_j C) \subseteq \gamma_{2j} C \qquad (j \geq 2). \tag{112}$$

To see this, span $\gamma_j C$ by left-normed words of length at least j. Apply $p([u, v]) = [u, v, v, u]$ to their last brackets, then use the square's polar formula for sums. If $c = \operatorname{cl} H$, then $\operatorname{cl} C \leq c+1$

and $\gamma_{c+1}C \subseteq K$ has square zero. It lies in K_0, so $\operatorname{cl} A = c$. The central stem property gives $d(A) = d(H)$. If H is abelian, $N(H) = 0$ and the same conclusion is simply $A = H$.

For a central extension $E \to H$ with kernel Z, the ordinary homology transgression is $\tau : H_2^{\mathrm{CE}}(H; k) \to Z$. A cycle with transgression zero lifts to an actual cycle in E. Naturality of λ therefore implies

$$N(E) = 0 \implies \ker \tau \subseteq \ker \lambda_H.$$

Hence $\dim Z \geq \operatorname{rank} \tau \geq n(H)$. At equality τ is onto and $\ker \tau = \ker \lambda_H$. After a kernel identification with the appropriate Frobenius twist of $N(H)$, it is the linearization of λ_H. The ordinary universal-coefficient isomorphism

$$H^2_{\mathrm{CE}}(H; Z) \simeq \operatorname{Hom}_k(H_2^{\mathrm{CE}}(H; k), Z)$$

then determines the extension class. This is the classical classification of central extensions over the fixed bottom and kernel; allowing a kernel isomorphism proves the equality assertion. Existence above proves that this particular ordinary class satisfies the Engel and sextic identities.

For clarity, the universal-coefficient fact requires no perfection: restrict a central cocycle to two-cycles and factor by boundaries. Every linear map on homology extends to the two-chain space over a field. Two extensions with the same restriction differ by a map factoring through the bracket, hence by a coboundary.

For the partial bound, the image of $H_2(E) \to H_2(H)$ is $\ker \tau$. Thus

$$\lambda_H(\ker \tau) \subseteq \pi N(E), \qquad n(E) \geq n(H) - \operatorname{rank} \tau \geq n(H) - \dim Z.$$

For attainment, let $Z = \ker(A \to H)$ and quotient A by a subspace $W \subseteq Z$ of dimension $n(H) - s$. Squaring on Z, followed by projection to H, is injective with image $N(H)$, so $p(Z) \cap Z = 0$. The same lifted bracket-sum calculation gives

$$N(A/W) = (p(W) + W)/W.$$

Its dimension is $\dim W = n(H) - s$. These central quotients still surject onto H, and retain its class and head rank. This proves Theorem 11.1.

J.2 Factorization and homogeneous repair

An ordinary map between $N = 0$ objects preserves the specified derived square: on a bracket this follows from the word q, and on bracket sums it follows from polarization and scalar homogeneity. For Proposition 11.2, take a finite free $N = 0$ source $F \to G \to H$ on the common head, with kernels $J \subseteq I \subseteq F'$. Put

$$I_0 = \{i \in I : p(i) \in I\}.$$

It is a vector space, contains $[I, F]$, and is therefore an ideal: $[I_0, F] \subseteq [I, F] \subseteq I_0$. It is square-stable by $p^2 = 0$. The construction above gives $A_H \simeq F/I_0$ over H. Since J is square-stable, $J \subseteq I_0$, which gives the required factorization.

For graded H, give CE chains their total weights. On a homogeneous two-cycle of weight j, every term in λ_H has weight $2j$. Additivity on cycles separates these weights. Linearize λ_H with a kernel coordinate in weight j for each image coordinate in weight $2j$. Extend this map to the chain space with homogeneous complements. It kills CE boundaries and defines a homogeneous central cocycle. By the equality assertion already proved, this is the minimum repair class. Two-chain weights start at two and $H_{2j} = 0$ for $2j > c$, proving the half-weight assertion.

We spell out the filtration step used below. For an epimorphism of nilpotent Lie algebras, $\pi(\gamma_j B) = \gamma_j A$. Thus the associated graded map is onto, and its kernel carries the induced filtration $K \cap \gamma_j B$, rather than the lower-central series of K itself. Central kernels and central flags remain central. Jacobi and the full polarized Engel and sextic identities pass to the associated graded; their individual coordinate degrees are less than four, so their full coefficient identities are valid over $k = \mathbb{F}_4$. By (112), define the descended square on a homogeneous class of weight j using its value modulo γ_{2j+1}, and extend by the quadratic polar formula. Changing a representative by γ_{j+1} changes its square by γ_{2j+1}. Thus a specified square descends to the associated graded derived algebra. Its polar, commutator identities descend. For the square-zero identity, homogeneous classes follow from the filtration. On sums use

$$p^2(u+v) - p^2(u) - p^2(v) = p([u,v]) + [p(u), p(v)] + [p(u) + p(v), [u,v]].$$

Its mixed degrees in the two inputs are $(2,2), (3,1), (1,3)$, all below four in each slot. Thus the mixed coefficient identities descend as well, proving $p^2 = 0$ on the whole graded derived algebra. Consequently $N = 0$ passes to the associated graded.

Suppose now a central extension $B \to A_H$ has class $c+1$. The subalgebra generated by lifts of a minimal head tuple still surjects onto A_H, and the omitted summand is central. It has the same lower-central terms from the second onward. Pass to its associated graded, discard kernel weights below $c+1$, and retain a scalar quotient of the nonzero top. We obtain a pure weight-$c+1$ cocycle α on the graded A_H. For $z \in Z = \ker(A_H \to H)$, the CE equation with two other entries gives

$$\alpha(z, A'_H) = 0.$$

The half-weight bound also gives $\alpha(z, (A_H)_1) = 0$. Consequently the lifted Z is central, and quotienting it leaves a central class-$(c+1)$ extension of H: its weights are below the surviving top. This contradicts the assumed rigidity of H. Minimum-cost uniqueness proves the last assertion of Theorem 11.3. This argument requires the displayed grading on H; it does not infer rigidity of an associated graded from rigidity of an arbitrary ungraded bottom.

Finally let $E \to H$ be as in (64). The map $\rho : Z \to H$, $z \mapsto p_E(z) \bmod Z$, has rank $n(H)$ by lifted bracket sums. Since $p_E(Z_c) = 0$,

$$\dim Z = n(H) + \dim \ker \rho \geq n(H) + \dim Z_c \geq n(H) + r_{\text{eff}}.$$

If $r_{\text{eff}} = 0$, centrality, the CE equation and weights imply $\alpha(Z, E) = 0$. The last extension descends to H, contradicting rigidity. This proves the quantitative statement.

J.3 Two-arrow obstruction and the general lower bound

The cocycle identities proved in Appendix I have the following consequence for $H = S_r$ or A_r: two head-preserving central arrows cannot raise its class to five. Here is the reduction, including ungraded arrows. Pass to the lower-central associated graded of the endpoint, retaining the induced central flag, and keep a scalar fifth-weight top in the last kernel. The first extension has no fifth-weight part because $Z^2_{\mathrm{CE},3E}(H)_5 = 0$. Its fourth-weight cocycle is a coboundary, so changing the section removes it. In the subalgebra generated by the common head, its remaining first kernel has weights at most three. If α is the last pure weight-five cocycle, centrality and CE give $\alpha(K, E') = 0$, and weights give $\alpha(K, E_1) = 0$. It therefore descends to a fifth-weight cocycle on H, which is zero. This is the desired contradiction.

For any endpoint counted by D_r, Proposition 11.2 gives $G \to A_r$. Write its kernel dimension as q, so that $\dim G - \dim S_r = m + q$. An ideal of dimension q in a finite nilpotent algebra admits a central flag of length at most q: choose a nonzero common adjoint-annihilated vector, quotient it, and repeat. If $q \leq 2$, this produces two central arrows over A_r, contrary to the preceding obstruction. Hence $D_r \geq m + 3$, independently of the length of the originally given tower.

J.4 The rank-four exception: a complete finite input

Suppose $r = 2$ and $q = 3$. Passing to the associated graded preserves $N = 0$, the given map to A_2, and its three-dimensional kernel K. Its central depth is at least three. On a three-dimensional nilpotent ideal any product of three head actions is zero. A nonzero twofold action therefore uses two degree-one heads and occupies three consecutive weights. Since the nonzero fifth term is in K, these weights must be $3, 4, 5$, each of dimension one.

Use complete simultaneous ordinary central covers preserving class four. The first cover of A_2 has dimension 23, with weight dimensions $(4, 6, 12, 1)$, and its next cover U has dimension 43, with dimensions $(4, 6, 20, 13)$. Completeness means that the added cochains together with the coboundaries span the entire CE/full-Engel equation kernel at each weight. Every hypothetical chain with kernel weights $3, 4, 5$ is a quotient of a scalar weight-five extension of U, by successive pullback. The top cochain space has 172 variables, equation rank 144, and dimension 28.

Put $J = \ker(U \to A_2)$. For a top cocycle α write

$$R_4(h)(v) = \alpha(v, h) \quad (v \in J_4), \qquad R_3(h, k)(u) = \alpha([u, h], k) \quad (u \in J_3).$$

In the hypothetical quotient both action matrices have rank one: there are nonzero head covectors b, c and nonzero functionals u, v such that

$$R_4(h) = c_h v, \qquad v \circ \operatorname{ad} h = b_h u.$$

Each normalized head covector has $(4^4 - 1)/3 = 85$ possibilities. For every one of the $85^2 = 7225$ pairs, these are linear equations on the full 28-dimensional cochain space over $\mathbb{F}_4$. Exact elimination gives rank 28 in every case, excluding a nonzero α.

This is the finite computer-assisted part of the sharp minimum. `verification/central_repair/` contains the actual cover table, the complete field-wide certificate, and a separate matrix generator and opposite-pivot replay. The replay checks every weight of both covers for completeness, reconstructs the top cochain kernel, and tests all 7225 full $\mathbb{F}_4$ systems. It does not enumerate only binary cochain values. Thus $q = 3$ is impossible at $r = 2$.

J.5 Exterior coupling and all-rank attainment

Let $T = k[t_1, t_2, t_3]/(t_1^2, t_2^2, t_3^2)$, and let the abelian algebra with basis a_1, a_2, a_3 act on T by multiplication. Write $b = 1$, $v_i = t_i$, $e_{ij} = t_i t_j$, and $q = t_1 t_2 t_3$. On this semidirect product define a central cocycle by

$$\alpha(a_i, -) = 0, \qquad \alpha(u, v) = [t_1 t_2 t_3](uv) \quad (u, v \in T).$$

It is alternating because u^2 is scalar. The CE equation with entries a_i, u, v is twice the top coefficient of $t_i uv$; all other cases vanish. The resulting twelve-dimensional algebra E has the additional central coordinate z and brackets

$$[a_i, b] = v_i, \quad [a_i, v_j] = e_{ij} \ (i \neq j), \quad [a_i, e_{jk}] = q \quad (\{i, j, k\} = \{1, 2, 3\}),$$

$$[b, q] = [v_1, e_{23}] = [v_2, e_{13}] = [v_3, e_{12}] = z.$$

Give these coordinates weights $1, 2, 3, 4, 5$. For $x = a + u$, $y = a' + u'$ in the semidirect product, the square of multiplication by t_a is zero and

$$[y, x, x] = t_a t_{a'} u \quad \text{modulo the center}, \qquad \alpha(t_a t_{a'} u, u) = [t_1 t_2 t_3](t_a t_{a'} u^2) = 0.$$

This proves the full Engel identity. Class five gives $\mathcal{R} = 0$. Only weight-two cycles can contribute to $N(E)$, and its head relations are the actual commuting $a_i \wedge a_j$. Thus $N(E) = 0$, by the doubled-weight formula for λ. The same argument applies to the quotients used next.

Set

$$M_9 = E/\langle e_{12}, q, z\rangle, \quad E_{10} = E/\langle q, z\rangle, \quad E_{11} = E/kz.$$

Define $A_r \to M_9$ by

$$A_1, A_2, A_3 \mapsto a_1, a_2, a_3, \quad B_3 \mapsto b, \qquad U_{13}, U_{23}, V_3 \mapsto v_1, v_2, v_3, \qquad T_{13}, T_{23} \mapsto e_{13}, e_{23},$$

and send the other displayed coordinates to zero, including all C_{ij}. Every bracket follows from (66); the missing products $[A_1, U_{23}]$ and $[A_2, U_{13}]$ correspond to $e_{12} = 0$ in the target. Take the actual fiber products

$$C_r = A_r \times_{M_9} E_{10}, \quad B_r = A_r \times_{M_9} E_{11}, \quad G_r = A_r \times_{M_9} E.$$

The shared head word $[A_1, [A_2, B_3]]$ vanishes in A_r and equals e_{12} in E. Its successive brackets with A_3 and B_3 give q, z. Hence the shared heads generate the entire kernel over A_r, of dimension three, and the whole fiber product. This proves the table (69), exact dimension and head rank. The classes are $4, 4, 5$.

Products and Lie subalgebras of $N = 0$ objects retain the specified derived square: on a derived bracket the square is the word q, and polarization keeps the square of a bracket sum in the same derived algebra. Thus these fiber products have $N = 0$. Since A_r is metabelian and $E'' = kz$, $G_r'' = kZ$, nonzero by $[U_{13}, T_{23}] = Z$. Binary indecomposability follows from the head-rigidity argument for S_r in Appendix I. A decomposition of G_r would give commuting complementary head subspaces of S_r; one must be zero. The corresponding nilpotent summand then has zero abelianization and is zero.

J.6 The rank-four attaining seed and preparation

For $r = 2$ use the seven-dimensional seed with weights $(1, 1, 2, 3, 3, 4, 5)$, coordinates x, y, c, p, q, w, z, and brackets

$$[x, y] = c, \quad [x, c] = p, \quad [y, c] = q, \quad [x, q] = [y, p] = w, \quad [x, w] = [c, p] = z.$$

The only nonautomatic Jacobi triples are (x, y, c) and (x, y, p); their two nonzero terms cancel. For $v = ax + by + gc + dp + eq + fw + hz$ direct calculation gives

$$\begin{aligned} [x, v, v] &= abp + b^2 q + bgw + (bd + g^2 + ae)z, \\ [y, v, v] &= a^2 p + abq + agw, \\ [c, v, v] &= agz, \quad [p, v, v] = abz, \quad [q, v, v] = a^2 z. \end{aligned}$$

The corresponding expressions for w, z are zero, and another bracket with v makes every displayed expression zero. This proves the complete three-Engel identity. It has class five and $N = 0$: its weight-two head bracket is injective, so no weight-two CE cycle contributes to λ.

Its quotient on x, y, c is the Heisenberg algebra H_3. Map $A_2 \to H_3$ by $A_1 \mapsto x$, $B_2 \mapsto y$, $U_{12} \mapsto c$, and all other coordinates to zero. The pullback with the seed adds p, q, w, z, all generated by shared head words vanishing in A_2. It has dimension $12 + 7 - 3 = 16$, class five, $N = 0$, and $G_2'' = kz$. It attains $D_2 = 1 + 4 = 5$ and completes the cost proof. Only this rank-four use of the two-index attachment is needed.

For $r \geq 3$, let $\widetilde{C}_r$ be the algebra obtained by adding only the central cubic P to S_r. Its head relations and fourth-weight λ-values are unchanged, so $n(\widetilde{C}_r) = m$. A class-five central extension of it would give two arrows over S_r, which have been excluded. The map $B_r \to \widetilde{C}_r$ has central kernel the m degree-two C_{ij} and the degree-four Q; the last scalar extension pairs Q with B_3. Hence $r_{\text{eff}} = 1$ and the cost $m + 1$ attains (64). At $r = 2$ the same argument uses both cubic directions p, q, then the degree-four w, then z. A single cubic preparation retaining

$n = m$ and attaining the rank-one bound there would give total cost $m + 3$, contradicting the rank-four exclusion. Zero preparation is excluded by the two-arrow obstruction at every rank.

The constructed three-arrow towers have first kernels $m + 1$ for $r \geq 3$ and $m + 2$ for $r = 2$, followed by two scalar kernels. Two arrows cannot suffice, and a minimum repair alone as the first arrow has source A_r by equality uniqueness; three more arrows are necessary and are supplied by the constructions. This proves Theorems 13.1 and 13.2.

K Exact inputs and reproducibility

The proof of two-point freeness uses the full central-extension certificate for the displayed sixteen-dimensional model. The checker reconstructs all coefficient equations on inputs of weights one through four, verifies rank 244 among 272 unknowns, checks the 28-dimensional kernel and the 28-dimensional section image, and hence proves the zero extension quotient. The certificate is supplied in `verification/chain_rank_certificates/`.

The group `verification/ordinary_reconstruction/` supplies seven universal ordinary Engel identity DAGs, the 21-dimensional sextic detector, the 45-dimensional class-four factor, and twelve local central-rigidity certificates. The checker reconstructs 3903 independent original CE/Engel equations and verifies complementary coboundary dimensions with an independent pivot convention. Every basis Jacobi triple and all 37191 cubic coefficients of the sextic detector are checked. These finite inputs have exactly the roles specified in Appendices H and I; the quotient, PBW, homology and arbitrary-rank gluing arguments are proved there.

Additional controls reconstruct the global-stability and rank/class interfaces, the central obstruction and fixed quadratic sectors. The group `verification/central_repair/` adds the finite inputs for Theorem 13.1: the complete 23- and 43-dimensional preserving covers, the 28-dimensional top kernel, and all 7225 projective head pairs over $\mathbb{F}_4$. Its independent matrix generator verifies that the added cochains and coboundaries span each full coefficient kernel. Opposite-pivot field elimination replays every exclusion system.

The same group checks actual minimum-repair tables, the seven- and twelve-dimensional attaining seeds, the literal three-index families at $r = 3, 4, 5$, and the prepared bottoms and attaining covers. These are controls of the finite inputs and explicit constructions; the arbitrary-rank fiber-product proofs, equality argument and arbitrary-tower reductions are in Appendix J. The portable entry point is

```
python verification/run_replay.py --no-write
```

The option `--output-dir` chooses an isolated report directory. The selected rank checker uses SymPy 1.14.0; other selected controls use the Python standard library. The supplied report records the actual interpreter and exact task coverage. No whole-paper proof-assistant formalization is asserted.

References

[1] Y. Bahturin and A. Olshanskii, Nilpotent algebras, implicit function theorem, and polynomial quasigroups, *Journal of Algebra* **632** (2023), 154–193. doi:10.1016/j.jalgebra.2023.05.024. Author manuscript: `https://arxiv.org/abs/2208.09527`.

[2] V. Drensky, Weak polynomial identities and their applications, *Communications in Mathematics* **29** (2021), 291–324. doi:10.2478/cm-2021-0022.

[3] D. Groves, *Problems in Lie rings and groups*, D.Phil. thesis, University of Oxford, 2000. Oxford Research Archive.

[4] M. K. Kinyon, K. Kunen, J. D. Phillips, and P. Vojtěchovský, The structure of automorphic loops, *Transactions of the American Mathematical Society* **368** (2016), no. 12, 8901–8927. doi:10.1090/tran/6622. Author manuscript: `https://arxiv.org/abs/1210.1642`.

[5] G. Hochschild, Cohomology of restricted Lie algebras, *American Journal of Mathematics* **76** (1954), no. 3, 555–580. doi:10.2307/2372701.

[6] C. Schneider, *Sophus: Computing in nilpotent Lie algebras*, GAP package manual, version 1.27, 2022, Chapter 1. `https://gap-packages.github.io/sophus/doc/chap1.html`.

[7] X. Liu, Y. Chen, Y. Xu, and Z. Sun, Triple-cycle permutations over finite fields of characteristic two, *International Journal of Foundations of Computer Science* **30** (2019), no. 2, 275–292. doi:10.1142/S0129054119500059.

[8] W. Lu, X. Wu, Y. Wang, and X. Cao, A general construction of regular complete permutation polynomials, *Designs, Codes and Cryptography* **91** (2023), 2627–2647. doi:10.1007/s10623-023-01224-6.

[9] A. Mileva and S. Markovski, Shapeless quasigroups derived by Feistel orthomorphisms, *Glasnik Matematički* **47(67)** (2012), 333–349. doi:10.3336/gm.47.2.09.

[10] J. Pakianathan and E. Yalçın, Quadratic maps and Bockstein closed group extensions, *Transactions of the American Mathematical Society* **359** (2007), 6079–6110. doi:10.1090/S0002-9947-07-04346-2.

[11] A. A. Premet, On Cartan subalgebras of Lie p-algebras, *Mathematics of the USSR-Izvestiya* **29** (1987), no. 1, 145–157. English original.

[12] H. Strade, *Lie algebras of small dimension*, preprint, 2006. `https://arxiv.org/abs/math/0601413`.

[13] G. Traustason, Engel Lie-algebras, *The Quarterly Journal of Mathematics* **44** (1993), no. 3, 355–384. doi:10.1093/qmath/44.3.355.

[14] G. Traustason, A polynomial upper bound for the nilpotency classes of Engel-3 Lie algebras over a field of characteristic 2, *Journal of the London Mathematical Society* **51** (1995), no. 3, 453–460. doi:10.1112/jlms/51.3.453.

[15] X. Xu, X. Zeng, and S. Zhang, Regular complete permutation polynomials over $\mathbb{F}_{2^n}$, *Designs, Codes and Cryptography* **90** (2022), 545–575. doi:10.1007/s10623-021-00992-3.

[16] X. Zhang, *Nilpotent Algebras with Maximal Class in Congruence Modular Varieties*, Ph.D. thesis, University of Manitoba, 1998. University of Manitoba repository.

[17] J. Deré, The structure of underlying Lie algebras, *Linear Algebra and its Applications* **581** (2019), 471–495. doi:10.1016/j.laa.2019.07.032. Author version: `https://arxiv.org/abs/1901.10222`.

[18] Z. Araghi Rostami, M. Parvizi, and P. Niroomand, The Bogomolov multiplier of Lie algebras, *Hacettepe Journal of Mathematics and Statistics* **49** (2020), 1190–1205. doi:10.15672/hujms.455076.

[19] U. Jezernik and J. Sanchez, Irrationality of generic quotient varieties via Bogomolov multipliers, *International Mathematics Research Notices* **2024** (2024), no. 1, 284–330. doi:10.1093/imrn/rnad012. Author version: `https://arxiv.org/abs/1811.01851`.

[20] B. Li, *Extremal dimensions and critical isomorphism growth of metabelian three-Engel Lie algebras*, companion manuscript, 2026. Published baseline: doi:10.5281/zenodo.23874228.

[21] B. Li, *Central-depth stability and optimal central realizations of three-Engel Lie algebras*, companion manuscript, 2026. Published baseline: doi:10.5281/zenodo.23874234.

[22] B. Li, *Quadratic maps satisfying $f \circ f = f + \mathrm{id}$: nilpotence, Lie reconstruction, and growth*, unified research manuscript, 2026. The 209-page baseline collection: doi:10.5281/zenodo.23874173. Earlier public version: doi:10.5281/zenodo.23852642.